

SHEEP FARMING

**FIELD AWARENESS AND
VETERINARY MANAGEMENT GUIDE
- VOLUME 2**



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Feedback

The e-book titled *“Sheep Farming – Field Awareness and Veterinary Management Guide”*, authored by Dr. M. S. Saravanan, Deputy Director of the Animal Husbandry Department, Virudhunagar District, Tamil Nadu, represents a well-conceived and comprehensive contribution to the field of small ruminant health management and livestock extension. It is an exceptionally well-structured field guide, thoughtfully designed to meet the needs of veterinarians, students, and livestock farmers alike.

The document systematically integrates the fundamental principles of sheep production with contemporary veterinary practices, providing a balanced and practical synthesis of disease prevention, flock health management, and sustainable production strategies. It covers a wide range of essential topics, including disease management, vaccination schedules, orthopaedic conditions, parasitic infestations, antimicrobial and anthelmintic resistance, and sustainable farming practices in a clear, concise, and highly practical manner.

A notable strength of this work lies in its structured and methodical presentation. The author has effectively organized diverse subject areas into clearly defined sections, supported by appropriate pictorial representations, which significantly enhance readability and facilitate ease of understanding. The emphasis on evidence-based, field-oriented approaches ensures that the content remains both scientifically robust and operationally relevant.

The division of the e-book into Part I and Part II further enhances its pedagogical value, allowing for systematic learning among farmers, students, and practitioners. The content is presented in a simple yet highly organized format, incorporating clear headings, key messages, and actionable insights. This approach makes the guide particularly effective for field-level application, especially in rural and resource limited settings.

The e-book is especially commendable for addressing emerging global concerns such as antimicrobial resistance and sustainable livestock systems. The discussion on integrated parasite management and responsible drug use reflects a progressive and forward-looking perspective aligned with current international veterinary and public health priorities. The incorporation of the One Health approach further strengthens the work by emphasizing the interconnectedness of animal, human, and environmental health.

Additionally, the use of structured formats such as bullet points, clinical signs, treatment protocols, and preventive strategies enhances its utility as a quick reference guide for busy field veterinarians and livestock handlers. The emphasis on preventive care, hygiene, early diagnosis, and stakeholder collaboration underscores a modern and responsible approach to livestock management.

From an educational standpoint, this e-book serves as a valuable reference for veterinary practitioners, students, and extension personnel. Its clarity of expression, logical organization, and practical orientation make it an effective tool for capacity building and knowledge dissemination at the grassroots level.

Overall, this work constitutes a meaningful and impactful resource that contributes significantly to improving sheep health management practices and promoting sustainable livestock production. It holds considerable relevance for both academic dissemination and practical implementation, not only within India but also in similar agro-climatic regions globally. Overall, it stands as an excellent extension and training resource with strong potential for field-level impact in veterinary science and animal husbandry.

I extend my hearty congratulations to the author for his commendable work for the benefit of sheep farming community.

With regards,

A handwritten signature in black ink, appearing to read 'Uma Rani', with a horizontal line underneath.

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The Art of Disease Management in a Sheep Farm: A Preface

The stewardship of a sheep farm is a delicate balance between ancient pastoral wisdom and the rigorous precision of modern veterinary science. It is not merely a task of clinical intervention but an intricate art form where the shepherd's observation meets the clinician's diagnostic acumen. Success in this domain is measured not by the diseases we cure, but by the outbreaks we prevent.

The Philosophy of Prevention

Disease management is built upon the foundation of the Five Freedoms of Animal Welfare. When we ensure freedom from hunger, thirst, discomfort, pain, and fear, we are doing more than practicing ethics; we are fortifying the biological defences of the flock. A sheep that is nutritionally secure and environmentally comfortable is a sheep that possesses the innate resilience to withstand pathogens.

The One Health Perspective

In the modern era, the sheep farm does not exist in isolation. We must operate under the One Health paradigm, recognizing that the health of our livestock is inextricably linked to the health of the environment and the human communities that depend on them. Managing zoonotic risks and ensuring the judicious use of antimicrobials are responsibilities that elevate the veterinarian from a farm consultant to a guardian of public health.

Precision and Tradition

The future of small ruminant health lies in the integration of Precision Farming. By utilizing data-driven insights—from genetic selection of hardy indigenous breeds to the strategic monitoring of grazing patterns—we can move away from "blanket treatments" toward targeted, effective interventions. This is particularly vital in the conservation of our native germplasm, where maintaining genetic diversity is our best insurance against emerging disease threats.

The Call to Action

This work is dedicated to the stakeholders of the veterinary profession: the farmers, the field clinicians, and the students. Sharing this knowledge is the highest form of professional service. It is my hope that these pages serve as a practical compass for those navigating the complexities of flock health, ensuring that our livestock thrive as productive, healthy, and respected members of our agricultural ecosystem.

How to Assess the Health Status of a Sheep Flock

FieldvetDrMSS



1 General Behaviour

- ✓ Healthy sheep is alert and active
- ✓ Moves with the flock
- ✓ Grazes normally

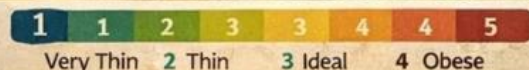
- ⚠ Warning signs: Isolation from flock
- Depression or weakness
 - Abnormal posture



Ideal flock BCS: 2.5-3.5

2 Body Condition Score (BCS)

- ✓ Assess fat and muscle around the loin area.
- ✓ Scale 1 ~ 3.5

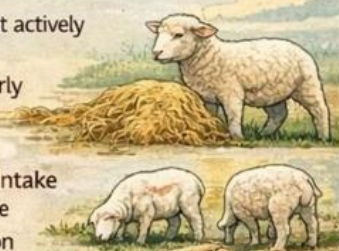


3 Appetite and Feeding Behaviour

- ✓ Healthy sheep eat actively during grazing
- ✓ Ruminates regularly

⚠ Warning signs:

- Reduced feed intake
- Refusal to graze
- Slow rumination



4 FAMACHA Eye Test

- ✓ Check the lower eyelid colour to detect anaemia by Haemonchus worms.



⚠ Ideal flock BCS: 2.5-3.5

5 Mobility and Gait

- ✓ Observe walking pattern.
- ✓ Healthy sheep
 - Walk normally
 - No lameness

⚠ Problems include:

- Foot rot
- Hoof overgrowth
- Joint infections



6 Coat and Skin Condition

- ✓ Healthy sheep have smooth and shiny wool/hair
- ✓ Clean skin without lesions

⚠ Warning signs:

- Reduced feed intake
- Refusal to graze
- Slow rumination



7 Respiration and Nasal Signs

- ✓ Normal respiration: 12 ~ 20 breaths per minute
- ✓ Warning signs:
 - Coughing
 - Nasal discharge
 - Rapid breathing



8 Lamb Health Indicators

- ✓ Healthy flocks typically show:
 - Low mortality (<5%)
 - Low disease incidence
- ✓ Low disease incidence

⚠ Warning signs:

- Weak lambs.
- scours
- High lamb mortality



Preventive Health Measures

- ✓ Regular vaccination
- ✓ Strategic deworming
- ✓ Good nutrition
- ✓ Clean water supply
- ✓ Veterinary supervision



Regular Observation is the Best Tool to Maintain a Healthy Sheep Flock.

FieldvetDrMSS

#FieldVet

Section 1: Health Management Basics

1. How to Assess the Health Status of a Sheep Flock



Importance of Flock Health Assessment

Regular health monitoring helps to:

- Detect diseases early
- Reduce mortality and economic losses
- Improve growth and reproduction
- Maintain overall flock productivity

Daily observation is the first step in flock health management.



Key Indicators to Assess Sheep Flock Health



1 General Behaviour

Healthy sheep:

- Are alert and active
- Move with the flock
- Graze normally

Warning signs:

- Isolation from flock
- Depression or weakness
- Abnormal posture



2 Body Condition Score (BCS)

Assess fat and muscle around the loin area.

- Scale 1–5

Score & Condition

- 1—Very thin
- 2—Thin
- 3—Ideal
- 4—Fat
- 5—Obese

Ideal flock BCS: 2.5–3.5



3 Appetite and Feeding Behaviour

Healthy sheep:

- Eat actively during grazing

- Ruminates regularly

Warning signs:

- Reduced feed intake
- Refusal to graze
- Slow rumination

4 FAMACHA Eye Test:

Used to detect anaemia caused by *Haemonchus* worms.

Check colour of lower eyelid:

Score - Colour-Condition

- 1 Red Healthy
- 2 Red-Pink Good
- 3 Pink Moderate
- 4 Pale Anaemia
- 5 White Severe infection

Animals with score 4–5 need treatment.

5 Coat and Skin Condition

Healthy sheep have:

- Smooth and shiny wool/hair
- Clean skin without lesions

Warning signs:

- Rough coat
- Wool loss
- Skin lesions or ticks

6 Mobility and Gait

- Observe walking pattern.

Healthy sheep:

- Walk normally
- No lameness

Problems include:

- Foot rot
- Hoof overgrowth
- Joint infections

7 Respiration and Nasal Signs

Normal respiration:

- 12–20 breaths per minute

Warning signs:

- Coughing
- Nasal discharge
- Rapid breathing

May indicate pneumonia or respiratory infection.

8 Faecal Condition

Normal faeces:

- Firm pellets

Abnormal signs:

- Diarrhoea
- Worm segments
- Blood or mucus

Often linked to parasites or digestive diseases.

9 Lamb Health Indicators

Monitor lambs for:

- Good suckling
- Active movement
- Normal growth

Warning signs:

- Weak lambs
- Scours
- High lamb mortality

10 Mortality and Morbidity Records

Healthy flocks typically show:

- Low mortality (<5%)
- Low disease incidence

Keep records for:

- Deaths
- Treatments

- Vaccinations



Preventive Health Measures

- Regular vaccination
- Strategic deworming
- Good nutrition
- Clean water supply
- Proper housing and hygiene
- Veterinary supervision



Key Message

“Regular Observation is the Best Tool to Maintain a Healthy Sheep Flock.”

Simple Diagnostic Equipment for Common Ailments of Sheep

(Field-Level & Farm Diagnostic Tools)

FieldvetDrMSS



1 Clinical Examination Tools

Stethoscope

- Used for:
 - Heart rate & rhythm
 - Lung sounds (pneumonia, bronchitis)
 - Ruminal movements

Diagnoses: Pneumonia, cardiac disorders, indigestion, rumen stasis



2 Basic Observation & Handling Tools

Torch / Headlamp

- Used for:
 - Oral cavity examination
 - Eye inspection
 - Wound examination

Diagnoses: Orf, FMD, eye infections, injuries

Diagnoses: Foot rot, foot abscess, laminitis



2 Basic Observation & Handling Tools

Torch / Headlamp

- Used for:
 - Oral cavity examination
 - Eye inspection
 - Wound examination

Diagnoses: Orf, FMD, eye infections, injuries



4 Hematological & Blood Testing Tools

Syringes, Vacutainers & Needles

- Blood sample collection

Diagnoses: Anemia, metabolic disorders, infectious diseases

Diagnoses: Anemia, parasitic infestations



3 Parasitic Disease Diagnostic Tools

Microscope

- Used for:
 - Identification
 - Fecal examination
 - Blood smear examination

Diagnoses: Gastrointestinal parasites, Coccidiosis, (Blood parasites (Anaplasmosis))



5 Digestive System Diagnostic Tools

Rumen Stomach Tube

- Used for:
 - Rumen fluid collection
 - Relief of bloat

Diagnoses: Rumen acidosis, Frothy bloat, indigestion



4 Reproductive & Urinary System Tools

Disposable Gloves

- Vaginal & rectal examination

Diagnoses: Pregnancy status, dystocia, uterine infections



8 Basic Supportive Diagnostic Tools

Weighing Tape

- Body weight estimation

Purpose: Drug dose calculation, growth monitoring



7 Skin & External Parasite Tools

Magnifying Lens

- Detection of lice, ticks, mites

Diagnoses: Ectoparasitic infestations, mange



8 Basic Supportive Diagnostic Tools

Weighing Tape

- Body weight estimation

Purpose: Drug dose calculation, growth monitoring



Key Message

Early diagnosis using simple tools reduces mortality, treatment cost, and production losses in sheep. #Fieldvet

2. Simple Diagnostic Equipment for Common Ailments of Sheep

(Field-Level & Farm Diagnostic Tools)

1. Clinical Examination Tools:

Stethoscope

Used for:

- Heart rate & rhythm
- Lung sounds (pneumonia, bronchitis)
- Ruminal movements

Diagnoses: Pneumonia, cardiac disorders, indigestion, rumen stasis

Digital Thermometer:

Used for:

- Measuring rectal temperature

Normal temperature: 38.5 – 40.0 °C

Diagnoses: Fever, septicemia, viral & bacterial infections

2. Basic Observation & Handling Tools:

Torch / Headlamp

Used for:

- Oral cavity examination
- Eye inspection
- Wound examination

Diagnoses: Orf, FMD, eye infections, injuries

Hoof Tester / Hoof Knife:

Used for:

- Hoof examination
- Detecting pain points

Diagnoses: Foot rot, foot abscess, laminitis

3. Parasitic Disease Diagnostic Tools:

Microscope

Used for:

- Fecal examination
- Blood smear examination

Diagnoses:

- Gastrointestinal parasites
- Coccidiosis
- Blood parasites (Anaplasmosis)

Fecal Examination Kit:

(Slides, coverslips, flotation solution)

Used for:

- Detection of worm eggs & oocysts

Diagnoses: Helminthiasis, coccidiosis

4. Hematological & Blood Testing Tools: **Syringes, Vacutainers & Needles****Used for:**

- Blood sample collection

Diagnoses: Anemia, metabolic disorders, infectious diseases

Hemoglobin Meter / PCV Tubes:**Used for:**

- Estimation of hemoglobin / packed cell volume

Diagnoses: Anemia, parasitic infestations

5. Digestive System Diagnostic Tools:**Rumen Stomach Tube****Used for:**

- Rumen fluid collection
- Relief of bloat

Diagnoses:

- Rumen acidosis
- Frothy bloat
- Indigestion

pH Paper:

Used for: Measuring rumen fluid pH

Normal rumen pH: 6.0 – 7.0

Diagnoses: Acidosis, alkalosis

6. Reproductive & Urinary System Tools:

Disposable Gloves

Used for:

- Vaginal & rectal examination

Diagnoses: Pregnancy status, dystocia, uterine infections

Urine Collection Container:

Used for:

- Urine examination

Diagnoses: Urolithiasis, urinary infections

7. Skin & External Parasite Tools:

Magnifying Lens

Used for:

- Detection of lice, ticks, mites

Diagnoses: Ectoparasitic infestations, mange

8. Basic Supportive Diagnostic Tools:

Weighing Tape

Used for:

- Body weight estimation

Purpose: Drug dose calculation, growth monitoring

 Record Book / Mobile App:

Used for:

- Recording temperature, treatments, vaccination, deworming

Importance: Disease monitoring & prevention planning

Key Message:

Early diagnosis using simple tools reduces mortality, treatment cost, and production losses in sheep.

Essential Vaccines for Sheep Health

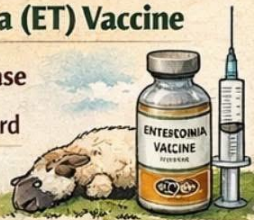
Protect Your Sheep from Deadly Diseases!

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1 Enterotoxaemia (ET) Vaccine

➔ Pulpy Kidney Disease

- Age: 3 months onward
- Booster: Annually (before monsoon)



2 Sheep Pox Vaccine

➔ Sheeppox Virus

- Age: 3 months onward
- Booster: Once yearly
- Important in endemic areas



3 PPR Vaccine

➔ Goat Plague (PPR)

- Age: 3-4 months onward
- Important in endemic areas



4 Foot and Mouth Disease (FMD) Vaccine

➔ Foot & Mouth Disease

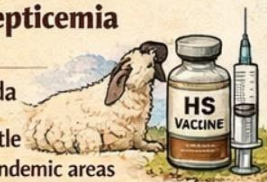
- Age: 4 months onward
- Booster: Every 6 months
- Govt: supported vaccination in many states



5 Hemorrhagic Septicemia (HS) Vaccine

➔ Pasteurella multocida

- More common in cattle but used in sheep in endemic areas



6 Black Quarter (BQ) Vaccine

➔ Clostridium chauvoei

- Used where outbreaks occur
- Booster: Annual



7 Anthrax Vaccine

➔ Bacillus anthracis

- Used only in endemic areas
- Booster: Annual



8 Bluetongue Vaccine

➔ Bluetongue Virus

- Recommended in high-risk southern & western states
- Booster: Annual (before vector season)



Suggested Annual Vaccination Calendar

(India - Semi-arid Region)

Jan-Feb	PPR	PPR (if due)
Feb-Mar	ET (Booster)	ET (Booster)
May-June	HS + BQ	HS + BQ
June-July	FMD	FMD
Before Monsoon	Sheep Pox	Sheep Pox
Vector season (regime specifics)	Bluetongue	Bluetongue

Important Field Guidelines:

- ✓ Vaccinate Only Healthy Animals
- ✓ Avoid Vaccinating during High Fever or Stress
- ✓ Maintain Cold Chain (2-8 °C)
- ✓ Deworm 2-3 weeks Before Vaccination (if required):



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Section 2: Vaccination & Preventive Care

3. Essential Vaccines for Sheep Health

Here is a practical field-oriented list of vaccines commonly used for sheep flocks in India, useful for flock health planning and extension education.

1 ☐ Enterotoxaemia (ET) Vaccine

- Protects against: *Clostridium perfringens* (Type D)
- Also called: Pulpy Kidney disease
- Age: 3 months onward
- Booster: Annually (before monsoon)

2 ☐ Sheep Pox Vaccine

- Protects against: Sheep pox virus
- Age: 3 months onward
- Booster: Once yearly
- Important in endemic areas

3 ☐ PPR Vaccine

- Protects against: Peste des petits ruminants virus
- Disease: PPR (Goat Plague)
- Age: 3–4 months onward
- Booster: Usually once in 3 years (as per state program)

4 ☐ Foot and Mouth Disease (FMD) Vaccine

- Protects against: Foot-and-mouth disease virus
- Age: 4 months onward
- Booster: Every 6 months
- Government-supported vaccination in many states

5 ☐ Hemorrhagic Septicemia (HS) Vaccine

- Protects against: *Pasteurella multocida*
- More common in cattle but used in sheep in endemic areas
- Annual vaccination before monsoon

6 ☐ Black Quarter (BQ) Vaccine

- Protects against: *Clostridium chauvoei*
- Used where outbreaks occur
- Annual vaccination

7 ☐ Anthrax Vaccine

- Protects against: *Bacillus anthracis*
- Used only in endemic areas
- Annual vaccination

8 ☐ Bluetongue Vaccine

- Protects against: Bluetongue virus
- Recommended in high-risk southern and western states
- Annual (before vector season)

Suggested Annual Vaccination Calendar (India – Semi-Arid Region):

Month & Vaccine

- Jan–Feb - PPR (if due)
- Feb–Mar - ET (Booster)
- May–June - HS + BQ
- June–July - FMD

Before Monsoon

- Sheep Pox
- Vector season (region-specific)
- Bluetongue.

Vaccination Schedule for Lambs:

- 3 months: ET + PPR
- 4 months: FMD
- 6 months: Sheep Pox
- Annual boosters as per protocol

⚠ Important Field Guidelines:

- ✓ Vaccinate only healthy animals
- ✓ Avoid vaccinating during high fever or stress
- ✓ Maintain cold chain (2–8°C)
- ✓ Deworm 2–3 weeks before vaccination (if required)
- ✓ Use sterile syringes.

PPR Vaccination Strategy for Sheep in Endemic Areas

(Peste des Petits Ruminants)

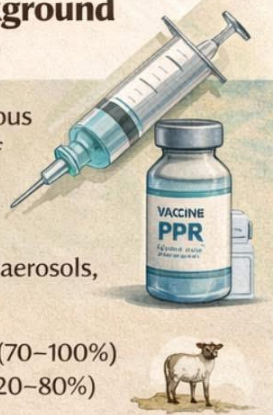
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Objective

- ✓ Prevent PPR outbreaks
- ✓ Achieve herd immunity > 80–90%
- ✓ Reduce mortality, economic losses & virus circulation
- ✓ Support PPR eradication programs

Disease Background (Endemic Areas)

- ✓ Highly contagious viral disease of sheep & goats
- ✓ Transmission: Direct contact, aerosols, secretions
- ✓ High morbidity (70–100%) and mortality (20–80%)
- ✓ Young animals most susceptible



Vaccination Schedule (Endemic Area)

1. Primary Vaccination

- Age: 3–4 months
- Avoid maternal antibody interference
- Vaccinate all susceptible sheep

2. Mass Vaccination

- Once yearly in endemic regions
- Cover entire flock at the same time

3. Revaccination

- Every 3 years (or as per national program)
- Annual vaccination in high-risk zones

Vaccination Schedule (Endemic Area)

1. Primary Vaccination

- Age: 3–4 months
- Avoid maternal antibody interference
- Vaccinate all susceptible sheep

2. Mass Vaccination

- Once yearly in endemic regions
- Cover entire flock at the same time
- Best period: **Before monsoon** / before animal movement

3. Revaccination

- Every 3 years (or as per national program)
- Annual vaccination in high-risk zones

Route & Dose

Route: Subcutaneous

- Dose: 1 ml per animal
- Use sterile syringes and needles



Special Groups

✦ Lambs (<3 months)

✦ **Newly introduced animals:**

✦ Vaccinate + quarantine (14 days)

✦ **Pregnant ewes:**

Vaccinate only if outbreak risk is high

✦ **Newly introduced animals:**
quarantine \ (4 days)

Key Message

- High vaccination coverage is the cornerstone of PPR control in endemic areas."

#Fieldvet



#PPRControl #VeterinaryCare #SheepHealth



4. PPR Vaccination Strategy for Sheep in Endemic Areas

(Peste des Petits Ruminants)

Objective:

- Prevent PPR outbreaks
- Achieve herd immunity $\geq 80\text{--}90\%$
- Reduce mortality, economic losses & virus circulation
- Support PPR eradication programs

Disease Background (Endemic Areas):

- Highly contagious viral disease of sheep & goats
- Transmission: Direct contact, aerosols, secretions
- High morbidity (70–100%) and mortality (20–80%)
- Young animals most susceptible

Vaccine Used:

- Live attenuated PPR vaccine
- Thermostable (preferred for field use)
- Provides long-term immunity (3–4 years)
- Safe in sheep & goats

Vaccination Schedule (Endemic Area):

1 Primary Vaccination

- Age: 3–4 months
- Avoid maternal antibody interference
- Vaccinate all susceptible sheep

2 Mass Vaccination

- Once yearly in endemic regions
- Cover entire flock at the same time
- Best period: Before monsoon / before animal movement

3 Revaccination

- Every 3 years (or as per national program)
- Annual vaccination in high-risk zones

Special Groups:

- Lambs (<3 months): Delay vaccination
- Pregnant ewes: Vaccinate only if outbreak risk is high
- Newly introduced animals: Vaccinate + quarantine (14 days)

Cold Chain & Handling:

- Maintain 2–8 °C
- Do not freeze vaccine
- Use reconstituted vaccine within 2 hours
- Protect from direct sunlight

Route & Dose:

- Route: Subcutaneous
- Dose: 1 ml per animal
- Use sterile syringes and needles

Do Not Vaccinate:

- Sick or febrile animals
- Severely stressed sheep
- During active outbreak (unless ring vaccination is advised)

Integrated Control Measures:

- Restrict animal movement
- Quarantine new stock
- Improve nutrition & biosecurity
- Early disease reporting
- Proper carcass disposal

Monitoring & Records:

- Maintain vaccination records
- Identify vaccinated animals (ear tag/marking)
- Monitor post-vaccination immunity
- Report adverse reactions (rare)

Key Message:

“High vaccination coverage is the cornerstone of PPR control in endemic areas.”



Vaccination Side Effects in SHEEP



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Common Reactions

- Swelling at injection site
- Mild fever
- Reduced feed intake



Moderate Reactions

- Large swelling
- Lameness
- Facial swelling



Severe Reactions

Anaphylaxis (Severe Allergy)

- Sudden collapse
- Breathing distress
- Frothing



Immediate Treatment:

- Adrenaline Injection
- Antihistamines
- Supportive Fluids



Abortion Risk

- Avoid in early pregnancy



Precautionary Measures

- Vaccinate only healthy sheep
- Keep vaccines cold (2-8°C)
- Use sterile needles
- Keep emergency medicines ready



Monitoring After Vaccination

- Observe animals for 30 minutes
- Recheck flock after 24 hours
- Record adverse reactions



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5. Vaccination side effects in Sheep flock

Vaccination in sheep flocks is essential for preventing major diseases, but mild to severe adverse reactions can occasionally occur. As a field veterinarian, understanding these reactions helps in planning safer flock-level immunization programs.

1. Common (Mild & Expected) Side Effects:

These are usually self-limiting and resolve within 24–72 hours:

◆ Local Reactions

- Swelling at injection site
- Warmth and mild pain
- Small granuloma or sterile abscess (especially with oily adjuvant vaccines)

◆ Systemic Reactions

- Mild fever (0.5–1.5°C rise)
- Lethargy
- Reduced feed intake for 1–2 days
- Temporary drop in milk yield (in lactating ewes)

2. Moderate Reactions:

- Large injection-site swelling (>5 cm)
- Persistent abscess formation
- Stiffness or lameness (if injected incorrectly)
- Mild hypersensitivity (urticaria, facial swelling)

These may require NSAIDs or antihistamines.

3. Severe (Rare but Critical) Reactions:

⚠ Anaphylaxis

- Occurs within minutes

Signs:

- Sudden collapse
- Dyspnea
- Frothing
- Pale mucous membranes
- Convulsions

Immediate Treatment:

- Adrenaline (Epinephrine) IM or IV
- Antihistamines

- Corticosteroids
- Supportive fluids

4 ☐☐ Abortion Risk:

Some vaccines (especially live vaccines) may cause abortion if:

- Given during early pregnancy
- Cold chain was compromised
- Animals were immunocompromised

Always verify gestational stage before vaccination.

5 ☐☐ Vaccine-Specific Reactions (Common in Sheep):

- Enterotoxaemia vaccine – Local swelling common
- PPR vaccine – Mild fever in some animals
- Sheep pox vaccine – Local pox lesion at injection site
- Brucella Rev-1 – May cause abortion if given to pregnant ewes

6 ☐☐ Risk Factors in Flocks

- Poor cold chain maintenance
- Vaccinating stressed animals (transport, heat stress, parasitism)
- Incorrect injection site (IM vs SC)
- Reusing needles
- Overdosing
- Mixing vaccines improperly

7 ☐☐ Preventive Measures for Flock Safety

- ✓ Vaccinate only healthy animals
- ✓ Avoid extreme heat hours (important in Indian field conditions)
- ✓ Maintain cold chain (2–8°C)
- ✓ Use sterile needles and change frequently
- ✓ Follow correct route (usually SC in sheep)
- ✓ Keep adrenaline ready during mass vaccination
- ✓ Avoid vaccinating heavily pregnant ewes unless recommended

8 ☐☐ Field Monitoring Protocol (Recommended)

For flock vaccination programs:

- Observe animals for 30 minutes post-vaccination
- Recheck flock after 24 hours
- Record batch number and manufacturer
- Maintain adverse event log

VACCINATION SCHEDULE FOR SHEEP — ASIA REGION

(General guideline – Adapt to local disease prevalence & govt programmes)

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Lamb Vaccination Schedule

Age	Vaccine	Disease Protected
Birth – 7 days	Tetanus Antitoxin	Tetanus (ff dam not vaccinated)
4–6 weeks	Enterotoxemia + Tetanus (1st Dose)	Pulpy Kidney, Tetanus
8–10 weeks	Enterotoxemia + Tetanus (Booster)	Clostridial Diseases
3 months	PPR Vaccine	Peste des Petits Ruminants
3–4 months	FMD Vaccine	Foot and Mouth Disease
3–4 months	Sheep Pox Vaccine	Sheep pox
Risk Areas	Orf Vaccine	Contagious Ecthyma

Adult Sheep Vaccination Schedule

- Enterotoxemia + Tetanus Once Yearly
- FMD Every 6 Months
- PPR Once Yearly*
- Sheep Pox Endemic Areas
- Anthrax Before Monsoon (Risk Zones)
- Orf As per Outbreak Risk

Pregnant Ewes (Pre-Lambing)

- Clostridial Booster 4–6 weeks before lambing
- Ensures Strong Colostral Immunity

Core Vaccines (Asia Region)

- PPR
- FMD
- Enterotoxemia
- Tetanus
- Sheep Pox (Endemic Areas)

Risk-Based Vaccines

- Anthrax
- Orf
- Bluetongue
- Enzootic Abortion

Important Guidelines



Vaccinate Healthy Animals



Keep Vaccines Cold (2–8°C)



No Deworming on Vaccination Day



Record Date & Batch No.

Asia-Specific Note

- Mass PPR & FMD Campaigns



- Consult Local Veterinarians

#FieldVet • Prevent Disease • Improve Productivity • Protect Livelihoods

6. Vaccination Schedule - Asia

(Field-Level Preventive Health Calendar)

Vaccination is a key component of flock health management in India. A proper schedule protects sheep from major infectious diseases that cause high mortality and economic loss.

Recommended Vaccination Schedule

Age / Time	Disease	Vaccination Schedule
1–2 months	Enterotoxaemia (Pulpy Kidney)	First vaccination
1–2 months	PPR (Peste des Petits Ruminants)	Single dose
2–3 months	Sheep Pox	First vaccination
3 months	Foot and Mouth Disease (FMD)	First dose
3–4 months	Anthrax	Vaccinate in endemic areas
3–4 months	Black Quarter (BQ)	Vaccinate in endemic areas
Pregnant ewes (1 month before lambing)	Enterotoxaemia	Booster to protect lambs
Every 6 months	FMD	Booster
Every year	Enterotoxaemia	Annual booster
Every year	Sheep Pox	Annual vaccination
Every 3 years	PPR	Revaccination
Every year	Anthrax	In endemic areas

Major Diseases Covered

1. Peste des Petits Ruminants (PPR)

- Highly contagious viral disease
- Causes high mortality in sheep and goats
- National control programme implemented in India

2 Enterotoxaemia (Pulpy Kidney)

- Caused by *Clostridium perfringens*
- Sudden death in well-fed lambs

3 Sheep Pox

- Viral disease-causing skin lesions and fever
- Leads to severe production loss

4 Foot and Mouth Disease (FMD)

- Highly contagious disease affecting cloven-hoofed animals
- Causes lameness, mouth lesions, and production loss

5 Anthrax

- Zoonotic disease affecting livestock and humans
- Vaccination recommended in endemic regions

6 Black Quarter (Blackleg)

- Bacterial disease affecting muscles
- Occurs mainly in young animals grazing contaminated soil

Important Vaccination Guidelines

- ✓ Vaccinate healthy animals only
- ✓ Maintain vaccine cold chain (2–8°C)
- ✓ Avoid vaccination during severe stress or extreme heat
- ✓ Use sterile needles and proper dosage
- ✓ Maintain vaccination records

Significance for Sheep Farmers

A proper vaccination program helps to:

- Reduce mortality in sheep flocks
- Improve meat production and flock productivity
- Prevent major disease outbreaks
- Increase farm profitability

Key Message

“Prevention through vaccination is the most economical and effective strategy in sheep flock health management.”

Vaccination Schedule for Sheep

– Europe Region

(Temperate climate · Intensive & semi-intensive systems
· High biosecurity standards)

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Core (Essential) Vaccinations

1. Clostridial Diseases (Multivalent ~6 to 8 way)

- Diseases covered:
 - Enterotoxemia (Pulpy kidney - *Cl. perfringens* type D)
 - Tetanus
 - Black disease
 - Blackleg
 - Malignant odema

🔍 Schedule: Lambs; 6–8 weeks
2nd dose: 4 weeks later

- Booster: Annually
- Pregnant ewes: 4~6 weeks before lambing

2 Pasteurellosis (*Mannheimia haemolytica* / *Bibersteinia tcheli*)

🌸 Pasteurellosis (*Mannheimia haemolytica* / *Bibersteinia*

- Importance: Pneumonia, sudden death in lambs

🔍 Schedule:

- Lambs: 4–6 weeks.
- Booster after 4 weeks
- Annual booster before cold / inrsj.

3 Footrot (*Dichelobacter nodosus*)

- Indication: Farms with chronic footrot

🔍 Schedule: Initial: 2 doses 4–6 weeks apart
Booster: 6–12 months
Best before wet season

4 Caseous Lymphadenitis (CLA)

- Indication: *Corynebacterium pseudotuberculosis*

🔍 Schedule: Lambs, 3–4 months
Booster annually (high-prevalence flocks)

Suggested Annual Vaccination Calendar

Spring	Key Vacines
• Spring	Clostridial, Bluetongue
Pre-Breeding	Toxoplasmosis, EAE
Autumn	Pasteurellosis, hovinot
Pre-Lambing	Clostridial booster

Important Guidelines

- ✓ Follow country-specific regulations (Eu nations vary)
- ✓ Maintain cold chain
- ✓ Vaccinate only healthy animals
- ✓ Record batch number & date
- ✓ Combine vaccination with biosecurity and nutrition

Important Guidelines

- ✓ Follow country-specific regulations
- ✓ (Eu nations vary)
- ✓ Maintain a cold chain:
- ✓ Vaccinate only healthy animals
- ✓ Record batch number & date



Key Message:

Strategic vaccination is the backbone of sheep health management in Europe, preventing major economic and reproductive losses.



Veterinary Preventive Medicine | Sheep Health Management | #Fieldvet

Strategic vaccination is the backbone of sheep health management in Europe, preventing major economic and reproductive losses.

7. Vaccination Schedule – Europe Region

(Temperate climate • Intensive & semi-intensive systems • High biosecurity standards)

Core (Essential) Vaccinations:

(Recommended for all sheep flocks across Europe)

1. Clostridial Diseases (Multivalent – 6 to 8 way)

Diseases covered:

- Enterotoxemia (Pulpy kidney – *Cl. perfringens* type D)
- Tetanus
- Black disease
- Blackleg
- Malignant oedema

Schedule:

- Lambs:
- 1st dose: 6–8 weeks
- 2nd dose: 4 weeks later
- Booster: Annually
- Pregnant ewes: 4–6 weeks before lambing

2. Pasteurellosis (*Mannheimia haemolytica* / *Bibersteinia trehalosi*):

Importance: Pneumonia, sudden death in lambs

Schedule:

- Lambs:
- 1st dose: 4–6 weeks
- Booster after 4 weeks
- Annual booster before cold / housing season

Region-Specific / Risk-Based Vaccinations:

3. Bluetongue (Serotype specific – EU regulated)

Vector: *Culicoides* midges

Status: Mandatory in endemic or outbreak zones

Schedule:

- As per national veterinary authority
- Usually annual before vector season (spring)

4. Footrot (*Dichelobacter nodosus*):

Indication: Farms with chronic footrot

Schedule:

- Initial: 2 doses 4–6 weeks apart
- Booster: 6–12 months
- Best before wet season

5. Caseous Lymphadenitis (CLA):

Cause: *Corynebacterium pseudotuberculosis*

Schedule:

- Lambs: 3–4 months
- Booster annually (high-prevalence flocks)

6. Enzootic Abortion of Ewes (EAE):

Cause: *Chlamydia abortus*

Schedule:

- Replacement ewe lambs: Before first mating
- Single dose (long-term immunity)

7. Toxoplasmosis:

Risk: Abortion storms

Schedule:

- Ewe lambs: At least 3 weeks before mating
- Single lifetime vaccination.

Optional / Special Situations:

- Orf (Contagious ecthyma)
- Only in infected flocks
- Q Fever
- Zoonotic risk farms
- Louping ill
- Tick-endemic upland regions.

Suggested Annual Vaccination Calendar:**Season & Key Vaccines**

- Spring —Clostridial, Bluetongue
- Pre-Breeding —Toxoplasmosis, EAE
- Autumn —Pasteurellosis, Footrot
- Pre-Lambing—Clostridial booster.

Important Guidelines

- Follow country-specific regulations (EU nations vary)
- Maintain cold chain
- Vaccinate only healthy animals
- Record batch number & date
- Combine vaccination with biosecurity and nutrition

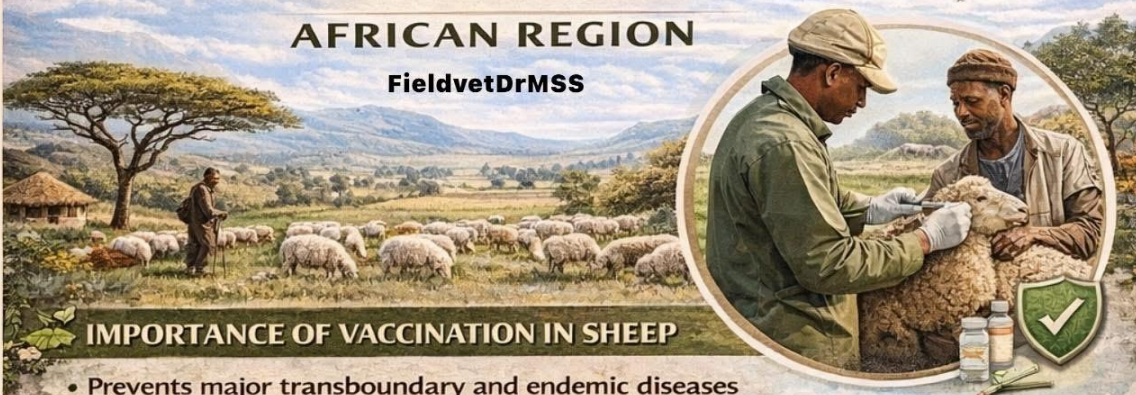
Key Message:

Strategic vaccination is the backbone of sheep health management in Europe, preventing major economic and reproductive losses.

VACCINATION SCHEDULE — FOR SHEEP —

AFRICAN REGION

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IMPORTANCE OF VACCINATION IN SHEEP

- Prevents major transboundary and endemic diseases
- Reduces mortality and production losses
- Essential for pastoral and semi-intensive systems
- Supports food security and livelihood sustainability

RECOMMENDED VACCINATION SCHEDULE

LAMBS (0–6 MONTHS)

Age	Vaccine	Disease Prevented
1–2 weeks	 Enterotoxaemia	Enterotoxaemia (Clostridium perfringens type C & D) 
4–6	PPR	Peste des Petits Ruminants
6–8	Sheep Pox	Sheep pox
8–10	Anthrax	Anthrax (endemic areas only)
10–12 weeks	FMD	Foot and Mouth Disease

PREGNANT EWES

- Enterotoxaemia: 4–6 weeks before lambing
- Avoid live vaccines in advanced pregnancy
- Ensures **passive immunity** to lambs through colostrum

REGION-SPECIFIC PRIORITY DISEASES

- | | |
|---|---|
| ☒ PPR | ☒ Sheep Pox |
| ☒ Sheep Pox | ☒ Enterotoxaemia |
| ☒ Anthrax  | ☒ Foot and Mouth Disease (selected regions) |

BEST TIME FOR VACCINATION

- ☒ Before rainy season
- ☒ Prior to animal movement or migration
- ☒ Avoid extreme heat and stress periods

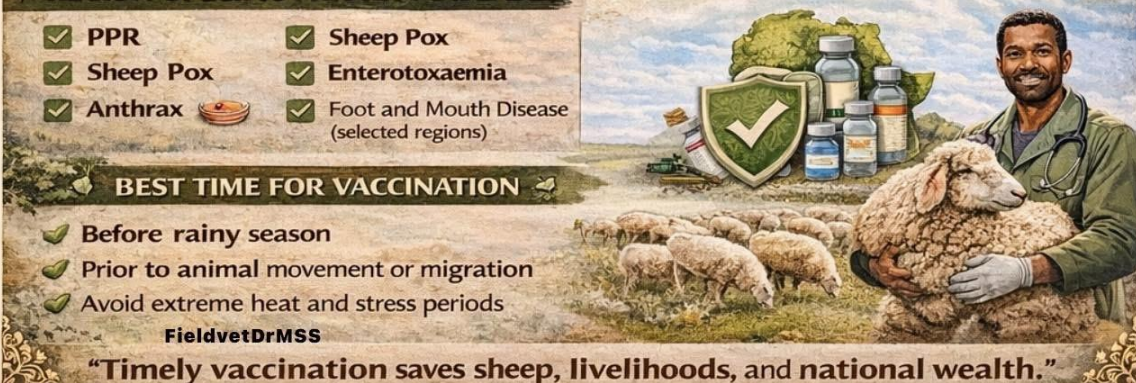
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ADULT SHEEP

Frequency	Vaccine	Remarks
Every 6–12 months	PPR	Annual in endemic zones
Every year	Sheep Pox	Before rainy season
Every year	Enterotoxaemia	Booster annually
Every year	Anthrax	High-risk / endemic areas
Every 6 months	FMD	Border & high-risk regions
Once / outbreak areas	Rift Valley Fever	As per government advisory

BEST TIME FOR VACCINATION

- ☒ Before rainy season
- ☒ Prior to animal movement or migration
- ☒ Avoid extreme heat and stress periods



“Timely vaccination saves sheep, livelihoods, and national wealth.”

8. Vaccination Schedule – African Region

(General Field Guideline for Flock Health Protection)

Vaccination is essential in African sheep production systems to prevent highly contagious and economically important diseases. The schedule may vary slightly by country depending on disease prevalence and veterinary regulations

Recommended Vaccination Schedule

Age / Time	Disease	Vaccine Schedule
At 1–2 weeks of age	Enterotoxaemia (Pulpy Kidney)	First vaccination
4–6 weeks	Enterotoxaemia booster	Booster dose
2–3 months	Sheep Pox	Single dose
3 months	Peste des Petits Ruminants (PPR)	Single vaccination
3–4 months	Anthrax	Annual vaccination in endemic areas
3–4 months	Black Quarter (Blackleg)	First dose
6 months	Brucellosis (female lambs only)	Single dose
Every year	PPR	Annual vaccination
Every year	Enterotoxaemia	Annual booster
Every year	Sheep Pox	Annual vaccination in risk areas
Every year	Anthrax	Annual vaccination in endemic zones

Important Diseases Covered

1. Peste des Petits Ruminants (PPR)

- Highly contagious viral disease
- Causes high mortality in sheep and goats
- Vaccination is very important across Africa

2 Enterotoxaemia (Pulpy Kidney Disease)

- Caused by *Clostridium perfringens*
- Often occurs in well-fed or fast-growing lambs

3 Sheep Pox

- Viral disease-causing skin lesions and severe production losses

4 Anthrax

- Zoonotic bacterial disease
- Occurs in endemic grazing areas

5 Black Quarter (Blackleg)

- Caused by *Clostridium chauvoei*
- Usually affects young animals

Vaccination Management Tips

- ✓ Vaccinate only healthy animals
- ✓ Maintain cold chain (2–8°C) for vaccines
- ✓ Avoid vaccination during severe stress or transport
- ✓ Follow booster schedules strictly
- ✓ Maintain proper vaccination records

Significance for Sheep Farmers

Proper vaccination helps to:

- Reduce mortality in flocks
- Improve meat and wool production
- Prevent disease outbreaks
- Protect farmer livelihood

Key Message

A well-planned vaccination program is the backbone of profitable sheep farming and flock health management.

Essential Vaccines for African Sheep Farming

Protect Your Sheep Flock from Deadly Diseases!

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1 Clostridial Diseases

→ Polyvalent vaccines (5-in-1, 7-in-1, 8-in-1)

Protect against:

- Enterotoxemia (Pulpy kidney)
- Tetanus
- Blackleg
- Black Disease
- Malignant Edema



2 Peste des Petits Ruminants (PPR)

→ Peste des petits ruminants virus

→ Highly endemic across Africa

→ Common in East & North Africa



3 Sheep Pox

→ Sheeppox virus

→ Common in East & North Africa



4 Rift Valley Fever (RVF)

→ Rift Valley fever virus

→ Periodic outbreaks (heavy rainfall years)

→ Important zoonotic impact



5 Brucellosis

→ *Brucella melitensis*

→ Vaccine: Rev.1 strain

→ Used in control programs



6 Pasteurellosis / Mannheimiosis

→ *Mannheimia haemolytica*

→ *Pasteurella multocida*

→ Important in intensive & feedlot systems



7 Contagious Caprine Pleuropneumonia

→ *Mycoplasma capricolum*
subspecies capripneumoniae

→ Primarily Goat disease, but relevant in mixed flocks



8 Anthrax

→ *Bacillus anthracis*

→ Endemic zones

→ Annual vaccination
high-risk districts



Regional Vaccine Focus

East Africa	PPR, Sheep pox, RVF, Clostridial
West Africa	PPR, Clostridial, Brucellosis
Southern Africa	Clostridial, Anthrax, Pasteurellosis
North Africa	PPR, Sheep pox, Brucellosis

FIELD GUIDELINES

- ✓ Vaccinate Lambs at 6–8 Weeks: Clostridial Vaccine
- ✓ Annual Booster for All Adults
- ✓ PPR Vaccine to Control Mortality in Endemic Areas
- ✓ High-Risk Areas: RVE Vaccine Before Rainy Season
- ✓ Follow National Disease Control Programs

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9. Core Vaccines Used in African Sheep Production

1 Clostridial Diseases (Core Vaccine – Widely Used):

Polyvalent vaccines (5-in-1, 7-in-1, 8-in-1)

Protect against:

- Clostridium perfringens (Enterotoxemia / Pulpy kidney)
- Clostridium tetani (Tetanus)
- Clostridium chauvoei (Blackleg)
- Clostridium novyi (Black disease)
- Clostridium septicum (Malignant edema)

 **Most essential vaccine in African small ruminant systems**

2 Peste des Petits Ruminants (PPR):

Disease: Peste des Petits Ruminants

Agent: Peste des petits ruminants virus

- Highly endemic across Sub-Saharan Africa
- Mass vaccination campaigns supported by AU-IBAR & FAO
- Critical for mortality control

3 Sheep Pox:

Disease: Sheep pox

Agent: Sheeppox virus

- Common in East and North Africa
- Used in endemic regions

4 Rift Valley Fever (RVF):

Disease: Rift Valley fever

Agent: Rift Valley fever virus

- Periodic outbreaks (heavy rainfall years)
- Vaccination in high-risk zones
- Important zoonotic significance

5 Brucellosis:

Disease: Brucellosis

Agent: Brucella melitensis

- Vaccine: Rev.1 strain
- Used in organized control programs

- Major public health relevance

6 □ Pasteurellosis / Mannheimiosis:

Agents:

- Mannheimia haemolytica
- Pasteurella multocida
- Important in intensive & feedlot systems
- Often seasonal (stress-associated)

7 □ Contagious Caprine Pleuropneumonia (where mixed flocks exist):

Agent: Mycoplasma capricolum

Primarily goat disease, but relevant in mixed sheep-goat systems.

8 □ Anthrax (Endemic Zones):

Disease: Anthrax

Agent: Bacillus anthracis

- Used in endemic pastoral regions
- Annual vaccination in high-risk districts

9 □ Foot-and-Mouth Disease (Selective Use):

Disease: Foot-and-mouth disease

Agent: Foot-and-mouth disease virus

- Sheep are minor hosts
- Used mainly in national control programs

Region & Priority Vaccines:

- East Africa - PPR, Sheep pox, RVF, Clostridial
- West Africa - PPR, Clostridial, Brucellosis
- Southern Africa - Clostridial, Pasteurellosis, Anthrax
- North Africa - PPR, Sheep pox, Brucellosis.

Typical Vaccination Schedule (General Field Guideline)

- Lambs: Clostridial at 6–8 weeks + booster
- Pre-breeding ewes: PPR, Brucellosis (as per policy)
- Annual: Clostridial booster
- High-risk seasons: RVF (before rainy season)

SHEEP VACCINATION SCHEDULE - NZ & AUSTRALIA

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NEW ZEALAND

Major products commonly used include:

- ✓ Multine' 5-in-1
- ✓ Covexin 10
- ✓ Nilvax Selenised
- ✓ Phenax Classic

1

Ewes (Breeding Females)

Pre-Lambing (4-6 weeks before lambing)

- ✓ 5-in-1 or 10-in-1 booster

Ensures strong colostral antibody transfer

Typical protocol:

- If previously vaccinated: Single annual booster
- If unvaccinated:
2 doses, 4-6 weeks apart (before first lambing)

2

Rams

Annual 5-in-1 or 10-in-1 booster

- ✓ Ideally 4 weeks before joining

3

Lambs

At Docking / Marking (4-8 weeks old)

- ✓ First 5-in-1 dose
- ✓ 4-6 Weeks Later
Second 5-in-1 dose
(critical for long-term immunity)
- ✓ Annual Booster



Optional (Risk-Based)

- ✓ Orf (Scabby mouth) –
Use **Phenax Classic** if disease present
- ✓ Selenium-containing vaccines in deficient regions



AUSTRALIA

Major vaccines include:

- ✓ Ultravac' 5-in-1
- ✓ Glanvac 6
- ✓ Websters 6-in-1
- ✓ Gudair

1

Ewes

4-6 Weeks Before Lambing

- ✓ Booster 5-in-1 or 6-in-1 annually

Ensures:

- ✓ Protection against pulpy kidney
- ✓ Tetanus
- ✓ Black disease
- ✓ Malignant oedema
- ✓ Blackleg

2

Rams

Annual booster before breeding season



Lambs

- ✓ Marking (6-8 weeks)
- ✓ First 5-in-1 / 6-in-1
- ✓ 4 Weeks Later
Second dose (mandatory)
- ✓ Annual Booster

Additional Vaccines

NZ	NZ	Australia
5-in-1	5-in-1	5-in-1 / 6-in-1
2 doses	2 doses	Annual, pre-lamb
Clostridial vaccines	Common	Common in endemic areas
Orf Based	Risk-based	Risk based

Healthy Flocks Start with Smart Vaccination.

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10. Vaccines in New Zealand and Australia

Sheep Vaccines – New Zealand

Clostridial disease vaccines

These protect against the major clostridial bacteria that cause tetanus, pulpy kidney (enterotoxaemia), blackleg, malignant oedema and black disease.

- Multine® 5-in-1 – Clostridial vaccine protecting against the five common clostridial diseases.
- Nilvax® Selenised – Specialist pre-lamb clostridial vaccine with selenium and immune stimulant.
- Covexin® 10 – Expanded 10-in-1 clostridial vaccine for broader protection (commonly used on farms with high risk).
- Lamb Vaccine (selenised or plain) – Short-term tetanus and some pulpy kidney protection for lambs, often used at docking/tailing.

Other sheep vaccines

- Phenax® Classic – Vaccine against scabby mouth (contagious ecthyma / orf) in sheep and goats.

(Note: Leptospirosis vaccines like Leptavoid-2® and Leptoshield® are registered for sheep in NZ, but these are generally cattle formulations also used off-label under vet guidance.)

Sheep Vaccines – Australia:

Clostridial disease vaccines

Australian sheep flocks use similar clostridial protective vaccines, often with 5- or 6-in-1 formulations:

Common products include:

- Ultravac® 5-in-1 – Standard clostridial disease vaccine.
- Glanvac® 6 B12 / Glanvac® 6S – Six-in-one vaccines with added protection and vitamin B12 variants.
- Websters® LV 5-in-1 / LV 6-in-1 – Multivalent clostridial vaccines available in different formulations.
- Guardian 6-in-1 – Clostridial vaccine offering protection against multiple clostridial agents.
- Tasvax 5-in-1 / 6-in-1 – Clostridial protection options from Cooper's Animal Health.

Combined vaccines

- Cydectin® Eweguard products – Combined vaccination and worm control vaccines (sheep vaccine with integrated parasite protection).

Other important sheep vaccines

- Gudair® (OJD vaccine) – Controls Ovine Johne’s Disease (OJD). Single dose typically for life (or long-term immune response).
- Scabby mouth / contagious ecthyma vaccine – Commonly used where scabby mouth is present or a local risk (available through veterinary suppliers).



General Notes on Use:

Vaccination Strategy

- Clostridial vaccines are typically given in two doses 4–6 weeks apart initially, with annual boosters.
- Vaccination should be timed relative to lambing, weaning, and stress periods for best maternal antibody transfer and lamb protection.
- OJD vaccination (e.g., Gudair) has specific protocols and handling precautions due to risk of human self-injection injury; consult a veterinarian.

Veterinary Advice:

This list is illustrative; actual vaccine choices, timing, and protocols should be decided with a local veterinarian or sheep health advisor, as registration and usage guidance vary between states/territories in Australia and regions in New Zealand.

New Zealand:

Major products commonly used include:

- Multine 5-in-1
- Covexin 10
- Nilvax Selenised
- Phenax Classic



1 ♀ Ewes (Breeding Females)

Pre-Lambing (4–6 weeks before lambing)

✓ 5-in-1 or 10-in-1 booster

→ Ensures strong colostral antibody transfer

Typical protocol:

- If previously vaccinated: Single annual booster
- If unvaccinated: 2 doses, 4–6 weeks apart (before first lambing)

2 Rams

- ✓ Annual 5-in-1 or 10-in-1 booster
- ✓ Ideally 4 weeks before joining

3 Lambs

At Docking / Marking (4–8 weeks old)

- ✓ First 5-in-1 dose

4–6 Weeks Later

- ✓ Second 5-in-1 dose (critical for long-term immunity)

Annual Booster

- ✓ At 12 months of age

Optional (Risk-Based)

- ✓ Orf (Scabby mouth) – Use Phenax Classic if disease present
- ✓ Selenium-containing vaccines in deficient regions

AU Sheep Vaccination Schedule – Australia

Common vaccines include:

- Ultravac 5-in-1
- Glanvac 6
- Websters 6-in-1
- Gudair

1 Ewes

4–6 Weeks Before Lambing

- ✓ Booster 5-in-1 or 6-in-1 annually

Ensures:

- Protection against pulpy kidney
- Tetanus
- Black disease
- Malignant oedema
- Blackleg

2 Rams

- ✓ Annual booster before breeding season

3 Lambs

Marking (6–8 weeks)

- ✓ First 5-in-1 / 6-in-1

4 Weeks Later

- ✓ Second dose (mandatory)

Annual Booster

- ✓ At 12 months

Additional Australian Vaccines:

Ovine Johne's Disease (OJD)

- ✓ Gudair – Single lifetime dose (usually at marking age)

⚠ Strict administration protocol due to operator safety concerns.

Scabby Mouth (Orf)

- ✓ Live vaccine where disease risk exists

Key Veterinary Considerations

- Always complete the primary two-dose course
- Maintain cold chain (2–8°C)
- Use sterile technique
- Observe meat & wool withholding periods
- Align vaccination with parasite control & reproductive management

Anthelmintic Resistance in Sheep Flocks in Asia



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Anthelmintic Resistance (AR)

Anthelmintic Resistance is the ability of parasitic worms to survive doses of deworming drugs that were previously effective.

It mainly affects gastro-intestinal nematodes (GINs) in sheep.

Rapidly developing resistance to many Drugs

Situation in Asia



Major Parasites Showing Resistance –

Common resistant gastrointestinal nematodes in sheep:



Haemonchus contortus

China

India

Anthelmintic Drugs Classes Affected

1 Benzimidazoles

- Albendazole
- Fenbendazole



2 Imidazothiazoles

3 Macrocytic lactones

- Ivermectin

- Moxidectin



4 Salicylanilides

Closantel, (still effective in some areas)

Several Asian countries have reported resistance to one or more major deworming Classes

Major Parasites Showing Resistance –

- ✗ **Haemonchus contortus**
- ✗ **Trichostrongylus spp.**
- ✗ **Oesophagostomum spp.**
- ✗ **Teladorsagia spp.**



These parasites, cause: Severe anemia, diarrhoea, Weight loss, Reduced wool and meat production, High lamb mortality.



Causes of Anthelmintic Resistance –

- ✓ Frequent deworming without diagnosis.
- ✓ Using the dewormer repeatedly
- ✓ Using the same dewormer repeatedly
- ✓ Lack of pasture flock management
- ✓ Treating the entire flock frequently



Detection Methods –

- ✓ Faecal Egg Count Reduction Test (FECRT)
- ✓ Egg Hatch Assay (EHA)
- ✓ Larval Development Test
- ✓ Molecular detection of resistance genes.



Strategies to Control Anthelmintic –

1 Targeted Selective Treatment (TST)

Treat only heavily infected animals.

2 Rotational use of dewormers.

3 Pasture management

- Rotational grazing
- Mixed grazing



Sustainable Parasite Control –



"Responsible Deworming Today Prevents Drug Failure Tomorrow!"

#FieldVet

Section 3: Parasitology & Resistance

11. Anthelmintic Resistance in Asia

What is Anthelmintic Resistance (AR)?

Anthelmintic Resistance is the ability of parasitic worms to survive doses of deworming drugs that were previously effective. It mainly affects gastro-intestinal nematodes (GINs) in sheep. The most important parasite involved is *Haemonchus contortus* (Barber's pole worm), which rapidly develops resistance to many drugs.

Situation in Asia:

Anthelmintic resistance is increasingly reported across many Asian countries.

India: Benzimidazole resistance in sheep nematodes reported in many flocks; resistance prevalence may exceed 70% in some regions.

- **China:** Sheep parasites show resistance to albendazole and ivermectin, with up to 85.7% of *H. contortus* populations resistant to ivermectin in some surveys.

- **Bangladesh:** Multiple drug resistance detected against albendazole, levamisole and ivermectin in sheep farms.

- **Malaysia & Southeast Asia:** High resistance to benzimidazoles, affecting about 50% of sheep farms.

Overall, several Asian countries have reported resistance to one or more major dewormer classes.

Major Parasites Showing Resistance:

Common resistant gastrointestinal nematodes in sheep:

- *Haemonchus contortus*
- *Trichostrongylus* spp.
- *Oesophagostomum* spp.
- *Teladorsagia* spp.

These parasites cause:

- Severe anemia
- Diarrhoea
- Weight loss
- Reduced wool and meat production
- High lamb mortality

Anthelmintic Drug Classes Affected:

Resistance has been reported against major drug groups:

1. Benzimidazoles

- Albendazole
- Fenbendazole

2. Imidazothiazoles

- Levamisole

3. Macrocyclic lactones

- Ivermectin
- Moxidectin

4. Salicylanilides

- Closantel (still effective in some areas)

Parasites in Asia have developed resistance to multiple drug families simultaneously in some regions.

Causes of Anthelmintic Resistance:

- Frequent deworming without diagnosis
- Underdosing of drugs
- Using the same dewormer repeatedly
- Treating the entire flock frequently
- Lack of pasture management
- Poor veterinary supervision

Repeated exposure to drugs selects resistant parasites, allowing them to dominate the population.

Detection Methods:

Veterinarians use several methods to detect resistance:

- Faecal Egg Count Reduction Test (FECRT)
- Egg Hatch Assay (EHA)
- Larval Development Test
- Molecular detection of resistance genes

Strategies to Control Anthelmintic Resistance:

1 **Targeted Selective Treatment (TST)**

Treat only heavily infected animals.

2 **Rotational use of dewormers**

Use different drug classes strategically.

3 Pasture management

- Rotational grazing
- Mixed grazing with cattle

4 FAMACHA system

Identify sheep with anemia caused by *Haemonchus*.

5 Integrated parasite management

- Nutritional supplementation
- Biological control (fungi)
- Breeding resistant sheep lines

Sustainable Parasite Control:

Future strategies include:

- Vaccine development
- Genetic resistance breeding
- Precision parasite monitoring
- Reduced dependence on chemical anthelmintics

Key Message:

“Responsible Deworming Today Prevents Drug Failure Tomorrow.”

Anthelmintic Resistance in Sheep Flocks in Africa



What is Anthelmintic Resistance (AR)?

Anthelmintic Resistance is the ability of parasitic worms to survive deworming drugs that were previously effective.

It mainly affects **gastro-intestinal nematodes** (GINs) in sheep and leads to treatment failure and heavy parasite burdens.

Major Parasites Showing Resistance –

Important resistant gastrointestinal nematodes include:



Haemonchus contortus (Barber's pole worm)

Trichostrongylus spp.

Oesophagostomum spp.

Teladorsagia spp.

* **Haemonchus contortus** is the most dominant and highly resistant parasite in African sheep flocks.

Among Worms **Haemonchus contortus** is the most dominant and highly resistant parasite in African sheep flocks.

Causes of Anthelmintic Resistance –

Major factors contributing to resistance:

- 1 **Frequent deworming** without diagnosis
- 2 **Underdosing** due to lack of weighing equipment
- 3 **Repeated use of the same drug**
- 4 **Treating the whole flock frequently**
- 5 **Poor pasture management.**
- 6 **Lack of veterinary supervision**

Underdosing and high treatment frequency are major drivers of resistance in African sheep farms.

Strategic Deworming Protects Sheep Health and Preserves Drug Effectiveness."



Situation in Africa



Anthelmintic Drug Classes Affected –

1 Benzimidazoles

- Albendazole
- Fenbendazole

2 Imidazothiazoles

- Levamisole
- Tetramisole

3 Macrocyclic lactones

- Ivermectin
- Moxidectin

4 Salicylanilides

- Closantel
- Rafoxanide

Resistance to **benzimidazoles** is the most widely reported in Africa.

Detection Methods –

- ✓ Faecal Egg Count Reduction Test (FECRT)
- ✓ Egg Hatch Assay (EHA)
- ✓ Larval Development Test
- ✓ Molecular detection of resistance genes



Sustainable Parasite Control –

1 Targeted Selective Treatment (TST)

Treat only animals with heavy parasite burdens.

2 Rotational Deworming

Use different drug classes strategically.

3 FAMACHA System

Identify sheep with anemia caused by Haemonchus infection.



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12. Anthelmintic Resistance in Africa

What is Anthelmintic Resistance (AR)?

Anthelmintic Resistance is the ability of parasitic worms to survive deworming drugs that were previously effective. It mainly affects gastro-intestinal nematodes (GINs) in sheep and leads to treatment failure and heavy parasite burdens.

Situation in Africa

Anthelmintic resistance has become a major constraint to sheep production in Africa, especially in tropical and subtropical regions.

- Resistance has been reported in South Africa, Zimbabwe, Ethiopia, Uganda, Kenya, Algeria, Mozambique and Sudan.
- Surveys show that many sheep farms harbor parasites resistant to at least one anthelmintic class.
- In South Africa, some studies reported *Haemonchus contortus* resistant on nearly all surveyed farms, indicating a severe situation.
- On communal farms in Limpopo Province, about 90% of sheep carried nematodes resistant to one or more drug classes.

Major Parasites Showing Resistance

Important resistant gastrointestinal nematodes include:

- *Haemonchus contortus* (Barber's pole worm)
- *Trichostrongylus* spp.
- *Oesophagostomum* spp.
- *Teladorsagia* spp.
- *Cooperia* spp.

Among them, *Haemonchus contortus* is the most dominant and highly resistant parasite in African sheep flocks.

Anthelmintic Drug Classes Affected

Resistance has been reported against several drug groups:

1 Benzimidazoles

- Albendazole
- Fenbendazole
- Oxfenbendazole

2 Midazothiazoles

- Levamisole
- Tetramisole

3 Macrocyclic Lactones

- Ivermectin
- Moxidectin
- Doramectin

4 Salicylanilides

- Closantel
- Rafoxanide

Resistance to benzimidazoles is the most widely reported in Africa.

Causes of Anthelmintic Resistance

Major factors contributing to resistance:

- Frequent deworming without diagnosis
- Underdosing due to lack of weighing equipment
- Repeated use of the same drug
- Treating the whole flock frequently
- Poor pasture management
- Lack of veterinary supervision

Underdosing and high treatment frequency are major drivers of resistance in African sheep farms.

Detection Methods

Veterinarians detect resistance using:

- Faecal Egg Count Reduction Test (FECRT)
- Egg Hatch Assay (EHA)
- Larval Development Test
- Molecular detection of resistance genes

These methods help determine drug efficacy and resistance levels.

Strategies to Control Anthelmintic Resistance

1 Targeted Selective Treatment (TST)

Treat only animals with heavy parasite burdens.

2 Rotational Deworming

Use different drug classes strategically.

3 FAMACHA System

Identify sheep with anemia caused by *Haemonchus* infection.

4 Pasture Management

- Rotational grazing
- Mixed grazing with cattle

5 Nutritional Support

Better nutrition improves resistance to parasites.

Sustainable Parasite Control

Future strategies include:

- Genetic selection of parasite-resistant sheep breeds
- Biological control (nematophagous fungi)
- Improved grazing systems
- Reduced reliance on chemical dewormers

Key Message

“Strategic Deworming Protects Sheep Health and Preserves Drug Effectiveness.”

STANDARD PROTOCOL FOR ANTHELMINTIC USE IN SHEEP

(Intensive / Semi-Intensive System)

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WHY DEWORM STRATEGICALLY IN INTENSIVE SYSTEMS?

- High stocking density – rapid parasite build-up
- Increased risk of anthelmintic resistance
- Subclinical infections reduce growth & fertility
- Efficient control improves feed conversion

⚠️ Routine Blind Deworming Accelerates Resistance

WHEN TO DEWORM (INTENSIVE SYSTEM)

✓ RISK-BASED DEWORMING SCHEDULE

Production Stage	Recommended Timing
Lambs	First dose at 6–8 weeks
Growing lambs	Every 6–8 weeks (based on risk)
Breeding stock	2–3 weeks before mating
Late pregnancy	Only if FEC / clinical signs indicate
Lactation	2–3 weeks post-lambing, if required

✓ Prefer Faecal Egg Count (FEC)-based decisions

✓ Use Targeted Selective Treatment (TST)

✓ TIME OF DAY FOR ADMINISTRATION

- ✓ Early morning
- ✓ Before feeding
- ✗ Avoid mid-day (heat stress)
- ✗ Avoid immediately after heavy feeding

✓ **Contraindicated in early pregnancy**

✗ **Avoid mid-day (heat stress)**

✗ **Avoid immediately after heavy feeding**

✗ **Along with vaccination (maintain 7–10 day gap)**

TIME / CONDITIONS TO AVOID DEWORMING

BENZIMIDAZOLES

Albendazole, Fenbendazole

- ✗ Contraindicated in early pregnancy
- ✗ Avoid repeated use without rotation
- ⚠️ Resistance common in intensive farms

IMIDAZOTHIAZOLES

Levamisole

- ✗ Avoid in Weak or stressed animals
- ✗ Avoid concurrent organophosphates
- ⚠️ Rapid adult nematode action

MACROCYCLIC LACTONES

Ivermectin, Doramectin, Moxidectil

- ✗ Avoid frequent repeat dosing
- ✗ Resistance develops rapidly in intensive actions:
- ⚠️ Limited effect on arrested larvae
- ⚠️ Overuse reduces long term efficacy

SALICYLANILIDES

Closantel, Oxytoclozanide

- ✗ Avoid overdose –risk of Blindness
- ✗ Avoid routine use in young lambs
- ⚠️ **Effective against:**
Haemonchus
Liver flukes
Blood-feeding parasites

PREFERRED CONTROL STRATEGY TARGETED SELECTIVE TREATMENT (TST)

Use: Faecal Egg Count (FEC)
Body Condition Score (BCS)
FAMACHA scoring

→ Treat only **high-risk** animals

RIGHT DRUG • RIGHT DOSE • RIGHT TIME • RIGHT ANIMAL

13. Standard Protocol for Anthelmintic Use

(Extensive Grazing System)

Why Deworm Strategically?

- Reduce production losses
- Prevent pasture contamination
- Delay anthelmintic resistance
- Improve flock health & fertility

👉 Avoid routine blind deworming

When To Deworm (Extensive System):

Strategic Seasonal Timing (India)

Season	Recommended Time	Purpose
Pre-monsoon	April–May	Reduce pasture contamination
Mid-monsoon	July–August	Peak parasite challenge
Post-monsoon	Sept–Oct	Reduce worm carryover
Winter	Only if FEC / signs present	Avoid unnecessary dosing.

Physiological Timing:

- Lambs: 6–8 weeks of age
- Growing lambs: Every 8–12 weeks (risk-based)
- Pre-breeding: 2–3 weeks before mating
- Lactation: After 2–3 weeks post-lambing (if required)

Time Of Day for Administration:

- ✓ Early morning or late evening
- ✓ Before grazing for better absorption
- ✗ Avoid mid-day (heat stress)

Time / Conditions to Avoid Deworming:

- ✗ First trimester of pregnancy
- ✗ Immediately after lambing (7–10 days)
- ✗ Severely weak, dehydrated, anaemic animals
- ✗ During disease outbreaks
- ✗ Along with vaccination (maintain 7–10 days gap)

Drug-Wise Anthelmintic Contraindications:

1. Benzimidazoles

(Albendazole, Fenbendazole, Oxfendazole)

- ✗ Contraindicated in first trimester of pregnancy
- ✗ Avoid high doses in breeding ewes

⚠ **Overuse → rapid resistance in grazing systems**

✓ Effective against:

- GI nematodes
- Liver flukes (Albendazole – adult stage)

2. Imidazothiazoles

(Levamisole)

✗ Avoid in:

- Weak, stressed, severely anaemic sheep
- Concurrent use with organophosphates

⚠ **Narrow safety margin – accurate dosing essential**

- ✓ Rapid action against adult nematodes

3. Macrocyclic Lactones

(Ivermectin, Doramectin, Moxidectin)

- ✗ Avoid unnecessary repeated use
- ✗ Resistance common in extensive grazing flocks

⚠ **Poor efficacy against arrested larvae in some regions**

⚠ **Environmental concern – affects dung fauna**

✓ Effective against:

- GI worms
- Lungworms
- Ectoparasites

4. Salicylanilides

(Closantel, Oxytoclozanide)

- ✗ Do NOT use in young lambs unless indicated
- ✗ Overdose may cause blindness

✓ Specific for:

- Liver flukes
- Blood-sucking parasites (Haemonchus)

Good Practices – Extensive System:

- ✓ Dose based on heaviest animal in the flock
- ✓ Use calibrated drenching gun
- ✓ Keep treated sheep off pasture for 12–24 hrs (if feasible)
- ✓ Rotate drug classes – not brands
- ✓ Maintain refugia (do not treat entire flock every time)

Preferred Approach:

Targeted Selective Treatment (TST)

Use:

- FAMACHA scoring
- Body condition scoring
- Faecal Egg Count (where available)

👉 Treat only animals that need treatment

Key Message:

“Right drug • Right time • Right animal • Right dose”

ZOONOTIC DISEASES OF SHEEP

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(Diseases transmissible
from Sheep to humans)

PARASITIC ZOOZOSES

BACTERIAL ZOOZOSES



Brucellosis (*Brucella melitensis*)

- Contact with aborted fetus, placenta
- Raw milk consumption



Anthrax (*Bacillus anthracis*)

- Handling infected carcasses, spores in soil
- Cutaneous/ pulmonary anthrax



Q Fever (*Coxiella burnetii*)

- Aerosols from birth fluids
- Inhalation of contaminated dust



Leptospirosis

- Urine-contaminated water
- Skin abrasions
- Human disease: Fever, jaundice, kidney failure



Listeriosis

- Contaminated silage,
- Raw milk



Toxoplasmosis

Transmission: Contaminated feed/water (cat feces)

Human disease: Abortion, congenital defects

Hydatidosis (*Echinococcus granulosus*)

Transmission: Ingestion of

- Ingestion of eggs from infected dog faeces

Human disease: Hydatid cysts (liver, lungs, brain)

⚠ Major public health concern



VIRAL ZOOZOSES



Orf (*Contagious Ecthyma*)

- Direct contact with lesions
- Painful pustular skin lesion



Rabies

(rare in sheep but zoonotic)

- Bite/saliva from infected sheep
- Fatal encephalitis



PARASITIC ZOOZOSES

Cryptosporidiosis

Transmission: Fecal-oral route

Human disease: Severe diarrhea

⚠ Dangerous in children & immunocompromised people

FUNGAL ZOOZOSES



Ringworm (*Dermatophytosis*)

- Direct contact with skin lesions
- Human disease: Circular itchy skin lesions

PREVENTION & CONTROL

- ✓ Wear gloves while handling sheep
- ✓ Proper disposal of aborted materials
- ✓ Avoid raw milk consumption
- ✓ Regular deworming of dogs
- ✓ Vaccination of sheep (where applicable)

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Section 4: Disease Transmission & Zoonosis

14. Zoonotic Diseases of Sheep

(Diseases transmissible from sheep to humans)

Bacterial Zoonoses:

1. Brucellosis (*Brucella melitensis*)

Transmission:

- Contact with aborted fetus, placenta
- Raw milk consumption

Human disease: Undulant fever

Risk group: Veterinarians, shepherds

2. Anthrax (*Bacillus anthracis*):

Transmission:

- Handling infected carcasses
- Spores in soil

Human disease: Cutaneous / pulmonary anthrax

⚠ Highly fatal if untreated

3. Q Fever (*Coxiella burnetii*):

Transmission:

- Aerosols from birth fluids
- Inhalation of contaminated dust

Human disease: Fever, pneumonia, hepatitis

4. Leptospirosis:

Transmission:

- Urine-contaminated water
- Skin abrasions

Human disease: Fever, jaundice, kidney failure

5. Listeriosis:

Transmission:

- Contaminated silage
- Raw milk

Human disease: Meningitis, abortion (pregnant women)

Viral Zoonoses

6 ☐☐ Orf (Contagious Ecthyma):

Transmission:

- Direct contact with lesions

Human disease: Painful pustular skin lesions

⚠ Common in shepherds & veterinarians

7 ☐☐ Rabies (rare in sheep but zoonotic):

Transmission:

- Bite/saliva from infected sheep

Human disease: Fatal encephalitis

Parasitic Zoonoses:

8 ☐☐ Hydatidosis (Echinococcus granulosus)

Transmission:

- Ingestion of eggs from dog feces

Human disease: Hydatid cysts (liver, lungs, brain)

⚠ Major public health concern

9 ☐☐ Toxoplasmosis:

Transmission:

- Contaminated feed/water (cat feces)

Human disease: Abortion, congenital defects

10 ☐☐ Cryptosporidiosis:

Transmission:

- Fecal–oral route

Human disease: Severe diarrhea

⚠ Dangerous in children & immunocompromised people

Fungal Zoonoses:

1 ☐☐☐ Ringworm (Dermatophytosis)

Transmission:

- Direct contact with skin lesions

Human disease: Circular itchy skin lesions

 Prevention & Control:

- ✓ Wear gloves while handling sheep
- ✓ Proper disposal of aborted materials
- ✓ Avoid raw milk consumption
- ✓ Regular deworming of dogs
- ✓ Vaccination of sheep (where applicable)
- ✓ Maintain farm biosecurity

 One-Line Summary:

“Sheep act as reservoirs for several zoonotic diseases including brucellosis, anthrax, Q fever, hydatidosis, orf, and toxoplasmosis, posing serious public health risks.”

CATTLE DISEASES COMMUNICABLE TO SHEEP

(Important in Mixed Grazing Systems) **FieldvetDrMSS**



VIRAL DISEASES

1. Foot and Mouth Disease (FMD)

- Aphthovirus
- Mild lesions, lameness
- Sheep as **silent carriers**

2. Bluetongue

- Bluetongue virus
- Oral ulcers, cyanotic tongue



BACTERIAL DISEASES

3. Brucellosis

- *Brucella abortus*
- Abortion, infertility

5. Tuberculosis

- *Mycobacterium bovis*
- Chronic wasting



7. Salmonellosis

- *Salmonella* spp.
- Vaccination, **Septicaemia**

4. Leptospirosis

- *Leptospira* spp.
- Abortion, weak lambs



6. Pasteurellosis

- *P. multocida*, *M. haemolytica*
- Pneumonia, sudden death



7. Salmonellosis

- *Salmonella* spp.
- Diarrhea, septicemia



FUNGAL & SKIN DISEASES

8. Ringworm

- *Trichophyton verrucosum*
- Circular alopecia

9. Dermatophilosis

- *Dermatophilus congolensis*
- Scabs, exudative dermatitis



KEY FIELD POINTS

- ✓ Mixed grazing increases transmission
- ✓ Cattle as reservoir hosts
- ✓ Vaccination & biosecurity vital
- ✓ Zoonotic risks - Health hazard

PREVENTION & CONTROL

- ✦ Separate sick animals immediately
- 💧 Avoid shared water sources
- 🏠 Maintain vaccination schedule
- 🏠 Improve housing hygiene



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15. Goat Diseases Communicable to Sheep

(Important in Mixed Small-Ruminant Farming Systems)

Viral Diseases:

1. Peste des Petits Ruminants (PPR)

- Agent: Morbillivirus
- Transmission: Direct contact, aerosols
- Sheep signs: Fever, oral erosions, diarrhea, pneumonia
-  Highly contagious, high mortality

2. Foot and Mouth Disease (FMD)

- Agent: Aphthovirus
- Transmission: Direct contact, aerosols, fomites
- Sheep signs: Mild lesions, lameness
- Sheep often act as silent carriers

3. Orf (Contagious Ecthyma)

- Agent: Parapoxvirus
- Transmission: Direct contact, fomites
- Sheep signs: Scabby lesions on lips, gums, teats
- Zoonotic: Yes

Bacterial Diseases:

4. Brucellosis

- Agent: *Brucella melitensis*
- Transmission: Aborted materials, uterine discharge
- Sheep signs: Abortion, infertility
- Zoonotic: Yes

5. Pasteurellosis

- Agent: *Mannheimia haemolytica*, *Pasteurella multocida*
- Transmission: Respiratory secretions
- Sheep signs: Pneumonia, sudden death

6. Caseous Lymphadenitis (CLA)

- Agent: *Corynebacterium pseudotuberculosis*
- Transmission: Wound contamination
- Sheep signs: Abscesses in lymph nodes
- Chronic economic losses

Parasitic Diseases:

7. Haemonchosis

- Agent: *Haemonchus contortus*
- Transmission: Grazing contaminated pasture
- Sheep signs: Anemia, bottle jaw, death

8. Fasciolosis

- Agent: *Fasciola* spp.
- Transmission: Snail-borne, wet grazing areas
- Sheep signs: Anemia, weight loss, sudden death.

Fungal & Skin Diseases:

9. Ringworm

- Agent: *Trichophyton* spp.
- Transmission: Direct contact
- Sheep signs: Circular alopecia
- Zoonotic: Yes

10. Dermatophilosis

- Agent: *Dermatophilus congolensis*
- Transmission: Wet conditions, skin trauma
- Sheep signs: Scabby dermatitis

⚠ Key Field Points:

- ✓ Goats and sheep share most pathogens
- ✓ Mixed housing increases respiratory & skin diseases
- ✓ Parasite burden rises in shared grazing
- ✓ Several diseases are zoonotic
- ✓ PPR is the most economically devastating disease

Prevention & Control:

- Vaccinate against PPR, FMD, Enterotoxaemia
- Separate sick animals immediately
- Avoid overcrowding
- Strategic deworming
- Improve ventilation & hygiene
- Quarantine newly purchased goats

One Health Importance: Brucellosis & Orf pose human risk, Small-ruminant disease control supports food security

DOG DISEASES COMMUNICABLE TO SHEEP

(Important in Farm, Pasture & Shepherd Dog Systems)

FieldvetDrMSS



VIRAL DISEASES

1. Rabies

- Lyssavirus
- Bite wounds, saliva

⚠️ **Fatal & zoonotic**



BACTERIAL DISEASES

2. Leptospirosis

- *Leptospira* spp.
- Urine-contaminated water/pasture
- Abortion, fever, weak lambs Yes

Zoonotic: Yes

3. Brucellosis (rare)

- *Brucella canis*
- Contaminated environment
- Reproductive failure (rare)



PARASITIC DISEASES (MAJOR IMPORTANCE)

4. Hydatidosis (Echinococcosis)

- *Echinococcus granulosus* Definitive host.
- Sheep role: Intermediate host
- Cysts in liver/lungs (usually subclinical) **Zoonotic:** Severe public health importance

5. Coenurosis (Gid / Sturdy)

- Dog fecal contamination
- Circling, head pressing, blindness

Economic loss: High



5. Toxocariasis

- *Toxocara canis*
- Eggs from dog feces

6. Toxocariasis

- *Toxocara canis*
- Ill-thrift (mainly lambs)



FUNGAL & ECTOPARASITIC DISEASES

7. Ringworm

- *Microsporum / Trichophyton* spp.
- Direct contact
- Circular alopecia



PREVENTION & CONTROL

- ✓ Regular deworming of dogs (Peraquantele based)
- ✓ Prevent dogs from eating sheep offal
- ✓ Dispose slaughter waste properly
- ✓ Vaccinate dogs against rabies



ONE HEALTH IMPORTANCE

- ✓ Hydatidosis & rabies pose serious human risk
- ✓ Control of dog diseases - protection of livestock & 100 humans

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16. Dog Diseases Communicable to Sheep

(Important in Farm, Pasture & Shepherd Dog Systems)

Viral Diseases:

1. Rabies

- Agent: Lyssavirus
- Transmission: Bite wounds, saliva
- Sheep signs: Aggression, paralysis, sudden death

 **Fatal & zoonotic**

Bacterial Diseases:

2. Leptospirosis

- Agent: Leptospira spp.
- Transmission: Urine-contaminated water/pasture
- Sheep signs: Abortion, fever, weak lambs
- Zoonotic: Yes

3. Brucellosis (rare)

- Agent: Brucella canis
- Transmission: Contaminated environment
- Sheep signs: Reproductive failure (rare)
- Zoonotic: Yes

Parasitic Diseases (Major Importance):

4. Hydatidosis (Echinococcosis)

- Agent: Echinococcus granulosus
- Dog role: Definitive host
- Sheep role: Intermediate host
- Sheep signs: Cysts in liver/lungs (usually subclinical)
- Zoonotic: Severe public health importance

5. Coenurosis (Gid / Sturdy)

- Agent: Taenia multiceps
- Transmission: Dog fecal contamination
- Sheep signs: Circling, head pressing, blindness
- Economic loss: High

6. Toxocariasis

- Agent: Toxocara canis
- Transmission: Eggs from dog feces

- Sheep signs: Ill-thrift (mainly lambs)

Fungal & Ectoparasitic Diseases:

7. Ringworm

- Agent: Microsporum / Trichophyton spp.
- Transmission: Direct contact
- Sheep signs: Circular alopecia
- Zoonotic: Yes

8. Mange (Sarcoptic / Demodectic)

- Transmission: Close contact
- Sheep signs: Pruritus, skin thickening (rare)

⚠ Key Field Points:

- Dogs act as definitive hosts for major sheep parasites
- Free-roaming dogs increase parasitic disease burden
- Sheep ingest infective stages from contaminated pasture
- Several diseases are zoonotic
- Shepherd & stray dogs are major risk factors

Prevention & Control:

- Regular deworming of dogs (Praziquantel-based)
- Prevent dogs from eating sheep offal
- Dispose slaughter waste properly
- Restrict stray dog access to farms
- Vaccinate dogs against rabies
- Maintain pasture hygiene

One Health Importance:

Hydatidosis & rabies pose serious human risk

Control of dog diseases = protection of livestock & humans

Diseases Transmitted from Ducks to Sheep

Risks in Integrated Duck-Sheep Farms

FieldvetDrMSS

1 Salmonellosis

→ *Salmonella enterica*

Transmission:

- Fecal contamination of pasture or water
- Shared ponds
- Contaminated feed areas

Clinical Signs in Sheep:

- Diarrhea
- Septicemia (especially lambs)
- Abortion in ewes



2 Leptospirosis (Environmental Link)

→ *Leptospira interrogans*

Ducks may act as mechanical carriers through contaminated water bodies

Transmission via:

- Urine-contaminated water
- Shared ponds or marshy grazing land



3 Campylobacteriosis

→ *Campylobacter jejuni*

Ducks are common reservoirs

In Sheep:

- Late-term abortion
- Weak lambs
- Stillbirth



4 Avian Influenza (Rare Cross-Species Event)

Sheep are not typical hosts, but experimental infections show possible seroconversion



5 Pasteurellosis (Indirect Environment Exposure)

Shared environments with bacteria-laden water may increase exposure risk.



6 Avian Influenza (Rare Cross-Species Event)

Sheep are not typical hosts but experimental infections show possible seroconversion



6 Cryptosporidiosis (Shared Water Source)

In Lambs:

- Profuse watery diarrhea
- Dehydration
- Reduced growth



Prevention Tips for Mixed Duck-Sheep Farms

- ✓ Separate water sources
- ✓ Avoid sheep grazing near duck ponds
- ✓ Regular manure management
- ✓ Control rodent populations
- ✓ Protect lambs from stagnant water
- ✓ Vaccinate sheep against endemic diseases

Direct duck to sheep disease transmission is uncommon, Shared contaminated water sources pose the greatest risk, Proper biosecurity measures and environmental hygiene are critical.

17. Diseases from Ducks to Sheep

1. Salmonellosis:

Causative agent: *Salmonella enterica*

Transmission

- Fecal contamination of pasture or water
- Shared ponds
- Contaminated feed storage areas

Clinical Signs in Sheep

- Diarrhea
- Septicemia (especially lambs)
- Abortion in ewes

Risk Factors

- Mixed duck–sheep systems
- Poor biosecurity
- Contaminated stagnant water

2. Leptospirosis (Environmental Link):

Causative agent: *Leptospira interrogans*

Ducks may act as mechanical carriers through contaminated water bodies.

Transmission

- Urine-contaminated water
- Shared ponds or marshy grazing land

Clinical Signs in Sheep

- Abortion storms
- Jaundice
- Fever
- Reduced fertility

3. Campylobacteriosis:

Causative agent: *Campylobacter jejuni*

Ducks are common reservoirs.

In Sheep

- Late-term abortion
- Weak lambs
- Stillbirth

Transmission occurs via:

- Contaminated water
- Fecal contamination of pasture

4 ☐ Avian Influenza (Rare Cross-Species Event):

Causative agent: Influenza A virus

Sheep are not typical hosts, but experimental infections have shown possible seroconversion.

⚠ **Natural transmission to sheep is rare and not a major field concern.**

5 ☐ Pasteurellosis (Indirect Environmental Exposure):

Causative agent: *Pasteurella multocida*

Though not directly transmitted from ducks to sheep, shared environments with high bacterial load may increase exposure risk.

6 ☐ Cryptosporidiosis (Shared Water Source):

Causative agent: *Cryptosporidium parvum*

Ducks can contaminate water with oocysts.

In Lambs:

- Profuse watery diarrhea
- Dehydration
- Reduced growth

Epidemiological Considerations

In integrated farming systems:

- Shared water sources are the major risk factor.
- Wet grazing areas increase pathogen survival.
- Ducks often serve as reservoir hosts, not primary disease amplifiers for sheep.

⊖ Diseases NOT Transmitted from Ducks to Sheep:

The following are species-specific and not transmitted:

- Duck viral hepatitis virus

- Newcastle disease virus
- Duck plague virus

Prevention in Mixed Duck–Sheep Farms

- ✓ Separate water sources
- ✓ Avoid sheep grazing near duck ponds
- ✓ Regular manure management
- ✓ Control rodent populations
- ✓ Vaccinate sheep against endemic diseases
- ✓ Improve drainage in low-lying grazing areas

Field Conclusion:

Direct transmission of duck-specific diseases to sheep is uncommon. However, shared environmental contamination (waterborne pathogens) poses the main risk. Proper biosecurity and environmental hygiene are critical in integrated livestock–poultry systems.

WATER-BORNE DISEASES OF SHEEP

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1. Leptospirosis

- Cause: *Leptospira* spp.
- Source: Stagnant water, ponds, urine-contaminated water
- Key signs: Fever, jaundice, abortions, hemoglobinuria
- Zoonotic: ☒ Yes

2. Colibacillosis (*E. coli* Diarrhoea)

- Cause: *Escherichia coli*
- Source: Contaminated drinking water
- Key signs: Profuse watery diarrhoea, dehydration, sudden death (lambs)
- Risk group: Neonatal lambs

3. Salmonellosis

Cause: *Campylobacter fetus*

- Source: Contaminated water
- Key signs: Fever, diarrhoea, abortions, septicemia
- Zoonotic: ☒ Yes

4. Cryptosporidiosis

- Cause: *Cryptosporidium parvum*
 - Source: Dirty water, overcrowding
 - Key signs: Watery diarrhoea, dehydration
 - Zoonotic: ☒ Yes
- Prevention: Provide clean, chlorinated or filtered drinking water
Avoid stagnant ponds and marshy grazing.

6. Coccidiosis

- Cause: *Eimeria* spp.
- Source: Contaminated water & wet pens, infected adults
- Key signs: Bloody diarrhoea, weight loss, weakness
- Risk group: Lambs

7. Giardiasis

- Cause: *Giardia* spp.
- Source: Untreated surface water
- Key signs: Chronic diarrhoea, poor growth
- Zoonotic: ☒ Yes

8. Listeriosis

- Cause: *Listeria monocytogenes*
- Source: Water contaminated with decaying organic matter
- Key signs: Circling, paralysis, abortions
- Zoonotic: ☒ Yes

10. Schistosomiasis

- Cause: *Schistosoma* spp.
 - Source: Snail-infested water
 - Key signs: Diarrhoea, anaemia, emaciation
- Prevention: Avoid stagnant ponds and marshy grazing.

Section 5: Disease Classification by Environment

18. Water-Borne Diseases

Contaminated water sources such as ponds, stagnant water, canals, and poorly maintained troughs can transmit several infectious diseases to sheep. These diseases cause diarrhoea, weakness, reduced productivity, and sometimes death, leading to economic losses for farmers

Major Water-Borne Diseases in Sheep

1 Leptospirosis

Cause: Leptospira bacteria

Source of infection: Water contaminated with urine of infected animals (especially rodents)

Clinical signs

- Fever
- Jaundice
- Abortion
- Weakness
- Significance
- Zoonotic disease (can infect humans)

2 Salmonellosis

Cause: Salmonella bacteria

Source: Contaminated water and feed

Clinical signs

- Severe diarrhea
- Fever
- Dehydration
- Abortion in pregnant ewes

Significance: High mortality in lambs

3 Coccidiosis

Cause: Eimeria species (protozoa)

Source: Water contaminated with feces

Clinical signs

- Bloody diarrhoea
- Weight loss
- Weakness

Significance: Affects young lambs severely

4 ☐ **Cryptosporidiosis**

Cause: Cryptosporidium protozoa

Source: Contaminated drinking water

Clinical signs

- Watery diarrhoea
- Dehydration
- Poor growth

Significance: Important cause of lamb diarrhoea

5 ☐ **Fasciolosis (Liver Fluke Disease)**

Cause: Liver fluke (Fasciola species)

Source: Grazing near waterlogged areas and snail-infested water bodies

- Clinical sign
- Anemia
- Bottle jaw
- Weight loss

Significance: Major economic loss in grazing sheep

6 ☐ **Blue-Green Algae Toxicity**

Cause: Toxins produced by cyanobacteria in stagnant water

Clinical signs

- Sudden illness
- Nervous signs
- Death in severe cases

Risk Factors

- Stagnant water sources
- Dirty water troughs
- Grazing near swamps or ponds
- Overcrowding
- Poor sanitation

🛡️ **Prevention and Control**

- ✓ Provide clean drinking water
- ✓ Clean water troughs regularly
- ✓ Avoid grazing in waterlogged areas

- ✓ Control rodents
- ✓ Regular deworming and health monitoring
- ✓ Improve farm sanitation

Field Significance

Water-borne diseases can cause:

- Diarrhea outbreaks in lambs
- Reduced growth and productivity
- Abortion in ewes
- Economic losses to farmers

Safe water supply is therefore a critical component of sheep health management.

✓ Key Message

“Clean water is essential for healthy sheep. Preventing water contamination helps control many infectious diseases in sheep flocks.”



Air-Borne Diseases of Sheep

Causes, Symptoms, and Prevention



What are Air-Borne Diseases?

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Air-borne diseases are infections transmitted through:

- ✓ **Aerosols** (coughing, sneezing)
- ✓ Poor ventilation
- ✓ Close confinement
- ✓ Dust laden air in sheds

They mainly affect the **RESPIRATORY SYSTEM** and cause significant economic losses.



Major Air-Borne Diseases of Sheep

1 Pneumonic Pasteurellosis

Causative agent: *Pasteurella multocida*,
Mannheimia haemolytica

Predisposing Factors:

- Sudden weather changes
- Overcrowding
- Transport stress

Symptoms

- High fever
- Nasal discharge
- Coughing
- Respiratory distress
- Sudden death (acute cases)



3 Contagious Caprine Pleuropneumonia-like Syndrome (CCPP-like)

Causative agent: *Mycoplasma sp*

- **Symptoms:**
- Severe coughing
- Rapid breathing
- Nasal discharge
- High mortality in outbreaks



- ✓ Strict quarantine
- ✓ Biosecurity

2 Sheep Pox

Causative agent: Sheep pox Virus

Symptoms

- Fever
- Papules & scabs on skin
- Nasal discharge
- Pneumonia in severe cases



Control

- Annual vaccination
- Strict quarantine
- Disinfection

4 Ovine Respiratory Complex

Mixed infection involving:

- Bacteria, Viruses,
- Mycoplasma

Symptoms

- Fever, Excess salivation
- Nasal discharge
- Breathing difficulty



Risk Factors for Air-Borne Diseases

- ✓ Poor ventilation
- ✓ Cold stress
- ✓ Overcrowding
- ✓ Nutritional deficiency
- ✓ Regular vaccination
- ✓ Clean & dry housing



Respiratory diseases spread faster in winter and rainy seasons -early detection saves flock.

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19. Air-Borne Diseases



What are Air-Borne Diseases?

Air-borne diseases are infections transmitted through:

- Aerosols (coughing, sneezing)
- Close confinement
- Poor ventilation
- Dust-laden air in sheds

They mainly affect the respiratory system and cause significant economic losses.

Major Air-Borne Diseases of Sheep:

1. Pneumonic Pasteurellosis

Causative agent: *Pasteurella multocida*, *Mannheimia haemolytica*

Predisposing factors:

- Sudden weather changes
- Overcrowding
- Transport stress

Symptoms:

- High fever
- Nasal discharge
- Coughing
- Respiratory distress
- Sudden death (acute cases)

Control:

- Timely vaccination
- Stress reduction
- Early antibiotic therapy

2. Sheep Pox:

Causative agent: Sheep pox virus

Transmission:

- Air-borne droplets
- Direct contact

Symptoms:

- Fever
- Papules & scabs on skin

- Nasal discharge
- Pneumonia in severe cases

Control:

- Annual vaccination
- Strict quarantine
- Disinfection

3 █ Contagious Caprine Pleuropneumonia-like Syndrome (CCPP-like):

Causative agent: Mycoplasma species

Symptoms:

- Severe coughing
- Rapid breathing
- Nasal discharge
- High mortality in outbreaks

Control:

- Isolation of affected animals
- Antibiotic therapy
- Biosecurity

4 █ Ovine Respiratory Complex:

Mixed infection involving:

- Bacteria
- Viruses
- Mycoplasma

Symptoms:

- Chronic coughing
- Reduced feed intake
- Poor growth
- Increased lamb mortality

Control:

- Improved housing ventilation
- Balanced nutrition
- Strategic deworming

5 █ Bluetongue (Air-borne spread in sheds):

Causative agent: Bluetongue virus

Transmission:

- Mainly vector-borne
- Aerosol spread possible in closed housing

Symptoms:

- Fever
- Excess salivation
- Nasal discharge
- Breathing difficulty

Control:

- Vaccination
- Vector control
- Housing management

** Risk Factors for Air-Borne Diseases:**

- Poor ventilation
- Overcrowding
- Dusty floors
- Cold stress
- Nutritional deficiency
- Mixed age groups

** Prevention & Control Measures:**

- ✓ Adequate ventilation
- ✓ Avoid overcrowding
- ✓ Regular vaccination
- ✓ Prompt isolation of sick sheep
- ✓ Clean & dry housing
- ✓ Stress-free handling

** Field Tip:**

Respiratory diseases spread faster in winter and rainy seasons—early detection saves the flock.

Disease Conditions of Sheep

Grazing in Seashore / Coastal Areas

(Coastal grasslands, salt marshes, sandy pastures)

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1 Parasitic Diseases

a. Fasciolosis (Liver Fluke)

- **Cause:** *Fasciola spp.*
- **Risk Factor:** Marshy coastal land, snail population
- **Symptoms:** Anemia, bottle jaw, weight loss, poor fertility
- **Diagnosis:** Fecal examination, liver lesions

 **Control:** Strategic flukicide use, avoid marsh grazing

b. Gastrointestinal Nematodes

- **Common Species:** *Haemonchus*, *Trichostrongylus*
- **Symptoms:** Pale mucosa, diarrhea, poor growth
- **Control:** FAMACHA scoring, rotational grazing

2 Mineral & Nutritional Disorders

a. Copper Deficiency

- **Cause:** High molybdenum & sulphur in coastal soils
- **Symptoms:** Faded wool, anemia, infertility, sway back

 **Control:** Sheep-specific mineral mixture.

b. Iodine Deficiency

- **Symptoms:** Goitre, weak lambs, reproductive failure

 **Control:** Iodized salt supplementation

Preventive Measures for Coastal Sheep Farming

- ✓ Regular deworming & fluke control
- ✓ Balanced mineral supplementation
- ✓ Clean fresh drinking water
- ✓ Vaccination (Clostridial & respiratory diseases)
- ✓ Avoid grazing in flooded / marshy areas
- ✓ Proper shelter against wind & rain

2 Mineral & Nutritional Disorders

a. Copper Deficiency

- **Cause:** High molybdenum & sulphur in coastal soils
- **Symptoms:** Faded wool, anemia, infertility, swayback in lambs
- **Control:** FAMACHA scoring, rotational grazing

 **Control:** Sheep-specific mineral mixture

3 Salt-Related Disorders

a. Salt Toxicity / Dehydration

- **Cause:** Excess salt intake + limited fresh water
- **Symptoms:** Excessive thirst, nervous signs, diarrhea

 **Control:** Continuous access to clean fresh water

4 Bacterial Diseases

- **Foot Rot:** *Dichelobacter nodosus*
- **Risk Factor:** Wet sand, muddy soil
- **Symptoms:** Lameness, foul smell, hoof lesions
- **Control:** Hoof trimming, zinc sulphate foot bath

5 Respiratory Diseases

a. Pneumonia

- **Risk Factor:** High humidity, strong coastal winds
- **Symptoms:** Coughing, nasal discharge, fever

 **Control:** Wind shelter, avoid overcrowding

20. Coastal & Marsh Grazing Diseases

1. Fasciolosis
2. Haemonchosis
3. Copper deficiency
4. Iodine deficiency
5. Salt toxicity
6. Foot rot
7. Pneumonia

Preventive Measures:

- Regular deworming & fluke control
- Balanced mineral supplementation
- Clean fresh drinking water
- Vaccination (clostridial & respiratory diseases)
- Avoid grazing in flooded / marshy areas
- Proper shelter against wind & rain.

Disease Conditions of Sheep

Grazing in Hilly Regions

(Hill slopes · Highland pastures · Forest margins · Cool & humid zones)

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1 Parasitic Diseases

a. Gastro-intestinal Nematodiasis

- Common parasites: *Haemonchus*, *Trichostrongylus*, *Ostertagia*
- Risk factors: Cool climate, high rainfall, lush pastures, humid zones
- Symptoms: Anaemia, bottle jaw, weight loss, bottle jaw

🔍 **Diagnosis:** Faecal egg count (FEC)
Control: Strategic deworming, rotational grazing

🌿 Fasciolosis (Liver Fluke)

- Cause: *Fasciola gigantica* / *F. hepatica*
- Risk factors: Marshy hill streams & snail habitats
- Symptoms: Anaemia, submandibular oedema, reduced productivity

🔍 **Control:** Flukicides, avoid wet grazing areas

2 Respiratory Diseases

a. Pneumonia

- Predisposing factors: Cold stress, fog, rain, wind
- Symptoms: Fever, nasal discharge, laboured breathing

🔍 **Control:** Shelter provision, timely antibiotic therapy

🌿 Maedi-Visna (in endemic areas)

- Symptoms: Progressive dyspnoea, weight loss

🔍 **Control:** Flock screening, marshy hill slopes

Preventive Measures

- ✓ Provide shelter against cold, rain & wind
- ✓ Strategic deworming & fluke control
- ✓ Regular hoof trimming & foot baths
- ✓ Vaccination (Bluetongue, bacterial diseases)
- ✓ Balanced nutrition with minerals
- ✓ Tick & vector control
- ✓ Avoid grazing in marshy hill slopes

2 Respiratory Diseases

a. Pneumonia

- Predisposing factors: Cold stress, fog, rain, wind
- Symptoms: Fever, nasal discharge, laboured breathing

🔍 **Control:** Shelter provision, timely antibiotic therapy

🌿 Maedi-Visna (in endemic areas)

- Symptoms: Progressive dyspnoea, disease

🔍 **Control:** Flock screening, culling of positives

4 Vector-Borne Diseases

a. Bluetongue

- Vector: "Culicoides" midges breed in moist hill areas
- Symptoms: Fever, oral lesions, lameness

🔍 **Control:** Vaccination, vector control

🌿 Tick-Borne Diseases

- Anaplasmosis, piroplasmosis (region specific)
- Symptoms: Fever, anaemia, weakness

🔍 **Control:** Regular acaricide application

6 Poisonous Hill Plants

Plant toxicities

- Examples: Bracken fern, Lantana, Phododendron
- Symptoms: Salivation, diarrhoea, neurological signs

🔍 **Control:** Controlled grazing, plant identification

Key Message:

Hilly region grazing predisposes sheep to parasitic, respiratory, foot and nutritional disorders. Preventive health management is essential.



21. Hilly Region Disease Conditions

(Hill slopes • Highland pastures • Forest margins • Cool & humid zones)

1. Parasitic Diseases:

Gastro-intestinal Nematodiasis:

- Common parasites: *Haemonchus*, *Trichostrongylus*, *Ostertagia*
- Risk factors: Cool climate, high rainfall, lush pasture
- Symptoms: Anaemia, diarrhoea, weight loss, bottle jaw
- Diagnosis: Faecal egg count (FEC)
- Control: Strategic deworming, rotational grazing

Fasciolosis (Liver Fluke):

- Cause: *Fasciola gigantica* / *F. hepatica*
- Risk factor: Marshy hill streams & snail habitats
- Symptoms: Anaemia, submandibular oedema, reduced productivity
- Control: Flukicides, avoid wet grazing areas

Lungworm Disease:

- Parasite: *Dictyocaulus filaria*
- Symptoms: Chronic cough, dyspnoea, poor growth
- Control: Anthelmintics, pasture management

2. Respiratory Diseases:

Pneumonia

- Predisposing factors: Cold stress, fog, rain, wind
- Symptoms: Fever, nasal discharge, laboured breathing
- Control: Shelter provision, timely antibiotic therapy

Maedi-Visna (in endemic areas):

- Nature: Chronic viral respiratory disease
- Symptoms: Progressive dyspnoea, weight loss
- Control: Flock screening, culling of positives

3. Foot & Musculoskeletal Disorders:

Foot Rot

- Cause: *Dichelobacter nodosus*
- Risk factor: Wet, muddy hill terrain
- Symptoms: Severe lameness, hoof lesions
- Control: Hoof trimming, zinc sulphate foot bath

Foot Abscess & Laminitis:

- Risk factor: Rocky slopes, constant climbing
- Symptoms: Lameness, swelling, reluctance to move
- Control: Proper hoof care, mineral supplementation

4. Vector-Borne Diseases:

Bluetongue

- Vector: Culicoides midges (breed in moist hill areas)
- Symptoms: Fever, oral lesions, lameness
- Control: Vaccination, vector control

Tick-Borne Diseases:

- Diseases: Anaplasmosis, piroplasmosis (region-specific)
- Symptoms: Fever, anaemia, weakness
- Control: Regular acaricide application

5. Nutritional & Metabolic Disorders:

Mineral Deficiencies

- Common deficiencies: Copper, Cobalt, Selenium
- Symptoms: Poor wool quality, anaemia, low immunity
- Control: Area-specific mineral mixture

Pregnancy Toxaemia:

- Risk factor: Cold stress, poor nutrition
- Symptoms: Depression, recumbency, ketosis
- Control: Adequate energy feeding in late gestation

6. Toxicities:

Poisonous Hill Plants

- Examples: Bracken fern, Lantana, Rhododendron
- Symptoms: Salivation, diarrhoea, neurological signs
- Control: Controlled grazing, plant identification

Preventive Measures:

- ✓ Provide shelter against cold, rain & wind
- ✓ Strategic deworming & fluke control
- ✓ Regular hoof trimming & foot baths
- ✓ Vaccination (Bluetongue, bacterial diseases)
- ✓ Balanced nutrition with minerals
- ✓ Tick & vector control
- ✓ Avoid grazing in marshy hill slopes

Key Message:

Hilly-region grazing predisposes sheep to parasitic, respiratory, foot and nutritional disorders—preventive health management is essential.



SHEEP POX + PASTEURELLOSIS

Mixed Infection in Sheep Farms

(Field Recognition • Treatment • Prevention)



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CAUSATIVE AGENTS

★ Sheep Pox

- Caused by **Sheeppox virus** (Family: Poxviridae)
- Highly contagious viral disease
- Causes systemic illness & immunosuppression



Pasteurellosis (Secondary Bacterial Pneumonia)

- *Pasteurella multocida*
- *Mannheimia haemolytica*
- Opportunistic respiratory pathogens
- Triggered by stress or viral infection



RISK FACTORS

- Overcrowding
- Poor ventilation
- Transport stress
- Nutritional deficiency
- Sudden weather changes
- Introduction of new animals without quarantine
- Lack of vaccination



MIXED INFECTION ALERT

- Skin pox lesions + severe pneumonia
- Rapid spread within flock
- Mortality >15~20%
- Poor response if antibiotics started late



RISK FACTORS



Sheep Pox Signs

- High fever (104, 106°F)
- Papules → pustules → thick scabs
- Lesions on lips, nostrils, eyelids, udder, perineum
- Enlarged lymph nodes
- Nasal discharge



Pasteurellosis Signs

- Sudden respiratory distress
- Dyspnea & open-mouth breathing
- Mucopurulent nasal discharge
- Coughing
- Frothy mouth (acute cases)
- Sudden death in peracute form



CLINICAL SIGNS

- Skin pox lesions + severe pneumonia
- Rapid spread within flock
- Mortality >15~20%
- Poor response if antibiotics started late



OUTBREAK ACTION PLAN

1. Immediate isolation
2. Start antibiotic therapy early
3. Provide supportive therapy to whole flock
4. Disinfect housing daily



PREVENTION (FIELD PROTOCOL)

- For Pasteurellosis Component
- Long-acting Oxytetracycline
- Enrofloxacin
- Ceftiofur
- NSAIDs (Flunixin / Meloxicam)



OUTBREAK ACTION PLAN

1. Immediate isolation
2. Start antibiotic therapy early
3. Provide supportive therapy to whole flock
4. Disinfect housing daily



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KEY MESSAGE

Sheep Pox weakens immunity.
Pasteurellosis causes fatal pneumonia.

➤ Early treatment • vaccination saves flocks.



Section 6: Infectious & Major Diseases

22. Sheep Pox + Pasteurellosis Mixed Infection

1. Diseases Involved

Sheep Pox

Caused by Sheeppox virus

Family:Poxviridae

Highly contagious viral disease of sheep.

Pasteurellosis

- Caused mainly by *Pasteurella multocida* and *Mannheimia haemolytica*
- Bacterial respiratory disease leading to pneumonia and septicemia.

2. Why Mixed Infection Occurs

Sheep pox weakens the animal's immune system and damages skin and respiratory tissues, allowing bacteria to invade.

Predisposing factors

- Stress due to transport or overcrowding
- Poor nutrition
- Sudden climate change
- Long-distance grazing migration
- Poor ventilation in sheep sheds
- Delay in vaccination

3. Clinical Signs in Mixed Infection

Skin Lesions (Sheep Pox)

- Fever (40–41°C)

Nodules and scabs on:

- Lips
- Nose
- Eyelids
- Ears
- Udder and tail region
- Thick skin lesions that later become crusts.

Respiratory Signs (Pasteurellosis)

- Severe pneumonia
- Nasal discharge

- Coughing
- Rapid breathing
- Depression and weakness

Systemic Signs

- Loss of appetite
- High mortality in lambs
- Sudden deaths in severe cases

4. Post-Mortem Findings

- Sheep Pox
- Nodular lesions in skin and lungs
- Enlarged lymph nodes
- Pox lesions in respiratory mucosa
- Pasteurellosis
- Fibrinous pneumonia
- Consolidated lung lobes
- Pleural fluid accumulation

5. Diagnosis

Clinical signs and flock history

Laboratory confirmation:

- PCR for Sheeppox virus
- Bacterial culture for *Pasteurella multocida*

Differential diagnosis includes:

- Goat pox
- Contagious ecthyma
- Bluetongue

6. Treatment (Supportive + Secondary Infection Control)

For bacterial pneumonia

- Broad spectrum antibiotics
- Oxytetracycline
- Enrofloxacin
- Ceftiofur

Supportive therapy

- Anti-inflammatory drugs
- Antihistamines
- Vitamin B complex

- Electrolytes and fluids

Wound care

- Antiseptic dressing for skin lesions.

(Note: No specific antiviral treatment for sheep pox.)

7. Prevention and Control

Vaccination

- Sheep pox vaccination in endemic areas.

Biosecurity

- Isolate affected animals
- Restrict animal movement
- Disinfect sheds and equipment

Management

- Avoid overcrowding
- Provide balanced nutrition
- Ensure adequate ventilation

Flock health monitoring

- Early detection and treatment of respiratory infections.

8. Economic Impact

Mixed infections cause:

- High mortality in lambs
- Reduced weight gain
- Wool damage
- Increased treatment costs
- Trade restrictions during outbreak

Key Field Message

“Sheep pox weakens the animal – Pasteurellosis kills the flock.”

Early vaccination, proper management, and rapid veterinary intervention are essential to prevent severe outbreaks.

FOOT AND MOUTH DISEASE

with **SECONDARY BACTERIAL INFECTION** in **SHEEP**



FMD Virus

Foot and Mouth Disease (FMD)

- Vesicles & Erosions
- Salivation
- Lameness



FMD + Secondary Infection

Clinical Presentation

	FMD Alone	FMD + Secondary Infection
- Lesions	- Clean Erosions	- Ulcers with Pus & Necrosis
- Lameness	- Mild	- Severe & Chronic
- Odor	- No Odor	- Foul Smell
- Healing	- Heals in 7-10 Days	- Delayed Recovery



Differential Diagnosis

- Foot Rot
- Bluetongue
- Orf (sore Mouth)



Treatment

- Antibiotics
- Antiseptics
- NSAIDs
- Supportive Care



Prevention

- Vaccination
- Biosecurity
- Movement Control



• Key Point •

- Persistent Lameness with Foul Odor & Pus

➔ **SUSPECT SECONDARY INFECTION!**



Economic Impact

- Weight Loss
- Decreased Wool
- Lamb Mortality



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23. Foot and Mouth Disease (FMD)

1 □ Primary Disease: Foot and Mouth Disease (FMD)

Causative Agent: Foot-and-mouth disease virus

Family: Picornaviridae

Affects: Sheep, goats, cattle, pigs, and other cloven-hoofed animals

Pathogenesis in Sheep:

Sheep often show mild or subclinical signs, making them important carriers in endemic areas (relevant in Indian small ruminant systems).

- Virus enters via inhalation or ingestion
- Replicates in pharyngeal epithelium
- Viremia → vesicle formation

Clinical Signs in Sheep

- Mild fever (often unnoticed)
- Lameness (most common sign)
- Vesicles/erosions in:
 - Interdigital space
 - Coronary band
 - Dental pad (rare compared to cattle)
 - Salivation (less obvious than cattle)
 - Sudden drop in body condition
 - In lambs: myocarditis ("tiger heart") → sudden death

2 □ Secondary Bacterial Infection:

Why Secondary Infection Occurs?

FMD causes:

- Ruptured vesicles
- Raw erosions in feet and mouth
- Compromised epithelial barrier

This predisposes to invasion by bacteria such as:

- *Fusobacterium necrophorum*
- *Dichelobacter nodosus*
- *Staphylococcus* spp.
- *Trueperella pyogenes*

Differential Diagnosis:

- Foot rot

- Bluetongue
- Contagious ecthyma (Orf)
- Sheep pox
- Vesicular stomatitis

Treatment Protocol:

 **Important:** There is no specific antiviral treatment for FMD. Management is supportive.

A. Supportive Care

- Soft bedding
- Isolation
- Electrolytes
- NSAIDs (e.g., Meloxicam)

B. Control of Secondary Bacterial Infection

• **Broad-spectrum antibiotics:**

- Oxytetracycline (long-acting)
- Penicillin + Streptomycin

• **Topical antiseptics:**

- 1% Potassium permanganate wash
- Povidone iodine
- 5% Copper sulfate footbath

C. Pain & Inflammation Control

- NSAIDs
- Avoid excessive movement

Prevention & Control:

Vaccination

- Regular vaccination with polyvalent FMD vaccine
- Booster as per endemic schedule (biannual in India)

Biosecurity

- Quarantine new animals (minimum 2–3 weeks)
- Disinfection of sheds
- Restrict movement during outbreaks

Herd-Level Strategy:

In endemic areas like India:

- Mass vaccination

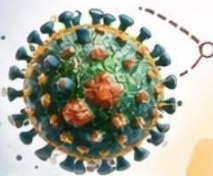
- Movement control
- Early reporting to veterinary authorities

 Economic Importance:

- Reduced weight gain
- Decreased wool production
- Lamb mortality
- Trade restrictions

 Key Veterinary Insight (Field Perspective):

In sheep, lameness may be the only visible sign of FMD. If lameness persists beyond 7–10 days with foul smell and pus, suspect secondary bacterial involvement.



Capripoxvirus

GOAT POX in SHEEP

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• Causative Agent: **Capripoxvirus** (Family: Poxviridae)

➤ • **Highly Contagious** • Major Loss in **Small Ruminants** ◀

EPIDEMIOLOGY

- Contagious **Viral Disease** of Sheep & Goats
- High **Fever**, Skin Lesions, **Mortality** in Lambs
- **Airborne + Contact** Transmission
- **Economic Loss** – Wool, Weight, Trade



TRANSMISSION



Incubation Period: 4 – 14 Days

CLINICAL SIGNS IN SHEEP

Macule → Papule → Vesicle → Pustule → Scab

Systemic Phase

- High Fever (40–41.5°C)
- Depression
- Anorexia
- Nasal & Eye Discharge
- Lymphadenopathy



Skin Lesions – Common Sites



Severe Cases

- Pneumonia
- Diarrhea
- Abortion
- Lamb Mortality

POSTMORTEM FINDINGS

- Nodules in Skin
- Pock Lesions in Lungs
- Enlarged Lymph Nodes
- Edema



DIFFERENTIAL DIAGNOSIS

- Orf (Contagious Ecthyma)
- Bluetongue
- Dermatophilosis
- Sheep Pox



TREATMENT (No Specific Antiviral)

- **Supportive Care**
- Antibiotics (Oxytetracycline)
- NSAIDs (Meloxicam)
- Antipyretics, Fluids
- Wound Care (Povidone Iodine)

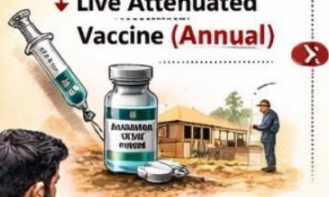


➤ **Isolation of Sick Animals**

PREVENTION & CONTROL

Vaccination

↓ Live Attenuated Vaccine (Annual)



Biosecurity

- ✔ Quarantine (21 – 30 days)
- ✔ Disinfection (Sodium Hypochlorite)
- ✔ Dispose Scabs/ Carcasses
- ✔ Movement Restriction



ECONOMIC IMPACT

- ↓ Weight Gain
- ↓ Wool Quality
- ↗ Skin Damage
- ↗ Mortality in Lambs
- Trade Restrictions



FIELD VETERINARY INSIGHT

- ↓ Sudden Fever + Generalized Nodular Skin Lesions
- ↓ Enlarged Lymph Nodes → **Strongly Suggests**

Capripox Infection



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Veterinary Knowledge. Rural Focus.

24. Goat Pox in Sheep

Causative Agent

Capripoxvirus

Family: Poxviridae

- Sheeppox and goatpox are caused by closely related Capripox viruses.
- Host specificity is common, but cross-infection can occur in mixed flocks.

🔑 Epidemiological Importance (India & Endemic Regions):

- Highly contagious viral disease
- Significant morbidity and mortality (especially lambs)
- Major economic loss in small ruminant production
- Important transboundary disease

Transmission:

- Direct contact (skin lesions, scabs)
- Aerosol droplets
- Contaminated equipment
- Insects (mechanical vectors)

Incubation period: 4–14 days

Clinical Signs in Sheep:

1 📋 Systemic Phase

- High fever (40–41.5°C)
- Depression
- Nasal & ocular discharge
- Enlarged superficial lymph nodes

2 📋 Skin Lesion Progression

Macule → Papule → Vesicle → Pustule → Scab

Common Sites:

- Face
- Ears
- Perineum
- Tail region
- Inner thighs
- Udder & scrotum

3 Severe Cases

- Pneumonia (nodular lung lesions)
- Diarrhoea
- Abortion
- High lamb mortality

Postmortem Findings:

- Nodules in skin
- Pock lesions in lungs
- Enlarged lymph nodes
- Edematous skin

Differential Diagnosis:

- Orf (Contagious ecthyma)
- Bluetongue
- Dermatophilosis
- Sheep pox (if goat-origin virus suspected)

Treatment:

 **No specific antiviral therapy.**

Supportive Care

- Broad-spectrum antibiotics (to prevent secondary bacterial infection)
- NSAIDs for fever
- Antihistamines if needed
- Fluid therapy in severe cases
- Isolation of affected animals

Prevention & Control:

Vaccination

- Live attenuated sheeppox/goatpox vaccine
- Annual vaccination in endemic areas

Biosecurity

- Strict quarantine (21–30 days)
- Disinfection of sheds
- Proper disposal of scabs & carcasses
- Movement restriction during outbreak

Economic Impact:

- Reduced weight gain

- Wool damage
- Skin quality deterioration
- Lamb mortality
- Trade restrictions

Field Veterinary Insight:

In endemic rural flocks, sudden fever + generalized nodular skin lesions + enlarged lymph nodes strongly suggest Capripox infection.

ORF IN LAMBS

ZOONOTIC WARNING

Orf can infect humans causing painful skin lesions.

- ✓ Wear gloves while handling infected animals
- ✓ Wash hands thoroughly after handling

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25. ORF (Contagious Ecthyma)

Causes • Symptoms • Diagnosis • Treatment • Control

What is Orf?

Orf is a highly contagious viral disease of sheep and goats, commonly affecting young lambs (1–6 weeks of age). It is a zoonotic disease (can infect humans).

Cause:

Causative agent:

 Orf virus (Parapoxvirus, Poxviridae family)

Source of infection:

- Infected sheep & goats
- Scabs in the environment (virus survives for months)

Mode of transmission:

- Direct contact with lesions
- Through abrasions on lips, gums, teats
- Contaminated feeders, fencing, bedding

Clinical Symptoms (Little Lambs):

Early Signs

- Fever (mild)
- Reduced suckling
- Depression

Typical Lesions

- Papules → vesicles → pustules → thick scabs

Lesions mainly on:

- Lips & mouth angles
- Gums & tongue
- Nose & nostrils
- Occasionally eyelids

Advanced / Complicated Cases

- Painful mouth → starvation
- Weight loss & poor growth
- Secondary bacterial infection
- Pneumonia or death (rare but possible)

Diagnosis:

Field Diagnosis

- Characteristic scabby lesions around mouth
- History of outbreak in flock
- Affects young lambs primarily

Laboratory Diagnosis (Confirmatory)

- PCR
- Virus isolation
- Electron microscopy

Usually field diagnosis is sufficient

Treatment:

No specific antiviral treatment available

Supportive & Symptomatic Treatment

- Topical antiseptics (povidone iodine, potassium permanganate)
- Antibiotic ointment to prevent secondary infection
- Broad-spectrum antibiotics (if severe secondary infection)
- NSAIDs for pain & fever
- Soft, palatable feed
- Assist feeding if lambs cannot suckle

Control & Prevention:

Biosecurity Measures

- ✓ Isolate affected lambs
- ✓ Disinfect feeders & water troughs
- ✓ Burn or safely dispose scabs
- ✓ Avoid handling lesions with bare hands

Vaccination:

- ✓ Live Orf vaccine (where endemic)
- ✓ Vaccinate only healthy animals
- ✓ Avoid vaccinating disease-free flocks

Flock Management:

- ✓ Avoid overcrowding

- ✓ Prevent injuries around mouth
- ✓ Maintain clean housing
- ✓ Ensure good nutrition & colostrum intake

 Zoonotic Warning:

Orf can infect humans causing painful skin lesions.

- ✓ Wear gloves while handling infected animals
- ✓ Wash hands thoroughly after handling

 Field Tip:

Orf is self-limiting but causes heavy production loss in lambs—early isolation prevents flock spread.

Tetanus in Sheep

Tetanus is preventable but often fatal — vaccination and wound hygiene are the only reliable safeguards.

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26. Tetanus In Sheep

Cause:

- Caused by *Clostridium tetani*, an anaerobic, spore-forming bacterium
- Spores present in soil, manure, dust

Infection enters through:

- Wounds and punctures
- Castration, docking, shearing wounds
- Lambing injuries
- Contaminated injections or surgical procedures
- Produces a potent neurotoxin (tetanospasmin)

Predisposing Factors:

- Poor wound hygiene
- Anaerobic (deep, closed) wounds
- Lack of vaccination
- Unsanitary husbandry practices

Symptoms:

- Incubation period: 3–21 days
- **Early signs:**
 - Stiff gait, reluctance to move
 - Muscle tremors
 - Difficulty opening mouth (lockjaw)
- **Progressive signs:**
 - Rigid extension of legs
 - Erect ears, wrinkled forehead
 - Tail raised stiffly
 - Hypersensitivity to sound and touch
- **Advanced stage:**
 - Opisthotonus (arched back)
 - Convulsions
 - Respiratory failure
 - High mortality rate (often >70%)

Diagnosis:

- **Primarily clinical diagnosis based on:**
 - History of recent wound or surgery
 - Characteristic muscle rigidity
 - Laboratory confirmation is rarely practical
- **Differential diagnosis:**
 - Hypocalcaemia
 - Strychnine poisoning
 - Polioencephalomalacia

Treatment:

(Treatment is often unrewarding once signs appear)

1. Neutralize toxin

- Tetanus antitoxin (if available) – early stage only

2. Eliminate bacteria

- Penicillin or broad-spectrum antibiotics

3. Wound management

- Locate and thoroughly clean wound
- Debridement and antiseptic dressing

4. Control muscle spasms

- Sedatives (diazepam, chloral hydrate)

5. Supportive care

- Quiet, dark environment
- Soft bedding
- Assisted feeding and hydration

Control & Prevention:

Most Important: Vaccination

Tetanus toxoid:

- First dose: 6–8 weeks of age
- Booster after 4 weeks
- Annual booster thereafter
- Vaccinate before castration, docking, or shearing

Management Practices:

- Maintain clean housing
- Use sterile instruments
- Proper wound disinfection
- Avoid unnecessary invasive procedures
- Immediate treatment of wounds

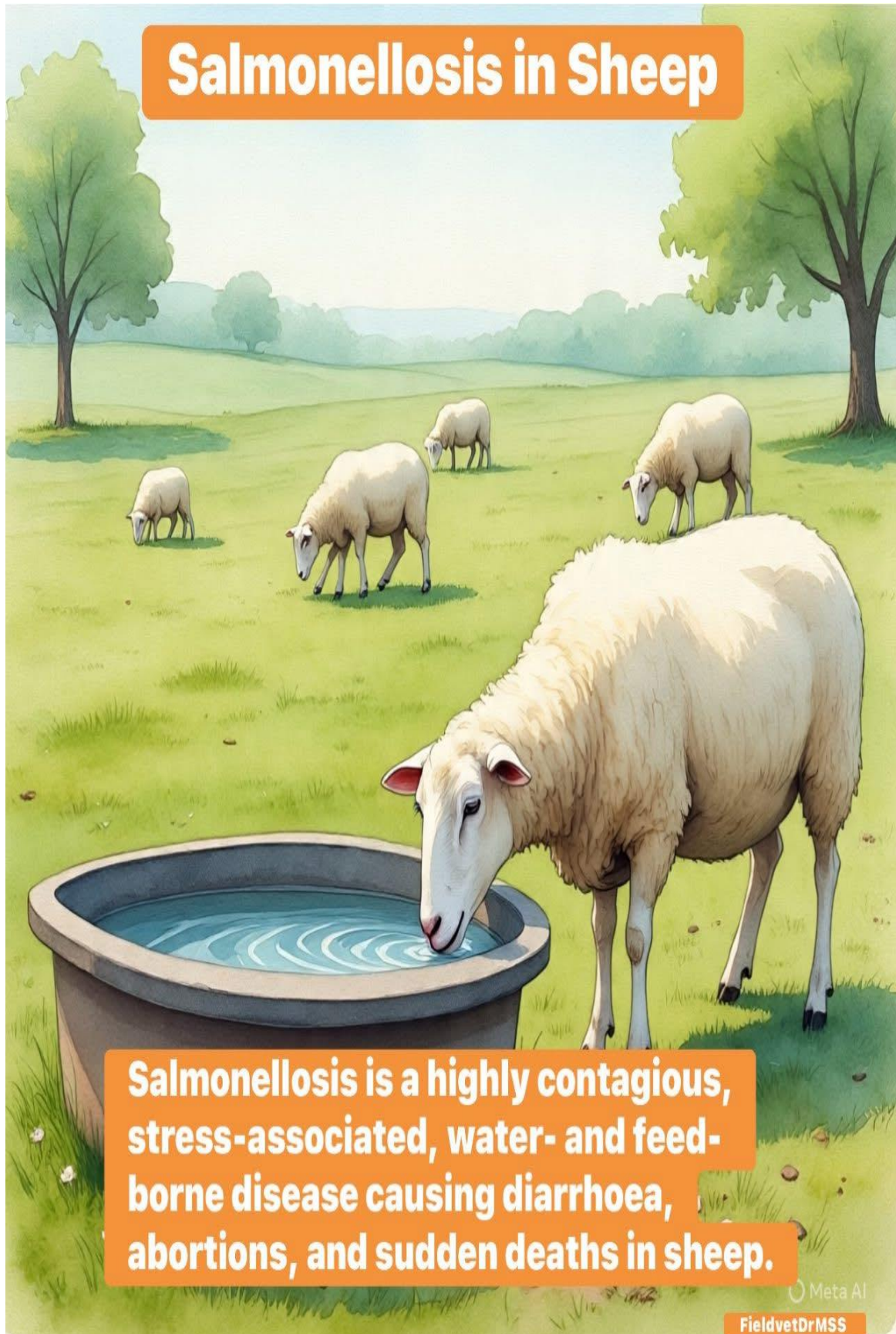
Public Health Importance:

- Zoonotic organism present in environment
- Humans at risk through contaminated wounds
- Proper hygiene and vaccination are essential.

Key Message:

Tetanus is preventable but often fatal — vaccination and wound hygiene are the only reliable safeguards

Salmonellosis in Sheep



Salmonellosis is a highly contagious, stress-associated, water- and feed-borne disease causing diarrhoea, abortions, and sudden deaths in sheep.

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27. Salmonellosis In Sheep

Cause:

- Salmonella spp.
- Common serotypes: S. Typhimurium, S. Dublin, S. Enteritidis
- Gram-negative, facultative intracellular bacteria

Source & Mode of Infection:

- Contaminated feed and drinking water
- Fecal contamination of sheds and water troughs
- Carrier animals shedding organisms
- **Stress factors:** transport, overcrowding, poor nutrition

Symptoms / Clinical Signs:

Adult Sheep

- High fever (40–41°C)
- Profuse watery or bloody diarrhoea
- Anorexia, depression
- Dehydration
- Abortions (especially late pregnancy)
- Septicemia → sudden death (acute cases)

Lambs

- Severe diarrhoea
- Rapid dehydration
- Weakness
- High mortality

Diagnosis:

- Fecal culture and isolation of Salmonella
- Blood culture in septicemic cases
- PCR for rapid confirmation
- **Post-mortem lesions:**
 - Enteritis
 - Enlarged spleen
 - Congested liver
- **Differential diagnosis:**
 - Colibacillosis
 - Clostridial enterotoxemia
 - Coccidiosis

Treatment:

⚠️ **Treat early; prognosis is poor in advanced septicemia**

Antibiotics (based on sensitivity testing):

- Enrofloxacin
- Ciprofloxacin
- Ceftriaxone
- Amoxicillin–clavulanate

Supportive Therapy:

- IV or oral fluid therapy
- Electrolytes
- NSAIDs (Flunixin meglumine, Meloxicam)
- Probiotics after antibiotic therapy

Control & Prevention:

- Provide clean, potable drinking water
- Avoid fecal contamination of feed and water
- Regular cleaning and disinfection of sheds and troughs
- Isolate sick animals immediately
- Reduce stress (transport, overcrowding)
- Proper carcass disposal
- Rodent control on the farm

Zoonotic Importance:

- Zoonotic disease ⚠️

Humans may get infected via:

- Direct animal contact
- Contaminated water or meat
- Use gloves and maintain personal hygiene

Key Field Message:

Salmonellosis is a highly contagious, stress-associated, water- and feed-borne disease causing diarrhea, abortions, and sudden deaths in sheep.



sheep Leptospirosis.

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28. Leptospirosis in Sheep

Cause

- Leptospira spp. (mainly *Leptospira interrogans*)
- Important serovars in sheep:
- Hardjo
- Pomona
- Icterohaemorrhagiae

Source of infection:

- Urine of infected animals (sheep, cattle, rodents, dogs)
- Contaminated water, pasture, mud

Mode of Transmission:

Entry through:

- Skin abrasions
- Mucous membranes
- Drinking contaminated water
- Contact with urine-soaked soil or bedding
- Zoonotic transmission to humans ⚠

Clinical Signs / Symptoms:

In Sheep

- Often subclinical
- Fever
- Reduced feed intake
- Lethargy
- Abortion (last trimester)
- Stillbirths / weak lambs
- Reduced fertility
- Occasional jaundice & hemoglobinuria
- Mastitis (rare)

In Lambs

- Fever
- Septicemia
- Sudden death (rare)

Diagnosis:

Field Diagnosis

History of:

- Abortion storms
- Wet grazing areas
- Rodent infestation

Laboratory Diagnosis

- MAT (Microscopic Agglutination Test) – Gold standard
- ELISA
- PCR (blood, urine)
- Culture (special media – time consuming)

Samples:

- Blood (acute stage)
- Urine (carrier animals)
- Aborted fetus / placenta

Treatment:

Antibiotics (Early treatment is critical)

- Oxytetracycline
- 10–20 mg/kg IM/IV
- Streptomycin
- 25 mg/kg IM (effective against carriers)
- Penicillin (supportive)

Supportive Therapy

- NSAIDs
- Fluids (if dehydrated)
- Vitamins

Treat entire flock if outbreak suspected.

Control & Prevention:

Farm-Level Control:

- ✓ Isolate affected animals
- ✓ Improve drainage & hygiene
- ✓ Rodent control (very important)
- ✓ Avoid stagnant water sources

Vaccination:

- Polyvalent Leptospira vaccine
- Annual booster in endemic areas

Biosecurity:

- ✓ Disinfect urine-contaminated areas
- ✓ Use gloves while handling animals
- ✓ Safe disposal of aborted materials

**Public Health Importance:**

- Zoonotic disease
- Causes fever, jaundice, kidney failure in humans
- High risk to:
 - Shepherds
 - Veterinarians
 - Farm workers

**One-Line Summary:**

“Leptospirosis in sheep is a zoonotic bacterial disease causing abortion, infertility, and fever, transmitted through urine-contaminated water and controlled by antibiotics, vaccination, and strict hygiene.”



Coccidiosis in Sheep

Coccidiosis is primarily a disease of lambs, causing diarrhoea, poor growth and economic losses.

good hygiene is the most effective control measure.

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good hygiene is the most effective control measure.

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29. Coccidiosis In Sheep

Cause:

- Eimeria species (protozoan parasites)
- **Important species:** E. ovinoidalis, E. crandallis
- Host-specific intestinal parasites

Source & Mode of Infection:

- Ingestion of sporulated oocysts from:
 - Contaminated drinking water
 - Feed troughs
 - Wet, dirty floors and bedding
- Favoured by:
 - Overcrowding
 - Poor hygiene
 - Stress (weaning, transport)
 - Wet and humid conditions

Symptoms / Clinical Signs:

Lambs (most affected)

- Profuse watery or bloody diarrhoea
- Tenesmus (straining)
- Dehydration
- Weight loss, stunted growth
- Rough hair coat
- Weakness, recumbency
- Death in severe cases

Adult Sheep

- Usually subclinical
- Act as carriers, shedding oocysts

Diagnosis:

- **Fecal examination:**
 - Oocyst detection by flotation technique
 - Oocyst count (OPG)

- **History of:**
 - Diarrhoea in lambs
 - Poor growth
 - Overcrowded housing
- **Post-mortem lesions:**
 - Thickened intestinal mucosa
 - Hemorrhagic enteritis
- **Differential diagnosis:**
 - Colibacillosis
 - Salmonellosis
 - Enterotoxemia

Treatment:

◆ Early treatment gives best results

Anticoccidial Drugs

- Amprolium
- Sulfa drugs (Sulfadimidine, Sulfaquinoxaline)
- Toltrazuril
- Diclazuril

Supportive Therapy

- Oral/IV fluids
- Electrolytes
- Multivitamins
- Probiotics

Control & Prevention:

- Avoid overcrowding
- Maintain dry, clean housing
- Regular cleaning and disinfection of pens
- Raise feed and water troughs
- Strategic prophylactic anticoccidial treatment in lambs
- Proper drainage to avoid wet floors

- Quarantine newly introduced animals

Zoonotic Importance:

- ✗ Not zoonotic.

Key Field Message:

Coccidiosis is primarily a disease of lambs, causing diarrhoea, poor growth and economic losses—good hygiene is the most effective control measure.

Worm Infestation-Induced Bacterial Diseases of Sheep

(Secondary Bacterial Infections Following Parasitism)

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Heavy **gastrointestinal** or external parasitic infestations weaken immunity, damage tissues, and create entry points for opportunistic **bacteria**. This leads to **secondary bacterial diseases**, increasing mortality and production losses.

1 Enterotoxemia (Pulpy Kidney Disease)

✓ **Primary Trigger:** Heavy gastrointestinal worm burden causing gut damage

✓ **Bacteria:** *Clostridium perfringens*

Mechanism: Worm-induced intestinal injury

- ✓ Sudden dietary changes • parasite Stress
- ✓ Rapid bacterial multiplication and toxin release

Signs: Sudden death
✓ Abdominal pain
✓ Neurological signs



3 Pasteurellosis (Secondary Pneumonia)

✓ **Predisposing Factor:** Stress • parasitic anemia

✓ **Bacteria:** *Mannheimia haemolytica*

Mechanism: Worm-induced immune suppression

- ✓ Reduced resistance to respiratory pathogens.

Signs: Fever
✓ Nasal discharge
✓ Respiratory distress



6 Foot Rot Complications

✓ **Predisposing Factor:** Weak immunity due parasitic load

Bacteria: *Dichelobacter nodosus*.



1 Pathophysiological Link

Worm Infestation leads to: • Nutritional deficiency

- ✓ Nutritional
- ✓ **Anemia**
- ✓ Protein loss
- ✓ Immune suppression
- **Anemia**
- Protein loss
- Immune suppression
- Tissue damage

Veterinary Prevention Strategy

- ✓ Strategic deworming (FAMACHA based)
- ✓ Pasture rotation
- ✓ Balanced mineral supplementation
- ✓ Vaccination against clostridial diseases
- ✓ Biosecurity & hygiene
- ✓ Early diagnosis and treatment

2 Clostridial Enteritis

✓ **Associated With:** Severe parasitism (especially *Haemonchus* spp).

✓ **Bacteria:** *Clostridium septicum*

Pathogenesis: • Mucosal damage from worms
✓ Bacterial invasion through damaged gut lining.

Clinical Signs: • Diarrhea
✓ Toxemia
✓ Sudden death in lambs



4 Foot Rot Complications

✓ **Predisposing Factor:** Weak immunity due parasitic load

Management:

- ✓ Regular hoof trimming
- ✓ Foot baths
- ✓ Strategic parasite control



5 Liver Abscess (Following Liver Fluke Damage)

✓ **Primary Parasite:** *Fasciola hepatica*

✓ **Secondary Bacteria:** *Mycoplasma* *avium*

Mechanism: • Pustules
✓ Crust formation
✓ Wool loss
✓ Secondary wound infections



6 Skin Pyoderma (Following External Parasites)

✓ **Primepsing Parasites:** use:

- Mites
- Ticks

Common Bacteria:

- ✓ Pustules ✓ Crust formation
- ✓ Wool loss • Secondary wound infections.



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Worm Infestation Induced Bacterial Diseases of Sheep

Section 7: Parasitic Infestations

30. Worm Infestation–Induced Diseases

(Secondary Bacterial Infections Following Parasitism)

Heavy gastrointestinal or external parasitic infestations weaken immunity, damage tissues, and create entry points for opportunistic bacteria. This leads to secondary bacterial diseases, increasing mortality and production losses.

1. Enterotoxemia (Pulpy Kidney Disease)

Primary Trigger: Heavy gastrointestinal worm burden causing gut damage

Bacteria: *Clostridium perfringens*

Mechanism:

- Worm-induced intestinal injury
- Sudden dietary changes + parasite stress
- Rapid bacterial multiplication and toxin release

Signs:

- Sudden death
- Abdominal pain
- Neurological signs

Prevention:

- Strategic deworming
- Regular vaccination
- Proper nutrition management

2. Clostridial Enteritis

Associated With: Severe parasitism (especially *Haemonchus* spp.)

Bacteria: *Clostridium septicum*

Pathogenesis:

- Mucosal damage from worms
- Bacterial invasion through damaged gut lining

Clinical Signs:

- Diarrhea
- Toxemia
- Sudden death in lambs

3 □ Pasteurellosis (Secondary Pneumonia):

Predisposing Factor: Stress + parasitic anemia

Bacteria: Mannheimia haemolytica

Mechanism:

- Worm-induced immune suppression
- Reduced resistance to respiratory pathogens

Signs:

- Fever
- Nasal discharge
- Respiratory distress

Control:

- Deworming program
- Reduce overcrowding
- Vaccination (where available)

4 □ Foot Rot Complications:

Predisposing Factor: Weak immunity due to parasitic load

Bacteria: Dichelobacter nodosus

Signs:

- Lameness
- Foul-smelling hoof lesions
- Reduced grazing

Management:

- Regular hoof trimming
- Foot baths
- Strategic parasite control

5 □ Liver Abscess (Following Liver Fluke Damage):

Primary Parasite: Fasciola hepatica

Secondary Bacteria: Trueperella pyogenes

Mechanism:

- Fluke migration damages liver tissue
- Bacterial invasion of necrotic areas

Signs:

- Weight loss
- Reduced productivity
- Condemnation at slaughter

6. Skin Pyoderma (Following External Parasites):**Predisposing Parasites:**

- Lice
- Mites
- Ticks

Common Bacteria: *Staphylococcus aureus***Signs:**

- Pustules
- Crust formation
- Wool loss
- Secondary wound infections

⚠ Pathophysiological Link:

Worm infestation leads to:

- Nutritional deficiency
- Anemia
- Protein loss
- Immune suppression
- Tissue damage

→ Creates ideal conditions for bacterial invasion.

Veterinary Prevention Strategy:

- ✓ Strategic deworming (FAMACHA-based)
- ✓ Pasture rotation
- ✓ Balanced mineral supplementation
- ✓ Vaccination against clostridial diseases
- ✓ Biosecurity & hygiene
- ✓ Early diagnosis and treatment

Lungworm Infestation in Sheep

(Verminous Pneumonia)

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CAUSES

Causative Parasites

- ✓ *Dictyocaulus filaria* (most common)
- ✓ *Protostrongylus rufescens*
- ✓ *Muellerius capillaris*

Mode of Infection

- Ingestion of infective larvae (L3) from pasture
- High risk in wet, cool climates
- Overgrazing & poor pasture rotation

PREDISPOSING FACTORS

- ✓ Rainy season & high humidity
- ✓ Young sheep and lambs
- ✓ Overstocking
- ✓ Poor nutrition
- ✓ Continuous grazing on same pasture

Severe Cases

- ✓ Pneumonia
- ✓ Secondary bacterial infection
- ✓ Death (rare but possible)

SYMPTOMS

Early Signs

- Frequent coughing (especially after exercise)
- Nasal discharge
- Mild fever
- Reduced appetite

Advanced Signs

- Persistent deep cough
- Laboured breathing (dyspnoea)
- Open-mouth breathing
- Weight loss and poor growth
- Exercise intolerance

DIAGNOSIS

- History of grazing in wet pastures
- Characteristic coughing
- Poor response to antibiotics alone

Laboratory Diagnosis

- Faecal examination (Baermann technique) – gold standard
- Detection of lungworm larvae

TREATMENT

Anthelmintics (Highly Effective)

- Albendazole
- Fenbendazole
- Ivermectin
- Levamisole
- Moxidectin (As per body weight and veterinary advice)

CONTROL & PREVENTION

- ✓ Pasture Management
- ✓ Rotational grazing
- ✓ Avoid wet & marshy grazing areas
- ✓ Mixed grazing with cattle

Strategic Deworming Based on faecal egg count

- ✓ Before rainy season
- ✓ Mid-rainy season

Lungworm infestation is a pasture associated disease – early diagnosis, and strategic deworming prevent economic losses and respiratory distress in sheep.

31. Lungworm Infestation

(Verminous Pneumonia)

Causes:

Causative Parasites

- Dictyocaulus filaria (most common)
- Protostrongylus rufescens
- Muellerius capillaris

Mode of Infection

- Ingestion of infective larvae (L3) from pasture
- High risk in wet, cool climates
- Overgrazing & poor pasture rotation

Predisposing Factors:

- Rainy season & high humidity
- Young sheep and lambs
- Overstocking
- Poor nutrition
- Continuous grazing on same pasture

Symptoms:

Early Signs

- Frequent coughing (especially after exercise)
- Nasal discharge
- Mild fever
- Reduced appetite

Advanced Signs

- Persistent deep cough
- Laboured breathing (dyspnoea)
- Open-mouth breathing
- Weight loss and poor growth
- Exercise intolerance

Severe Cases

- Pneumonia
- Secondary bacterial infection
- Death (rare but possible)

Diagnosis:**Field Diagnosis**

- History of grazing in wet pastures
- Characteristic coughing
- Poor response to antibiotics alone

Laboratory Diagnosis

- Faecal examination (Baermann technique) – gold standard
- Detection of lungworm larvae

Post-mortem Findings

- Adult worms in trachea & bronchi
- Consolidated lungs
- Verminous pneumonia lesions

Treatment:**Anthelmintics (Highly Effective)**

- Albendazole
- Fenbendazole
- Ivermectin
- Levamisole
- Moxidectin

(As per body weight and veterinary advice)

Supportive Therapy:

- Broad-spectrum antibiotics (for secondary infection)
- NSAIDs (to reduce inflammation)
- Good nutrition & rest

Control & Prevention:**Pasture Management**

- ✓ Rotational grazing
- ✓ Avoid wet & marshy grazing areas
- ✓ Mixed grazing with cattle

Strategic Deworming

- ✓ Before rainy season
- ✓ Mid-rainy season
- ✓ Based on faecal egg count

Management Practices

- ✓ Avoid overstocking
- ✓ Improve nutrition
- ✓ Quarantine & deworm new animals

Differential Diagnosis:

- Pasteurellosis
- PPR
- Mycoplasma pneumonia
- Maedi-Visna (chronic cases).

Key Message:

Lungworm infestation is a pasture-associated disease—early diagnosis and strategic deworming prevent economic losses and respiratory distress in sheep.

Oestrus ovis Infestation in Sheep

(Nasal Bot Fly Infestation / Oestrosis)

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CAUSES

Causative Agent

- **Oestrus ovis** (sheep nasal bot fly)

Mode of Infestation

- Adult female fly deposits live larvae (L1) around nostrils
- Larvae migrate to nasal passages and frontal sinuses
- Develop through L1 → L2 → L3, then expelled to soil.

SYMPTOMS

- **Early / Mild Infestation**
- Frequent sneezing
- Head shaking
- Nasal irritation
- Serous nasal discharge



Severe Infestation

- Respiratory distress
- Open-mouth breathing
- Head pressing
- Nervous signs (rare – due to sinus involvement)
- Reduced productivity



SYMPTOMS

- **Early / Mild Infestation**
- Frequent sneezing
- Head shaking
- Nasal irritation
- Serous nasal discharge



SYMPTOMS

Moderate Infestation

- Thick muco-purulent nasal discharge
- Noisy breathing
- Weight loss



DIAGNOSIS

- ✓ Seasonal occurrence
- ✓ Sneezing and nasal discharge
- ✓ History of fly exposure



Laboratory / Confirmatory Diagnosis

- Visualization of larvae during sneezing
- Endoscopic examination (advanced facilities)

TREATMENT

Antiparasitic Drugs

- ✓ Ivermectin
- ✓ Doramectin
- ✓ Moxidectin

(Highly effective against all larval stages)



Fly Control Red 1915

- ✓ Insecticide sprays or pour-on formulations
- ✓ Avoid grazing during peak fly activity (midday)

Strategic Deworming

- ✓ Broad-spectrum antibiotics
- ✓ Improve nutrition & rest
- ✓ Improve grazing at rest



ECONOMIC IMPORTANCE

- ✓ Reduced weight gain
- ✓ Lower wool yield production
- ✓ Increased susceptibility to respiratory distress diseases

DIFFERENTIAL DIAGNOSIS

- ✓ Pasteurellosis
- ✓ Lungworm Infestation
- ✓ Nasal foreign bodies



Oestrus ovis infestation is a seasonal parasitic disease – strategic anthelmintic treatment and fly control effectively prevent respiratory distress and production losses in sheep

32. Oestrus ovis Infestation

(Nasal Bot Fly Infestation / Oestrosis)

Causes

Causative Agent

- Oestrus ovis (sheep nasal bot fly)

Mode of Infestation

- Adult female fly deposits live larvae (L1) around nostrils
- Larvae migrate to nasal passages and frontal sinuses
- Develop through L1 → L2 → L3, then expelled to soil

Risk Factors

- Warm, dry climate
- Summer and post-monsoon season
- Extensive grazing systems
- Poor fly control

Symptoms

Early / Mild Infestation

- Frequent sneezing
- Head shaking
- Nasal irritation
- Serous nasal discharge

Moderate Infestation

- Thick muco-purulent nasal discharge
- Noisy breathing (snoring sound)
- Reduced feed intake
- Weight loss

Severe Infestation

- Respiratory distress
- Open-mouth breathing
- Head pressing
- Nervous signs (rare – due to sinus involvement)
- Reduced productivity

Diagnosis

Clinical Diagnosis

- Seasonal occurrence

- Sneezing and nasal discharge
- History of fly exposure

Laboratory / Confirmatory Diagnosis

- Visualization of larvae during sneezing
- Endoscopic examination (advanced facilities)

Post-Mortem Findings

- Larvae in nasal cavity & frontal sinuses
- Mucosal inflammation
- Secondary bacterial infection

Treatment

Antiparasitic Drugs

- ✓ Ivermectin
- ✓ Doramectin
- ✓ Moxidectin

(Highly effective against all larval stages)

Supportive Treatment

- Broad-spectrum antibiotics (if secondary infection)
- NSAIDs (reduce inflammation)
- Improve nutrition & rest

Control & Prevention

Fly Control

- ✓ Insecticide sprays or pour-on formulations
- ✓ Avoid grazing during peak fly activity (midday)

Strategic Deworming

- ✓ End of fly season
- ✓ Before breeding season
- ✓ Whole-flock treatment

Management Practices

- ✓ Maintain flock hygiene
- ✓ Avoid overcrowding
- ✓ Quarantine & treat new animals

Economic Importance

- Reduced weight gain

- Lower wool yield
- Reduced milk production
- Increased susceptibility to respiratory diseases

Differential Diagnosis

- Pasteurellosis
- Lungworm infestation
- Nasal foreign bodies
- Sinusitis

Key Message

Oestrus ovis infestation is a seasonal parasitic disease—strategic anthelmintic treatment and fly control effectively prevent respiratory distress and production losses in sheep.



IMPORTANT TICK INFESTATION IN SHEEP

Major Species, Diseases & Control Methods

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COMMON TICK SPECIES AFFECTING SHEEP

- 1 Rhipicephalus (Boophilus) spp.**
- One-host tick
 - Heavy infestation + anemia, poor growth
 - Vector of *Babesiosis*, *Anaplasmosis*

- 3 Haemaphysalis spp.**
- Three-host tick
 - Common in hilly and forest areas
 - Vector of *Theileria*, *Babesia*

- 2 Hyalomma spp.**
- Long-legged, aggressive tick
 - Causes severe irritation and wounds

- 4 Ixodes spp.**
- Found in cooler climates
 - Vector of *Theileria*, *Babesia*

TICK-BORNE DISEASES IN SHEEP

- + **Babesiosis** – Fever, anemia, hemoglobinuria
- + **Theileriosis** – Lymph node enlargement, emaciation
- + **Anaplasmosis** – Progressive anemia, weakness
- + **Tick paralysis** – Ascending paralysis, sudden death
- + **Secondary infections** – Myiasis, dermatitis



⚠️ Rotate acaricides to prevent resistance.

CONTROL METHODS OF TICKS IN SHEEP

- 1 CHEMICAL CONTROL (Acaricides)**
- Commonly used drugs
 - Synthetic pyrethroids: *Cypermethrin*, *Deltamethrin*
 - Organophosphates: *Chlorpyrifos*
 - Macrocytic lactones: *Ivermectin*, *Doramectin*

- 2 MANAGEMENTAL CONTROL**
- Regular inspection of flock
 - Quarantine new animals (2-3 weeks)
 - Avoid overstocking
 - Maintain clean sheds & dry bedding

⚠️ Rotate Acaricides to prevent resistance

PASTURE MANAGEMENT

- Rotational grazing
- Pasture spelling (tick life cycle break)
- Bush and weed control

Avoid grazing in heavily infested areas during peak season

BIOLOGICAL & ECOLOGICAL METHODS

- Encourage natural predators (birds)
- Use of tick resistant sheep breeds
- Reduce chemical overuse to protect ecosystem

KEY FIELD POINTS

Effective tick control in sheep requires an **integrated approach** combining acaricides, good management, and pasture hygiene to reduce disease, improve productivity, and minimize economic losses.



33. Tick Infestation in Sheep

Major Species, Diseases & Control Methods

Common Tick Species Affecting Sheep

1. Rhipicephalus (Boophilus) spp.

- One-host tick
- Heavy infestation → anemia, poor growth
- Vector of Babesiosis, Anaplasmosis

2. Hyalomma spp.

- Long-legged, aggressive tick
- Causes severe irritation and wounds
- Vector of Theileriosis

3. Haemaphysalis spp.

- Three-host tick
- Common in hilly and forest areas
- Vector of Theileria, Babesia

4. Ixodes spp.

- Found in cooler climates
- Vector of Tick paralysis, louping ill

Tick-Borne Diseases In Sheep

- Babesiosis – Fever, anemia, hemoglobinuria
- Theileriosis – Lymph node enlargement, emaciation
- Anaplasmosis – Progressive anemia, weakness
- Tick paralysis – Ascending paralysis, sudden death
- Secondary infections – Myiasis, dermatitis

Effects of Tick Infestation

- Blood loss → Anemia
- Reduced wool quality and weight gain
- Skin damage → leather defects
- Stress and reduced immunity
- Economic losses to farmers

Control Methods of Ticks in Sheep

1 Chemical Control (Acaricides)

Commonly used drugs

- Synthetic pyrethroids: Cypermethrin, Deltamethrin
- Organophosphates: Chlorpyrifos
- Macrocyclic lactones: Ivermectin, Doramectin

Application methods

- Dipping
- Spraying
- Pour-on
- Injectable (systemic)

Rotate acaricides to prevent resistance

2 Managemental Control

- Regular inspection of flock
- Quarantine new animals (2–3 weeks)
- Avoid overstocking
- Maintain clean sheds and dry bedding
- Strategic shearing to expose ticks

3 Pasture Management

- Rotational grazing
- Pasture spelling (tick life cycle break)
- Bush and weed control
- Avoid grazing in heavily infested areas during peak season

4 Biological & Ecological Methods

- Encourage natural predators (birds)
- Use of tick-resistant sheep breeds
- Reduce chemical overuse to protect ecosystem

5 Integrated Tick Management (Best Practice)

- ✓ Chemical + Management + Pasture control
- ✓ Strategic treatment during peak seasons
- ✓ Farmer education & surveillance

Key Field Points

- Treat entire flock, not individual animals
- Follow correct dose & interval
- Observe withdrawal period
- Monitor for acaricide resistance

Conclusion

Effective tick control in sheep requires an integrated approach combining acaricides, good management, and pasture hygiene to reduce disease, improve productivity, and minimize economic losses.

Mite Infestation (Mange) IN SHEEP

Common Causative Mites

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Psoroptes ovis
(Sheep Scab Mite)



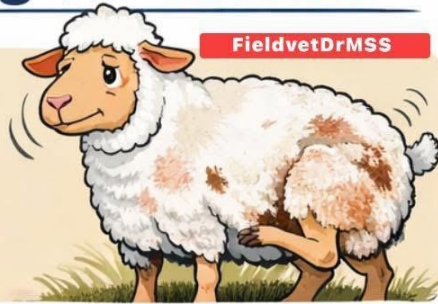
Sarcoptes scabiei



Chorioptes bovis



Demodex ovis



Clinical Signs

Psoroptic Mange (Sheep Scab)



• Intense Itching & Wool Loss



• Crusty Lesions

• Weight Loss

Sarcoptes Mange



• Thickened Skin

• Hair Loss on Face & Ears

• Severe Itching

Chorioptic Mange



• Lesions on Feet & Tail

• Crusting & Scaling

Demodectic Mange



• Skin Nodules



Diagnosis

Clinical Exam

- ▶ Itching & Lesions
- ▶ Flock History



Treatment & Control

Systemic Treatment

- ▶ Ivermectin, Doramectin, Moxidectin
- ▶ Repeat Every 7-14 Days

Topical Treatment

- ▶ Lime Sulfur Dip
- ▶ Amitraz or Pyrethroid Sprays
- ▶ Treat **Entire Flock**



Wool Damage



Weight Loss



Secondary Infections



Reduced Production



Quarantine
New Animals



Regular
Flock Checks



Disinfect
Housing



Disinfect
equipment

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34. Mite Infestation (Mange)



Common Causative Mites

1. *Psoroptes ovis* – Causes psoroptic mange (Sheep scab)
2. *Sarcoptes scabiei* – Causes sarcoptic mange
3. *Chorioptes bovis* – Causes chorioptic mange (foot and tail mange)
4. *Demodex ovis* – Causes demodectic mange (rare in sheep)

Clinical Signs

1. Psoroptic Mange (Sheep Scab)

- Intense pruritus (severe itching)
- Sheep rubbing against objects
- Wool loss (patchy alopecia)
- Thick crust formation
- Moist exudative lesions
- Weight loss, anemia in severe cases

2. Sarcoptic Mange

- Severe itching
- Thickened, wrinkled skin
- Hair loss on face, ears, axilla, groin
- Reduced feed intake

3. Chorioptic Mange

- Mild itching
- Lesions on feet, tail, scrotum
- Crusting and scaling

4. Demodectic Mange

- Nodules in skin
- Usually non-pruritic
- Rare and often localized



Diagnosis

Clinical Diagnosis

- History of intense itching and flock spread
- Typical lesion distribution
- Seasonality (common in winter & overcrowding)

Laboratory Diagnosis

- Deep skin scraping (especially for Sarcoptes & Demodex)
- Superficial skin scraping (Psoroptes, Chorioptes)
- Microscopic examination
- Crust examination

Response to treatment (therapeutic diagnosis)

Treatment

Systemic Treatment (Most Effective)

- Ivermectin – 0.2 mg/kg SC (repeat after 7–14 days)
- Doramectin – 0.2 mg/kg SC
- Moxidectin – As per label dose

(Repeat dose is essential to kill newly hatched mites)

Topical Treatment

- Lime sulfur dips
- Amitraz spray (0.025–0.05%)
- Synthetic pyrethroids (cypermethrin spray)

 **Treat entire flock, not just affected animals.**

Control & Prevention

- Quarantine new animals (minimum 2–3 weeks)
- Regular flock inspection
- Avoid overcrowding
- Proper nutrition to improve immunity
- Shearing management
- Disinfection of housing and equipment
- Strategic deworming & ectoparasite control schedule

Economic Importance

- Wool damage
- Reduced body weight gain
- Secondary bacterial infections
- Decreased reproductive performance

Trade restrictions in severe outbreaks (especially psoroptic mange)

BEST TREATMENT FOR FOOT ROT IN SHEEP

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1. Causes

- *Dichelobacter nodosus* (primary bacterium)
- *Fusobacterium necrophorum* (secondary infection)
- Favouring conditions: wet; muddy, unhygienic flooring

2. Clinical Signs

- Lameness, reluctance to **Walk**
- Swollen, painful interdigital space
- Foul smell from feet
- Loss of body condition
- Underrunning of hoof wall

3. Best Treatment Protocol

A. Foot Bathing (Essential Step)

Use any one of the following solutions:

- 10% Zinc Sulfate (most effective)
- 5% Formalin (effective but irritant; use with caution)
- Copper Sulfate 5–10%

Routine:

- Stand sheep in solution for 10–15 minutes
- Repeat every 5–7 days for 3–4 sessions

B. Systemic Antibiotics

Oxytetracycline (Long-acting)

- Penicillin–Streptomycin combination
- Amoxicillin

C. Topical Antibiotics

Apply after cleaning and drying:

- Oxytetracycline spray
- Iodine-based antiseptics
- Chlorhexidine solution (0.5%)

D. Hoof Trimming

Remove loose, underrun horn

- Expose infected pockets for aeration
- Avoid excessive trimming to prevent bleeding

E. Pain Management

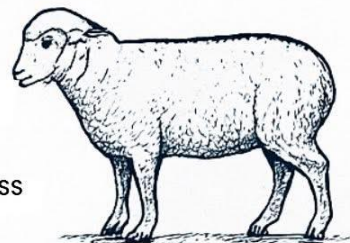
NSAIDs: Meloxicam / Flunixin

Reduces pain and improve mobility

4. Control & Prevention

- Regular hoof trimming (every 3–4 months)
- Keep flooring dry, avoid muddy areas
- Quarantine and inspect new animals
- Routine foot baths in rainy season

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Section 8: Clinical Conditions & Disorders

35. Foot Rot Treatment Protocol

1. Causes

- *Dichelobacter nodosus* (primary bacterium)
- *Fusobacterium necrophorum* (secondary infection)
- Favouring conditions: wet, muddy, unhygienic flooring

2. Clinical Signs:

- Lameness, reluctance to walk
- Foul smell from feet
- Underrunning of hoof wall
- Swollen, painful interdigital space
- Loss of body condition

3. Best Treatment Protocol:

A. Foot Bathing (Essential Step)

Use any one of the following solutions:

- 10% Zinc Sulfate (most effective)
- 5% Formalin (effective but irritant; use with caution)
- Copper Sulfate 5–10%

Routine:

- Stand sheep in solution for 10–15 minutes
- Repeat every 5–7 days for 3–4 sessions

B. Systemic Antibiotics (Severe Cases):

- Oxytetracycline (Long-acting)
- Penicillin–Streptomycin combination
- Amoxicillin

(Administer under veterinary supervision)

C. Topical Antibiotics:

Apply after cleaning and trimming:

- Oxytetracycline spray
- Iodine-based antiseptics
- Chlorhexidine solution (0.5%)

D. Hoof Trimming:

- Remove loose, underrun horn

- Expose infected pockets for aeration
- Avoid excessive trimming to prevent bleeding

E. Pain Management:

- NSAIDs: Meloxicam / Flunixin
- Reduces pain and improves mobility

4. Control & Prevention:

- Regular hoof trimming (every 3–4 months)
- Keep flooring dry; avoid muddy areas
- Quarantine and inspect new animals
- Routine foot baths in rainy season
- Culling chronically infected animals

5. Prognosis:

- Good with early treatment
- Chronic cases may need repeated therapy or culling.

BEST TREATMENT PROTOCOL FOR CASEOUS LYMPHADENITIS(CLA) IN SHEEP



"CLA control depends more on biosecurity, vaccination, and culling than on treatment."

Meta AI

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36. Caseous Lymphadenitis (CLA)

Causes • Symptoms • Diagnosis • Treatment • Control

1. Cause:

- *Corynebacterium pseudotuberculosis* → chronic, contagious disease affecting lymph nodes and internal organs.

2. Symptoms:

- Firm, enlarged lymph nodes (parotid, prescapular, prefemoral)
- Abscesses with thick “cheesy” green pus
- Weight loss, reduced wool quality
- Coughing / respiratory distress (internal CLA)
- Reduced milk and meat productivity

3. Diagnosis:

- Physical examination of lymph nodes
- Needle aspiration of abscess → characteristic pus
- Bacterial culture & PCR
- ELISA serology for flock screening
- Ultrasound for internal abscesses

4. Best Treatment Protocol:

A. Abscess Management (External CLA)

1. Isolate the affected animal
2. Clip wool & disinfect area (iodine/chlorhexidine)
3. When abscess is mature:
 - Lance at lowest point
 - Drain pus completely
 - Flush cavity with iodine/chlorhexidine
4. Collect and burn the pus & contaminated material
5. Keep animal in isolation until wound heals

B. Antibiotic Therapy:

CLA is difficult to eradicate because organism survives inside cells.

Useful options (under veterinary supervision):

- Procaine Penicillin + Streptomycin
- Tulathromycin
- Long-acting Oxytetracycline

- Rifampicin (rarely used)
- Ceftiofur (limited benefit)

Antibiotics reduce progression but do not completely cure chronic CLA.

C. Internal CLA:

- Long-term antibiotics may reduce severity
- Supportive care (fluids, NSAIDs)
- Cull chronic, severely affected animals

5. Control & Prevention (Most Important):

A. Biosecurity

- Quarantine all new animals for 30 days
- Check for abscesses before introduction
- Purchase from CLA-free flocks

B. Shearing Management

- Sterilize shearing blades between sheep
- Reduce shearing cuts → common source of infection
- Clean shearing area regularly

C. Sanitation

- Disinfect equipment (ear-taggers, castration tools)
- Burn or deep bury abscess material
- Avoid overcrowding

D. Vaccination

- CLA vaccines (where available) reduce severity and new infections
- Annual vaccination recommended for endemic flocks

E. Culling

- Remove repeat offenders or sheep with multiple abscesses
- Improves long-term flock resistance and reduces spread

Key Message

“CLA control depends more on biosecurity, vaccination, and culling than on treatment.”

OBSTETRICAL DISORDERS OF SHEEP

Causes – Symptoms – Diagnosis – Treatment – Control

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1. Dystocia (Difficult Lambing)

Causes

- Oversized fetus
- Malpresentations (breech, head back, limb flexion)
- Incomplete cervical dilation
- Uterine inertia (primary or secondary)

Symptoms

- Prolonged labour (>1 hour without progress)
- Straining without delivering
- Visible abnormal presentation

Diagnosis

Vaginal examination

Assess fetal position, size and dilation

Treatment: Correct malpresentation

Gentle traction with lubrication, Oxytocin for uterine inertia, Caesarean section if needed

3. Vaginal / Uterine Prolapse

Causes

- Late gestation abdominal pressure
- Hypocalcemia
- Over-conditioning
- Very short tail docking

Symptoms

- Tissue protruding from vulva

Diagnosis

Visual examination to identify type

Treatment: Clean and replace prolapsed tissue. Retention sutures (Bühner stitch)

Correct mineral imbalance

4. Retained Placenta (Retained Fetal Membranes)

Causes

- Difficult Lambing
- Selenium / vitamin E deficiency
- Infections

Symptoms

Placenta not expelled within 6–8 hours.

Foul smell, fever in severe cases

2. Ring Womb (Failure of Cervical Dilation):

Causes: Hormonal imbalance

Older ewes, Mineral deficiencies

Symptoms: Active labour with closed cervix. No progress in delivery
No progress in lambing

Diagnosis: Vaginal exam showing undilated cervix

Treatment: Gentle manual stretching

Prostaglandins or oxytocin inj

Caesarean section if unresponsive

Control: Culling repeat cases

Balanced late-pregnancy nutrition, prevent any stress

5. Pregnancy Toxemia (Lambing Paralysis) – Obstetrical impact

Causes

- Twin/triplet pregnancies
- Negative energy balance

Symptoms

Weakness, recumbency, Unable to lamb normally

Treatment: Propylene glycol

IV glucose, Induce lambing / C-section if ewe worsens as per vet guidance

Control

- High energy feed in late pregnancy
- Monitor high-risk ewes

7. Abortions (Obstetrical Failure)

Causes

- Infectious agents (Brucella, Chlamydia, Toxoplasma, Campylobacter)
- Nutritional deficiencies
- Stress

37. Obstetrical Disorders

(Causes – Symptoms – Diagnosis – Treatment – Control)

1. Dystocia (Difficult Lambing)

Causes

- Oversized fetus
- Malpresentations (breech, head back, limb flexion)
- Incomplete cervical dilation
- Uterine inertia (primary or secondary)

Symptoms

- Prolonged labour (>1 hour without progress)
- Straining without delivering
- Visible abnormal presentation

Diagnosis

- Vaginal examination
- Assess fetal position, size and dilation

Treatment

- Correct malpresentation
- Gentle traction with lubrication
- Oxytocin for uterine inertia
- Caesarean section if needed

Control

- Use rams with moderate birth weights
- Good late-pregnancy nutrition
- Avoid breeding very young or very old ewes

2. Vaginal / Uterine Prolapse:

Causes

- Late gestation abdominal pressure
- Hypocalcemia
- Over-conditioning
- Very short tail docking

Symptoms

- Tissue protruding from vulva
- Straining, discomfort

Diagnosis: Visual examination to identify type

Treatment

- Clean and replace prolapsed tissue
- Retention sutures (Bühner stitch)
- Correct mineral imbalance

Control

- Avoid overfeeding
- Adequate minerals (Ca, P)
- Proper tail docking length

3. Ring Womb (Failure of Cervical Dilation):

Causes

- Hormonal imbalance
- Older ewes
- Mineral deficiencies

Symptoms

- Active labour with closed cervix
- No progress in lambing

Diagnosis

- Vaginal exam showing undilated cervix

Treatment

- Gentle manual stretching
- Prostaglandins or oxytocin
- Caesarean section if unresponsive

Control

- Culling repeat cases
- Balanced late-pregnancy nutrition

4. Retained Placenta (Retained Fetal Membranes):

Causes

- Difficult lambing
- Selenium / Vitamin E deficiency
- Infections

Symptoms

- Placenta not expelled within 6–8 hours
- Foul smell, fever in severe cases

Diagnosis

- Clinical signs + vaginal exam

Treatment

- Do NOT manually pull placenta
- Oxytocin or prostaglandin
- Antibiotics if metritis occurs

Control

- Adequate Se + Vitamin E nutrition
- Hygienic lambing practices

5. Pregnancy Toxemia (Lambing Paralysis) – Obstetrical Impact:

Causes

- Twin/triplet pregnancies
- Negative energy balance

Symptoms

- Weakness, recumbency
- Unable to lamb normally

Diagnosis

- Clinical signs + ketone testing

Treatment

- Propylene glycol
- IV glucose
- Induce lambing / C-section if ewe worsens

Control

- High-energy feed in late pregnancy
- Monitor high-risk ewes

6. Uterine Torsion (Rare in Sheep):

Causes

- Rolling in late pregnancy
- Large single fetus

Symptoms

- Labour without cervical dilation
- Restlessness, abdominal pain

Diagnosis

- Vaginal exam showing twist
- Manual palpation

Treatment

- Rolling ewe using plank method
- C-section if correction fails

Control

- Avoid slippery flooring
- Close monitoring in late gestation

7. Abortions (Obstetrical Failure):

Causes

- Infectious agents (Brucella, Chlamydia, Toxoplasma, Campylobacter)
- Nutritional deficiencies
- Stress

Symptoms

- Premature expulsion of fetus
- Vaginal discharge
- Retained placenta

Diagnosis

- History + clinical exam
- Laboratory testing (PCR, serology)

Treatment

- Supportive
- Antibiotics depending on pathogen

Control

- Vaccination (where available)
- Biosecurity and hygiene
- Rodent control

Key Message

“Early detection protects both ewe and lamb.”

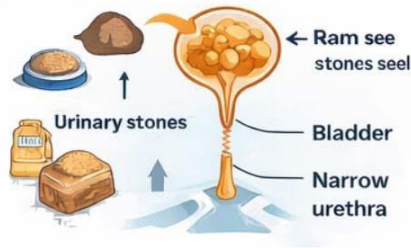
Urolithiasis in Ram

(Urinary Calculi / Obstructive Uropathy)

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Causes

- **Urinary stone formation** (phosphate, calcium carbonate, silica)
- **High-concentrate, low-roughage feeding**
- **Imbalanced Calcium:** Phosphorus ratio (Ca:P < 2:1)
- **High phosphorus diets** (grain-based rations)
- **High magnesium intake**
- **Alkaline urine**
- **Inadequate water intake**
- **Early castration** → narrow urethra
- **Pelleted feeds** with high mineral content
- **Sudden dietary changes**
- **Cold weather** → reduced water consumption.



Symptoms

- **Straining to urinate** (dysuria)
- **Repeated unsuccessful attempts** to pass urine
- **Scanty or absent urine flow**
- **Dribbling** of blood-tinged urine
- **Stretching posture** (like urination stance)
- **Kicking** or biting abdomen
- **Swollen prepuce or penis**
- **Restlessness & depression**
- **Loss of appetite**
- **Grinding of teeth** (pain)
- **Abdominal distension** (bladder rupture)
- **Sudden death** (due to uroperitoneum)

Diagnosis

- **Clinical Diagnosis**
- History of concentrate feeding
- **Observation** of painful urination attempts
- **Palpation of urethra** (urethral process, sigmoid flexure)
- **Rectal palpation** (distended bladder)



Imaging

- **Ultrasonography**
 - Distended bladder
 - Free fluid
- **Radiography** (limited use)
 - Mineralized calculi

Laboratory Diagnosis

Serum biochemistry:

- ↑ Blood urea nitrogen
- ↑ Creatinine
- ↑ Potassium



Urine examination (if available)

- Crystals
- Alkaline pH



Treatment

Emergency Management

- **Pain relief:** NSAIDs (Meloxicam / Flunixin)
- **IV fluids** to correct dehydration & electrolytes
- **Antispasmodics** (Butylscopolamine)

Specific Treatment

- **Urethral Process Amputation**
 - First-line treatment in rams
 - Catheterization / Hydropulsion (at uddigce)
 - **Surgical Options** (best for breeding ram)
 - Perineal urethrostomy (salvage procedure)
 - Bladder repair if rupture detected early
- **Medical Management**
 - **Urine acidifiers** (Ammonium chloride)
 - Dietary correction

Key Message

Urolithiasis is a nutritional and management disease – prevention is more effective than treatment.

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38. Urolithiasis in Ram

(Urinary Calculi / Obstructive Uropathy)

Causes

- Urinary stone formation (phosphate, calcium carbonate, silica).
- High-concentrate, low-roughage feeding.
- Imbalanced Calcium: Phosphorus ratio ($\text{Ca:P} < 2:1$).
- High phosphorus diets (grain-based rations).
- High magnesium intake.
- Alkaline urine.
- Inadequate water intake.
- Early castration → narrow urethra.
- Pelleted feeds with high mineral content.
- Sudden dietary changes.
- Cold weather → reduced water consumption.

Symptoms:

- Straining to urinate (dysuria).
- Repeated unsuccessful attempts to pass urine.
- Scanty or absent urine flow.
- Dribbling of blood-tinged urine.
- Stretching posture (like urination stance).
- Kicking or biting abdomen.
- Swollen prepuce or penis.
- Restlessness and depression.
- Loss of appetite.
- Grinding of teeth due to pain.
- Abdominal distension (bladder rupture).
- Sudden death due to uroperitoneum.

Diagnosis:

Clinical Diagnosis

- History of concentrate feeding.
- Observation of painful urination attempts.
- Palpation of urethra (urethral process, sigmoid flexure).
- Rectal palpation (distended bladder).

Laboratory Diagnosis

- Serum biochemistry:
 - ↑ Blood urea nitrogen (BUN)

- ↑ Creatinine
- ↑ Potassium
- Urine examination (if available):
 - Crystals
 - Alkaline pH

Imaging

- Ultrasonography:
 - Distended bladder
 - Free abdominal fluid (rupture)
- Radiography (limited use):
 - Mineralized calculi

Treatment:

Emergency Management

- Pain relief: NSAIDs (Meloxicam / Flunixin).
- IV fluids to correct dehydration & electrolytes.
- Antispasmodics (Butylscopolamine).

Specific Treatment

1. Urethral Process Amputation

- First-line treatment in rams.
- Relieves obstruction at distal urethra.

2. Catheterization / Hydropulsion

- Flushing of small calculi.
- Requires sedation.

3. Surgical Treatment

- Tube cystostomy – best for breeding rams.
- Perineal urethrostomy – salvage procedure.
- Bladder repair if rupture detected early.

4. Medical Management

- Urine acidifiers: Ammonium chloride.
- Dietary correction.

5. Antibiotics

- To prevent secondary urinary tract infection.

Control & Prevention:

- Maintain Ca:P ratio = 2:1.
- Increase roughage; reduce concentrates.
- Avoid excess phosphorus & magnesium.
- Provide clean, abundant drinking water.
- Add salt (1–3%) in ration to increase water intake.
- Use ammonium chloride (0.5–1%) as urine acidifier.
- Avoid early castration (delay >3 months).
- Avoid sudden feed changes.
- Encourage grazing rather than stall feeding.
- Monitor mineral content of feed and water.

Key Message:

Urolithiasis is a nutritional and management disease—prevention is more effective than treatment.

JOINT ILL (SEPTIC ARTHRITIS) IN SHEEP

Causes • Symptoms • Diagnosis • Treatment • Control

1. DEFINITION

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Joint ill is infectious inflammation of joints seen mainly in young lambs, caused by bacteria entering the bloodstream—most commonly through the navel (umbilical infection) — and localizing in joints.



2. CAUSES

A. Predisposing Factors

- Poor navel hygiene after birth
- Dirty, wet lambing pens
- Delayed or absent navel dipping
- Weak immunity due poor colostrum intake
- Overcrowding and poor ventilation



2. CAUSES

A. Predisposing Factors

- Poor navel hygiene after birth
- Dirty, wet lambing pens
- Delayed or absent navel dipping
- Weak immunity due to poor colostrum intake
- Overcrowding and poor ventilation



4. SYMPTOMS (CLINICAL SIGNS)

Local Signs

- Hot, swollen, painful joints
- Commonly affected joints:
 - Knee (carpal)
 - Hock (tarsal), Stifle
 - Stifle
- Reluctance to stand or walk
- Lameness or recumbency.



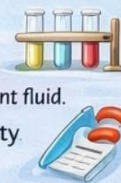
4. DIAGNOSIS

- Field Diagnosis
- History of poor lambing hygiene
- Swollen joints in young lambs (1–4 weeks)



• Clinical & Laboratory Diagnosis

- Joint palpation (pain, heat, fluctuation)
- Joint fluid aspiration: Turbid or purulent fluid.
- Bacterial culture & antibiotic sensitivity
- Blood tests (leukocytosis)



5. DIAGNOSIS

A. Medical Treatment (Early Cases)

- Systemic antibiotics (5–7 days or longer.)
- Penicillin-Streptomycin
- Amoxiellin
- Oxytetracycline



- NSAIDs for pain & inflammation.
- Supportive care
 - Fluids, Good nutrition
 - Warm, dry bedding



6. TREATMENT

A. Medical Treatment (Early Cases)

- Joint lavage with sterile saline
- Intra-articular antibiotics (veterinary supervision only)



c. Chronic Cases

- ✓ Poor prognosis
- ✓ Often require culling due to permanent joint damage



KEY MESSAGE

Joint ill is preventable—clean lambing practices and early treatment are the most effective control measures.

#SheepHealth • #NavelDipping • #LambCare • #VeterinaryPublicHealth • #Fieldvet

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39. Joint ILL in Sheep (Navel ILL/ Infectious Arthritis)

Joint ill is a common bacterial infection of joints in newborn lambs, usually occurring during the first few weeks of life. It is mainly associated with poor navel hygiene and contaminated lambing environments

Causes

Joint ill occurs when bacteria enter the body through the umbilical cord (navel) soon after birth.

Common bacteria involved:

- Streptococcus spp.
- Staphylococcus spp.
- Trueperella (Arcanobacterium) pyogenes
- E. coli

Mode of Infection

- Lamb is born in a dirty lambing area.
- Bacteria enter through the open umbilical cord.
- Bacteria enter the bloodstream (septicemia).
- Infection settles in joints, causing inflammation and swelling.

Clinical Signs

- Swollen, hot and painful joints
- Lameness or difficulty walking
- Lamb reluctant to stand
- Fever and dullness
- Reduced suckling
- Multiple joints may be affected (knees, hocks)

In severe cases:

- Joint deformity
- Septicemia and death

Diagnosis

- Clinical examination
- History of poor lambing hygiene
- Joint swelling and lameness in young lambs
- Joint fluid examination (if required)

Treatment

- Early treatment is very important.

- Antibiotics
 - Long acting penicillin
 - Oxytetracycline
 - Other antibiotics as advised by veterinarian
- Supportive care
- Anti-inflammatory drugs
- Proper nursing care
- Isolate affected lambs

Chronic cases may have permanent joint damage.

Prevention (Most Important)

- Maintain clean lambing pens
- Disinfect the navel immediately after birth using iodine solution
- Provide adequate colostrum within first 1–2 hours
- Keep bedding dry and hygienic
- Avoid overcrowding
- Ensure good flock hygiene

Field Veterinary Significance

- Causes high lamb mortality and poor growth
- Leads to economic losses in sheep farms
- Preventable with simple hygiene measures

Key Message:

Joint ill is largely preventable through proper lambing hygiene, navel disinfection, and early colostrum feeding.

ORTHOPAEDIC ISSUES OF SHEEP

Causes • Clinical Signs • Management • Prevention

WHY ORTHOPAEDIC HEALTH MATTERS

- Lameness is a major cause of production loss
- Leads to poor growth, reduced fertility, welfare issues
- Early diagnosis = better recovery & lower economic loss



COMMON ORTHOPAEDIC DISORDERS IN SHEEP

1. FOOT ROT

- **CAUSE:** *Dichelobacter nodosus* + *Fusobacterium necrophorum*
- **SIGNS:** Severe lameness, underrunning of hoof horn
- **MANAGEMENT:** Hoof trimming, footbath (2eSO₄ / CuSO₄), antibiotics
- **PREVENTION:** Dry flooring, regular foot care, biosecurity



2. FOOT SCALD (INTERDIGITAL DERMATITIS)

- **CAUSE:** Moist conditions, bacterial infection
- **SIGNS:** Mild-moderate lameness, reddened interdigital skin
- **MANAGEMENT:** Topical antiseptics, footbath



4. FRACTURES

- **CAUSE:** Poor hoof wear, soft soil, neglect
- **SIGNS:** Abnormal gait, uneven weight bearing
- **MANAGEMENT:** Corrective hoof trimming & trimming



5. JOINT ILL (SEPTIC ARTHRITIS)

- **Cause:** Bacterial infection via navel (lambs)
- **SIGNS:** Abnormal gait, uneven weight bearing
- **Management:** Antibiotics, joint lavage



6. OSTEOMYELITIS

- **CAUSE:** Extension of infection from wounds or joints
- **SIGNS:** joint swelling, stiffness
- **MANAGEMENT:** Anti-inflammatories, antibiotics if infectious



RISK FACTORS

- Wet & muddy housing
- Poor nutrition (Ca, P imbalance)
- Rough handling & transport
- Poor hygiene in lambing pens
- Lack of routine hoof care

ROLE OF VETERINARIANS

- ✓ Early lameness diagnosis
- ✓ Correct hoof trimming guidance
- ✓ Rational antibiotic use (AMR prevention)
- ✓ Farmer (education & flock health planning)

PREVENTION STRATEGIES

- ✓ Regular hoof trimming
- ✓ Clean & dry housing
- ✓ Balanced mineral nutrition
- ✓ Footbath protocols
- ✓ Prompt treatment of lame animals
- ✓ Biosecurity for new stock

ECONOMIC & WELFARE IMPACT

- ✓ Reduced treatment costs
- ✓ Improved mobility & productivity
- ✓ Better animal welfare
- ✓ Increased flock profitability



KEY MESSAGE

Healthy feet and joints are essential for productive and welfare-friendly sheep farming,

#SheepHealth • #Orthopaedics • #Lameness • #VeterinaryEducation • #Fieldvet

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40. Orthopaedic Issues in Sheep

Orthopaedic disorders in sheep affect bones, joints, tendons, and hooves, leading to lameness, poor mobility, reduced grazing ability, and economic losses in sheep farming. Early detection and proper management are essential for flock health.

Common Orthopaedic Problems in Sheep

1. Foot Rot

Cause: Bacterial infection (mainly *Dichelobacter nodosus* and *Fusobacterium necrophorum*)

Signs

- Severe lameness
- Foul-smelling hoof lesions
- Separation of hoof horn

Significance

- Reduced grazing
- Weight loss
- Decreased productivity

2. Foot Abscess

Cause: Bacterial infection entering through hoof injuries.

Signs

- Swollen hoof
- Sudden severe lameness
- Heat and pain in the hoof

Significance

- Difficulty walking
- Reduced feed intake

3. Joint Ill (Septic Arthritis in Lambs)

Cause: Bacterial infection entering through the navel in newborn lambs.

Signs

- Swollen joints
- Lameness
- Fever and weakness

Significance

- High lamb mortality
- Permanent joint deformities

4 □□ Fractures

Cause

- Trauma
- Accidents during handling
- Dog attacks
- Falls

Common sites

- Leg bones
- Pelvis

Significance

- Severe pain
- Reduced mobility
- May require euthanasia in severe cases

5 □□ Laminitis

Cause

- Sudden diet changes
- Grain overload
- Metabolic disturbances

Signs

- Reluctance to walk
- Stiff gait
- Painful hooves

Significance

- Chronic lameness
- Poor growth

6 □□ Tendon and Ligament Injuries

Cause

- Overexertion
- Slippery floors
- Trauma

Signs

- Limping
- Swelling around joints

- Abnormal gait

Significance

- Reduced grazing ability

Risk Factors

- Wet and muddy grazing lands
- Poor hoof care
- Rough terrain
- Nutritional imbalance
- Overcrowding
- Poor lambing hygiene

Prevention Strategies

- ✓ Regular hoof trimming
- ✓ Maintain dry bedding and clean pens
- ✓ Provide balanced nutrition with minerals
- ✓ Footbath using zinc sulphate or copper sulphate
- ✓ Good lambing hygiene
- ✓ Early treatment of lameness

Veterinary & Farm Significance

Orthopaedic problems can cause:

- Reduced grazing efficiency
- Weight loss and poor growth
- Lower meat and wool production
- Increased treatment costs

Early detection of lameness is a key indicator of flock health.

Key Message

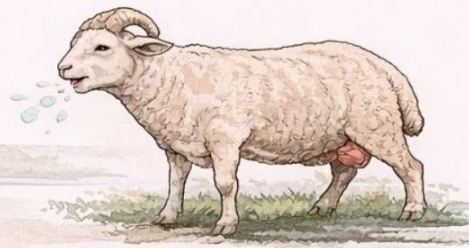
Healthy feet and joints are essential for productive sheep farming.

Regular hoof care, good hygiene, and early veterinary intervention prevent major losses.

Maedi in Sheep

(Ovine Progressive Pneumonia – OPP)

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↪ Cause

- Maedi–Visna virus (MVV)
- A lentivirus belonging to the family *Retroviridae*

1 Epidemiology

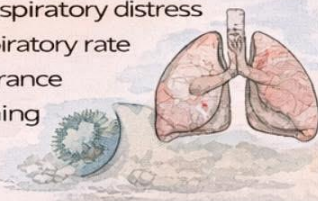
- Affects adult sheep (usually > 2 years)
- Chronic, slowly progressive disease
- More common in:
 - Intensively managed flocks
 - Closed housing
 - Poor ventilation
- Worldwide distribution

2 Pathogenesis

- Vertical transmission
 - Via colostrum and milk
- Horizontal transmission
 - Aerosol (close contact, –coughing)
 - Introduction of carrier animals
 - Long incubation period (months to years)

🦠 Clinical Signs

- **Maedi Form** – (Maedi Form - Respiratory)
 - Progressive respiratory distress
 - Increased respiratory rate
 - Exercise intolerance
 - Chronic coughing
 - Weight loss (emaciation)



🦠 Post-mortem Lesions

- Heavy, firm, non-collapsing lungs
- Greyish lung surface
- Interstitial pneumonia
- Enlarged mediastinal lymph nodes

🗉 Key Message

• Maedi is a slow, incurable viral disease early testing, and strict biosecurity are the only effective control measures. Differential Diagnosis is very important.

↪ Clinical Signs (Maedi Form – Respiratory)

- Progressive respiratory distress
- Increased respiratory rate
- Exercise intolerance
- Chronic coughing
- Weight loss (emaciation)
- Normal or subnormal temperature
- Joints



🦠 Post-mortem Lesions

- Heavy, firm, non-collapsing lungs
- Greyish lung surface
- Interstitial pneumonia
- Enlarged mediastinal lymph nodes

🦠 Diagnosis

↪ Field Diagnosis

- Chronic respiratory signs
- Progressive weight loss
- Poor response to antibiotics

🦠 Laboratory

- Serology (ELISA, AGID) – most common
- PCR for MVV
- Histopathology (interstitial pneumonia)

🦠 Differential Diagnosis

- Pasteurellosis
- CCPP-like disease
- Lungworm infestation
- PPR
- Chronic pneumonia



Economic Importance

- Progressive production losses
- Reduced body weight and wool yield
- Reduced milk production
- Premature culling in advanced stages

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Section 9: Advanced & Miscellaneous Conditions

41. Maedi in Sheep

Maedi-Visna (MVV) is a globally widespread, persistent viral disease in sheep, causing chronic pneumonia (Maedi) and neurological issues (Visna), leading to significant economic losses, with major exceptions being Australia and New Zealand.

It affects most sheep-keeping regions, including Europe, Asia, Africa, and the Americas, though prevalence varies, with higher rates in developed nations linked to intensive farming practices like pooled colostrum feeding, favoring transmission via milk, aerosols, and fluids.

Global Presence:

Widespread: Occurs in most sheep-farming countries, with reported cases across continents.

Exceptions:

Australia and New Zealand are considered largely free of MVV, notes ScienceDirect.com, and Dove Medical Press.

High Prevalence:

Asia (especially China) and Europe show high flock prevalence, with significant presence in North America, South America, and Africa.

Key Characteristics:

Causative Agent: Small Ruminant Lentivirus (SRLV).

Clinical Signs: Progressive lung disease (Maedi), arthritis, mastitis, and encephalitis (Visna).

Economic Impact: Causes substantial losses due to reduced production, involuntary culling, and trade restrictions.

Transmission & Control

Main Routes: Colostrum/milk (milk-borne) and contact with infected fluids (aerosols, respiratory exudates).

Control Challenges: No effective treatment or vaccine; long infection-to-seroconversion period; difficulty in early diagnosis.

Management: Emphasizes surveillance (serology), biosecurity, hygiene, and culling seropositive animals.

Other Names:

- Ovine Progressive Pneumonia (OPP) in the U.S.
- La bouhite (France)
- Graaff-Reinet disease (South Africa)
- Zwoegerziekte (Netherlands)

PAPILLOMATOSIS IN SHEEP

CAUSE

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- Ovine Papillomavirus (OPV)
- A DNA virus causing warts.

EPIDEMIOLOGY

- Young sheep (< 2 years)
- Poor hygiene, overcrowding
- Spread via contact & fomites.

PATHOGENESIS

- 1 Virus enters through skin abrasions.



- 2 Viral infection of epithelial cells.



- 3 Formation of papillomas.

CLINICAL SIGNS



Lips



Eyelid



Teats

- Cauliflower-like warts
- Usually self-limiting.

DIAGNOSIS

- Clinical Appearance
- PCR & Histopathology



DIFFERENTIAL DIAGNOSIS

- Contagious Ecthyma (Orf)
- Squamous Cell Carcinoma
- Fibroma & Abscess



TREATMENT

- Most Regress in 2–6 Months
- Improve Hygiene.
- Isolate Severe Cases.
- Surgical Removal (if needed).

PREVENTION

- Avoid Skin Trauma.
- Disinfect Equipment.
- Isolate Affected Sheep.
- Reduce Overcrowding.

ECONOMIC IMPACT



Milk Production



Secondary Infections



Carcass Losses

NOT ZOONOTIC!

- Does **NOT** infect humans.



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42. Papillomatosis in Sheep

1. Definition

Papillomatosis is a viral disease characterized by benign epithelial tumors (warts or papillomas) affecting the skin and mucous membranes of sheep. It is caused by Ovine Papillomavirus (OPV).

2. Etiology:

- **Causative agent:** Ovine Papillomavirus
- **Family:** Papillomaviridae
- **Type:** Non-enveloped, double-stranded DNA virus
- **Resistance:**
 - Resistant to environmental conditions
 - Survives for months in dry environments
 - Sensitive to disinfectants like formalin and sodium hypochlorite

3. Epidemiology:

- Occurs worldwide
- More common in young sheep (< 2 years)
- Higher incidence in:
 - Poor hygiene conditions
 - Overcrowded flocks
 - Animals with skin abrasions

Transmission

- Direct contact between animals
- Indirect transmission via:
 - Fomites (shearing equipment, tagging tools)
 - Contaminated housing
- Virus enters through skin wounds or abrasions

4. Pathogenesis:

1. Virus enters through micro-abrasions
2. Infects basal epithelial cells
3. Viral DNA integrates and causes:
 - Hyperplasia
 - Excessive keratinocyte proliferation
4. Formation of papillomas(warts)

 **Usually benign and self-limiting**

 **Rarely may undergo malignant transformation (very uncommon in sheep)**

5. Clinical Signs:

Incubation Period

- 1–3 months

Lesion Characteristics

- Cauliflower-like growths
- Pedunculated or sessile masses
- Rough, dry surface

Common Sites

- Lips
- Muzzle
- Eyelids
- Ears
- Teats and udder
- Occasionally genital region

General Condition

- Usually normal
- No systemic illness
- May interfere with:
- Grazing (if oral lesions severe)
- Milking (if teat lesions present)

6. Differential Diagnosis:

- Contagious ecthyma (Orf)
- Squamous cell carcinoma
- Fibroma
- Cutaneous abscess
- Dermatophytosis

Key differentiation:

Orf lesions are proliferative but accompanied by scab formation and inflammation, whereas papillomas are dry, firm, and non-inflammatory.

7. Diagnosis:

Clinical Diagnosis

- Based on appearance and age group
- Typical wart-like lesions

Laboratory Diagnosis (if required)

- Histopathology
- Hyperkeratosis
- Acanthosis
- Koilocytosis
- PCR for viral DNA (research/confirmation cases)

8. Treatment:

⚠ Most cases are self-limiting and regress within 2–6 months.

Field Management

- No specific antiviral therapy
- Improve hygiene
- Isolate severely affected animals

Surgical Removal

- Indicated when:
- Large obstructive growths
- Interference with feeding or milking
- Use aseptic technique

Autogenous Vaccine

- Prepared from wart tissue
- Occasionally used in persistent flock outbreaks

9. Control & Prevention:

- Avoid skin trauma during shearing
- Disinfect instruments
- Isolate affected animals
- Maintain proper nutrition
- Reduce overcrowding

No widely available commercial vaccine for sheep papillomatosis.

10. Economic Importance:

- Usually minor economic impact
- Losses occur when:
- Lesions interfere with suckling or milking
- Secondary bacterial infection develops
- Carcass condemnation (rare)

11. Zoonotic Significance:

✗ Not zoonotic

(Ovine papillomavirus does not infect humans)

12. Prognosis

- ✓ Excellent
- ✓ Most cases regress spontaneously
- ✓ Immunity develops after recovery

DIFFERENT TYPES OF TUMORS IN SHEEP

(Symptoms, Diagnosis, Differential Dx, Treatment & Control)

1 Squamous Cell Carcinoma (SCC)

Common Sites:

- Eyelids
- Vulva
- Perineum
- Skin (especially non-pigmented areas)



- **Symptoms:**
 - Clinical appearance
 - Ulcerative mass: • Biopsy (keratin pearls).
 - Non-healing wound
 - Progressive enlargement.
 - Weight loss in advanced cases.

- **Diagnosis:**
 - ▶ Early surgical excision
 - ▶ Clinical appearance. ▶ Cryotherapy (small lesions)
 - ▶ Biopsy (keratin pearls) ▶ Culling (advanced cases)

Most tumors are uncommon

3 Lymphoma (Enzootic/Sporadic)

Symptoms:

- Enlarged lymph nodes.
- Weight loss.
- Organ enlargement.



- **Diagnosis:**
 - ▶ Clinical examination,
 - ▶ Blood smear,
 - ▶ Lymph node biopsy.

- **Differential Diagnosis:**
 - ▶ Caseous lymphadenitis.
 - Caseous lymphadenitis. ▶ Abscess.
 - Abscess. ▶ Tuberculosis.

5 Fibroma / Fibrosarcoma

Symptoms:

- Firm, slow-growing skin mass.
- Usually solitary.



- **Diagnosis:**
 - ▶ Biopsy
 - ▶ Surgical Excision

2 Ovine Pulmonary Adenocarcinoma (Jaagsiekte)

Cause:

Jaagsiekte Sheep Retrovirus (JSRV)



Symptoms:

- Progressive respiratory distress.
- Chronic weight loss.
- Nasal discharge (frothy fluid when lifting hindquarters).
- Exercise intolerance.

- **Diagnosis:**
 - ▶ Clinical signs; ▶ Thoracic ultrasound,
 - ▶ Postmortem; Firm gray lung nodules.
 - ▶ PCR (confirmat).

- **Differential Diagnosis:**
 - ▶ Major effective disease in some regions.

- Maedi-Viřna.
- Chronic pneumonia Test and cull: Separate young stock
- Lung Abscess. Maintain closed flock
- Strict biosecurity

4 Papilloma (Warts)

- Ovine Papillomavirus

Symptoms:

- Firm, slow-growing skin
- Usually solitary.



- **Diagnosis:**
 - ▶ Clinical appearance
 - ▶ Histopathology if required

- **Differential Diagnosis:**
 - ▶ Orf, ▶ Fibroma.
 - Orf, ▶ SCC.
 - Fibroma.

7 Ocular Tumors

- ▶ Includes: SCC. • Melanoma.

Symptoms:

- Eye mass: 1 • Clinical exam.
- Tearing. 1 • Biopsy,
- Blindness:



- ▶ Surgical removal if obstructive.

GENERAL DIAGNOSTIC & TREATMENT GUIDELINES

1. History & Age
2. Growth rate
3. Location
4. Palpation (imm acit), Biopsy (gold ctandard)
6. Histopatnology

TEST, BIOPSY & EARLY REMOVAL - KEYS TO SUCCESS

ECONOMIC IMPORTANCE

- Good Fisk- management:
- Avoid chronic irritation.
- Biosecurity (for viral tumors like OPA).
- Genetic selection.

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43. Tumours in Sheep

(Symptoms, Diagnosis, Differential Diagnosis, Treatment & Control)

Tumors in sheep are generally uncommon compared to dogs and cattle, but certain neoplasms have economic and clinical importance.

1. Squamous Cell Carcinoma (SCC):

Common Sites

- Eyelids
- Vulva
- Perineum
- Skin (especially non-pigmented areas)

Symptoms

- Ulcerative, proliferative mass
- Non-healing wound
- Bleeding lesion
- Progressive enlargement
- Weight loss in advanced cases

Diagnosis

- Clinical appearance
- Biopsy and histopathology (keratin pearls)

Differential Diagnosis

- Papilloma
- Orf (Contagious ecthyma)
- Chronic abscess
- Fibroma

Treatment

- Early surgical excision
- Cryotherapy (small lesions)
- Culling (advanced cases)

Control

- Avoid excessive UV exposure
- Select pigmented breeds
- Early detection and removal

2 ☐☐ Ovine Pulmonary Adenocarcinoma:(Jaagsiekte)

Cause

- Jaagsiekte Sheep Retrovirus (JSRV)

Symptoms

- Progressive respiratory distress
- Chronic weight loss
- Nasal discharge (frothy fluid on lifting hindquarters)
- Exercise intolerance

Diagnosis

- Clinical signs
- Thoracic ultrasound
- **Postmortem:** Firm gray lung nodules
- PCR (confirmation)

Differential Diagnosis

- Maedi-Visna
- Chronic pneumonia
- Lung abscess

Treatment

✗ No effective treatment

Control

- Test and cull
- Separate young stock
- Maintain closed flock
- Strict biosecurity

⚠ Major economic disease in some regions.

3 ☐☐ Lymphoma (Enzootic/ Sporadic):

Symptoms

- Enlarged lymph nodes
- Weight loss
- Weakness
- Organ enlargement

Diagnosis

- Clinical examination

- Blood smear
- Lymph node biopsy

Differential Diagnosis

- Caseous lymphadenitis
- Abscess
- Tuberculosis

Treatment

✗ Usually not economical, Culling recommended

Control

- Remove affected animals
- Improve herd health management

4 Papilloma (Warts):

Cause

- Ovine Papillomavirus

Symptoms

- Cauliflower-like skin growths
- Common in young sheep
- Usually non-painful

Diagnosis

- Clinical appearance
- Histopathology if required

Differential Diagnosis

- Orf
- Fibroma
- SCC

Treatment

- Usually self-limiting
- Surgical removal if obstructive

Control

- Improve hygiene
- Disinfect equipment
- Avoid skin trauma

5 ☐☐ Fibroma / Fibrosarcoma:

Symptoms

- Firm, slow-growing skin mass
- Usually solitary

Diagnosis

- Biopsy

Differential Diagnosis

- Papilloma
- Abscess
- SCC

Treatment

- Surgical excision

Control

- Early removal

6 ☐☐ Melanoma:

More common in pigmented breeds

Symptoms

- Dark pigmented mass
- May ulcerate

Diagnosis

- Biopsy

Differential Diagnosis

- Pigmented SCC
- Hematoma

Treatment

- Surgical excision

Control

- Genetic selection

7 ☐☐ Ocular Tumours:

Includes: SCC, Melanoma

Symptoms

- Eye mass
- Tearing
- Blindness

Diagnosis

- Clinical exam
- Biopsy

Treatment

- Surgical removal
- Enucleation (advanced cases)

General Diagnostic Approach to Tumors in Sheep:

1. History & Age
2. Growth rate
3. Location
4. Palpation (firm vs soft)
5. Biopsy (gold standard)
6. Histopathology

General Treatment Principles:

- Early detection is critical
- Surgical excision is main treatment
- Advanced tumors → culling often economical
- No routine chemotherapy in sheep

Control & Prevention:

- Good flock management
- Avoid chronic irritation
- Biosecurity (for viral tumors like OPA)
- Genetic selection
- Remove affected animals early

Economic Importance:

- Reduced productivity
- Carcass condemnation
- Mortality (OPA)
- Treatment cost

CAPTURE MYOPATHY IN SHEEP

A Stress-Induced, Potentially Fatal Condition

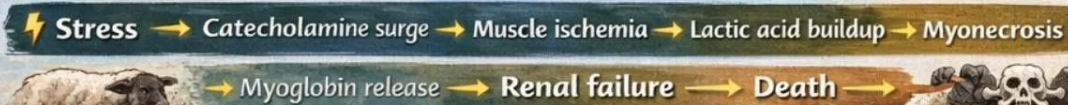


What is Capture Myopathy?

Capture Myopathy is a metabolic muscle disorder caused by severe stress, fear, struggle, and exhaustion during handling, restraint, transport, chasing or shearing.

★ Results in muscle damage, metabolic acidosis, kidney failure & sudden death.

Pathophysiology (Simple Flow)



Clinical Signs

- ✓ Rapid breathing (tachypnoea)
- ✓ Muscle tremors
- ✓ Hyperthermia
- ✓ Anxiety, panic
- ✓ Stiff or reluctant movement



Clinical Signs

- ✓ Severe muscle rigidity
- ✓ Recumbency
- ✓ Dark / coffee-colored urine (myoglobinuria)
- ✓ Collapse
- ✓ Sudden death



ANIMAL WELFARE MESSAGE

Capture Myopathy is **PREVENTABLE**.
 Stress kills sheep silently.

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Risk Factors in Sheep

- ✓ Rough handling & prolonged restraint
- ✓ Chasing during catching or transport
- ✓ Heat stress & dehydration
- ✓ Long-distance transport
- ✓ Overcrowding
- ✓ Poor handling facilities
- ✓ Temperamental or wild sheep
- ✓ Pregnant or debilitated animals

Laboratory Findings

- ✓ ↑ Creatine Kinase (CK)
- ✓ ↑ AST
- ✓ Metabolic acidosis
- ✓ Myoglobinuria
- ✓ Electrolyte imbalance
- ✓ Urinalysis



Diagnosis

- ✓ History of recent stress or exertion
- ✓ Clinical signs
- ✓ Blood biochemistry
- ✓ Urinalysis



Emergency Management (Field Level)

- ✓ Stop all handling immediately
- ✓ Early intervention improves outcome
- ✓ Survivors may have permanent muscle damage:



Stress + Struggle + Heat = Death Risk



44. Capture Myopathy in Sheep

Capture myopathy is a stress-induced muscular disorder that occurs when sheep experience extreme physical exertion, fear, or prolonged handling. It is commonly seen during capture, transport, chasing, or handling of wild or semi-managed animals.

What is Capture Myopathy?

Capture myopathy is a metabolic muscle disease caused by excessive stress leading to:

- Muscle damage
- Metabolic acidosis
- Circulatory failure

It can occur in both domestic sheep and wild sheep species.

Causes

Common triggering factors include:

- Prolonged chasing during herding
- Rough handling or restraint
- Long-distance transportation
- Extreme fear or panic
- Overcrowding in holding pens
- High environmental temperatures

These stresses cause excessive muscle activity and oxygen depletion, leading to muscle degeneration.

Pathophysiology

During severe stress:

- Adrenaline and stress hormones increase
- Muscles consume large amounts of energy
- Oxygen deficiency develops
- Lactic acid accumulates in muscles
- Muscle fibers undergo necrosis

This leads to systemic metabolic disturbances.

Clinical Signs

Signs may appear immediately or within several hours to days.

- Weakness and reluctance to move
- Muscle stiffness
- Rapid breathing
- Elevated body temperature

- Dark-colored urine (myoglobinuria)
- Recumbency
- Sudden death in severe cases

Diagnosis

Diagnosis is based on:

- History of recent stress or capture
- Clinical signs
- Elevated muscle enzymes (CK, AST)
- Post-mortem findings of muscle degeneration

Post-Mortem Findings

- Pale or darkened skeletal muscles
- Muscle hemorrhage
- Kidney damage due to myoglobin
- Evidence of metabolic acidosis

Prevention

Prevention is more effective than treatment.

Key preventive measures:

- ✓ Gentle handling practices
- ✓ Avoid prolonged chasing of sheep
- ✓ Provide rest during transportation
- ✓ Maintain adequate space in holding pens
- ✓ Avoid handling during extreme heat
- ✓ Gradual movement of flocks

Treatment (Supportive)

Treatment success is often limited but may include:

- Fluid therapy
- Anti-inflammatory drugs
- Sedation to reduce stress
- Vitamin E and Selenium supplementation
- Cooling and rest

Early veterinary intervention improves survival chances.

Key Message for Shepherds

Minimize stress during handling and transport to prevent capture myopathy.

Stress management is essential for animal welfare and flock productivity.

Sudden Death of Sheep

Time of Occurrence & Diagnostic Significance

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1 Sudden Death in Early Morning (Night–Early Dawn)

Likely Causes

- Enterotoxemia (Pulpy kidney disease)
- Hypocalcemia
- Hypomagnesemia
- Acute bloat
- Snake bite
- Predator attack



Key Clue: Death after heavy feeding → Enterotoxemia

3 Sudden Death During Morning Hours

Likely Causes

- Enterotoxemia
- Acute pneumonia
- Pasteurellosis
- Electrocution
- Toxic plant ingestion during grazing



Key Clue: Group deaths after grazing → Poisoning

3 Sudden Death During Afternoon (Hot Hours)

Likely Causes

- Heat stroke
- Acute bloat
- Nitrate / Nitrite poisoning
- Water deprivation
- Exertional stress



Key Clue: Hot weather + exertion → Heat stress

Key Message:

The time of **sudden** death provides vital diagnostic clues – proper history taking is as important as post-mortem examination.

2 Sudden Death During Morning Hours

Likely Causes

- Enterotoxemia
- Acute pneumonia
- Pasteurellosis
- Electrocution
- Toxic plant ingestion during grazing



Key Clue: Group deaths after grazing → Poisoning

4 Sudden Death During Evening

Likely Causes

- Acute enterotoxemia
- Anaphylactic reaction (drug/vaccine)
- Traumatic injuries
- Electrocution



Key Clue: Death within minutes-hours after injection → Anaphylaxis

5 Sudden Death During Night

Likely Causes

- Snake bite
- Predator attack
- Acute bloat
- Enterotoxemia
- Poisoning



Key Clue: Swelling at bite site → Snake bite

Time-Based Diagnostic Approach (Feld's Tip)

Time of Death	Most Likely Cause
Night/ Early morning	Enterotoxemia, bloat
After grazing	Poisoning
Hot afternoon	Heat stroke
Post vaccination	Anaphylaxis
Any time bleeding	Anthrax

45. Sudden Death of Sheep

(Time of Occurrence & Diagnostic Significance)

1. Sudden Death in Early Morning (Night–Early Dawn):

Likely Causes

- Enterotoxemia (Pulpy kidney disease)
- Hypocalcemia
- Hypomagnesemia
- Acute bloat
- Snake bite
- Predator attack

Diagnostic Significance

- Animals appear normal the previous evening
- Often found dead without struggle
- Full rumen, rapid post-mortem changes
- Strong association with high concentrate feeding

 **Key Clue:** Death after heavy feeding → Enterotoxemia

2. Sudden Death During Morning Hours:

Likely Causes

- Enterotoxemia
- Acute pneumonia
- Pasteurellosis
- Electrocution
- Toxic plant ingestion during grazing

Diagnostic Significance

- Death soon after grazing or movement
- Froth at nostrils in respiratory causes
- Multiple animals affected → Infectious or toxic cause

 **Key Clue:** Group deaths after grazing → Poisoning

3. Sudden Death During Afternoon (Hot Hours):

Likely Causes

- Heat stroke
- Acute bloat
- Nitrate / Nitrite poisoning
- Water deprivation

- Exertional stress

Diagnostic Significance

- Occurs during high ambient temperature
- Panting, open-mouth breathing before death
- Dark or chocolate-colored blood (nitrate poisoning)

 **Key Clue:** Hot weather + exertion → Heat stress

4. Sudden Death During Evening:

Likely Causes

- Acute enterotoxemia
- Anaphylactic reaction (drug/vaccine)
- Traumatic injuries
- Electrocution

Diagnostic Significance

- Often follows vaccination, injection, or handling
- History of recent treatment is crucial

 **Key Clue:** Death within minutes–hours after injection → Anaphylaxis

5. Sudden Death During Night:

Likely Causes

- Snake bite
- Predator attack
- Acute bloat
- Enterotoxemia
- Poisoning

Diagnostic Significance

- Bite marks, bleeding, swelling
- Signs of struggle or enclosure damage
- Often single-animal death

 **Key Clue:** Swelling at bite site → Snake bite

6. Sudden Death at Any Time of Day:

Possible Causes

- Anthrax
- Acute septicemia
- Clostridial diseases
- Lightning strike

- Severe toxicosis

⚠ Important Warning:

- Do NOT open carcass if anthrax suspected
- Signs: bleeding from natural orifices, no rigor mortis

ANTIMICROBIAL RESISTANCE (AMR)



— IN SHEEP FLOCKS —



A Growing Threat to **Animal & Public Health**

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WHAT IS AMR ?

- When bacteria stop responding to drugs (Antibiotics/Dewormers)



IMPACTS

- ↓ Treatment Failure
- ↓ Higher Mortality
- ↓ Drug Residues in Meat/Milk
- ↓ Economic Losses
- ↓ Risk to Human Health



CAUSES OF AMR



Overuse & Misuse of Drugs



Incorrect Dose & Duration



No Lab Diagnosis



Poor Farm Hygiene



Uncontrolled Deworming



COMMON DISEASES & RESISTANT BACTERIA



Pneumonia



Diarrhoea



Mastitis



Foot Rot



E. coli, Salmonella, Staph., Pasteurella



PREVENTION & CONTROL



Use Antibiotics Wisely (Vet Prescription Only)



Lab Culture & Sensitivity Test



Vaccination & Biosecurity



Good Nutrition & Hygiene



Regular Deworming (Balanced)

ONE HEALTH — Protect Animals • Humans • Environment



USE ANTIMICROBIALS CAREFULLY —

SAVE THEIR POWER!

FieldvetDrMSS



46. Antimicrobial Resistance (AMR)

What is Antimicrobial Resistance (AMR)?

Antimicrobial Resistance occurs when bacteria, parasites, or other microbes evolve to survive drugs that were once effective. In sheep farming, this mainly results from misuse or overuse of antibiotics and antiparasitic drugs. AMR threatens animal health, farm productivity, and public health under the One Health framework.

Why AMR is Important in Sheep Farming

- Reduced effectiveness of antibiotics and dewormers
- Increased treatment failures
- Higher mortality in lambs
- Increased treatment costs
- Risk of drug residues in meat and milk
- Spread of resistant microbes to humans and environment

Common Causes of AMR in Sheep Flocks

- Overuse of antibiotics
- Routine use without diagnosis
- Incorrect dosage
- Under-dosing promotes resistant bacteria
- Incomplete treatment courses
- Frequent prophylactic use
- Antibiotics used for prevention instead of treatment
- Use of the same drug repeatedly
- Uncontrolled use of dewormers
- Leading to anthelmintic resistance

Diseases in Sheep Where AMR is Emerging

- Pneumonia complex
- Enteritis / diarrhoea in lambs
- Mastitis
- Foot rot
- Caseous lymphadenitis

Common bacteria involved:

- Escherichia coli
- Pasteurella multocida
- Staphylococcus aureus
- Salmonella spp.

Signs That AMR May Be Present

- Antibiotic treatment fails repeatedly
- Disease reappears soon after treatment
- Higher number of animals affected
- Slow recovery of treated animals

How to Prevent AMR in Sheep Farms

1 Use Antibiotics Responsibly Only under veterinary prescription

Use correct dose and duration

2 Laboratory Diagnosis

Perform culture and sensitivity tests whenever possible.

3 Improve Farm Biosecurity

- Isolate sick animals
- Quarantine new animals

4 Vaccination Programs

Vaccinate against diseases such as:

- Sheep pox
- Enterotoxaemia
- Pasteurellosis

5 Good Nutrition

Balanced feeding improves immune resistance

6 Hygienic Management

- Clean sheds
- Dry bedding
- Proper ventilation

Reduce Dependence on Antibiotics

Alternative strategies include:

- Vaccination
- Probiotics
- Mineral supplementation
- Improved grazing management
- Integrated parasite management

AMR and One Health

Antimicrobial resistance spreads between:

- Animals
- Humans
- Environment

Responsible antimicrobial use protects:

- ✓ Animal health
- ✓ Human health
- ✓ Ecosystem health

Key Message

“Use Antimicrobials Carefully Today to Protect Their Power Tomorrow.”

Role of Veterinarians

Veterinarians help by:

- Diagnosing diseases accurately
- Advising correct drug use
- Monitoring drug resistance
- Educating farmers on judicious antimicrobial use

Conclusion:

Toward a Resilient Future in Sheep Husbandry

The management of disease in sheep is a continuous journey that transitions from the microscopic level of the laboratory to the expansive reality of the grazing field. It is a discipline that requires equal parts vigilance, empathy, and scientific rigor. As we conclude this exploration, it is clear that the "art" of management lies in the ability to harmonize technical protocols with the unique rhythms of the flock.

The Pillars of Sustained Health

- **Proactive over Reactive:** The shift from treating sick individuals to managing the health of the entire population is the hallmark of a successful enterprise. Through strategic vaccination, rigorous biosecurity, and integrated parasite management, we reduce the burden of disease before it can take hold.
- **Welfare as a Diagnostic Tool:** Adhering to the Five Freedoms is not a luxury but a clinical necessity. High welfare standards directly correlate with lower cortisol levels and stronger immune responses, making animal rights a cornerstone of animal productivity.
- **The Power of Indigenous Resilience:** Our native breeds carry centuries of adapted immunity within their DNA. Protecting and promoting these indigenous genetic resources is our most sustainable defense against the changing landscapes of climate and disease.

A Vision for the "One Health" Practitioner

As we look forward, the role of the veterinarian and the farm manager must continue to evolve. By embracing Precision Livestock Farming and data-driven diagnostics, we can ensure that every intervention is as targeted and minimal as possible, safeguarding our environment from chemical runoff and our communities from antimicrobial resistance.

Ultimately, the goal of disease management is to create a system where sheep can thrive in their natural state, supported by a framework of "One Health" that protects the animal, the owner, and the ecosystem alike. Let this work serve as a reminder that in the service of these animals, our greatest tools are observation, education, and an unwavering commitment to the ethics of care.

Dedicated to the Advancement of Veterinary Knowledge.